BIOLOGY

STUDENT TEXTBOOK
GRADE 12

2023

FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA
MINISTRY OF EDUCATION

BIOLOGY STUDENT TEXTBOOK GRADE 12

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Biology

STUDENT TEXTBOOK
GRADE 12

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Unit 1: Applications of Biology

Unit learning outcome

At the end of this unit, the will be able to:

- Define biotechnology.
- Describe conservation of natural resources
- Describe fuud and nutrition security
- Explain conscious citizen and global citizen
- Describe biotechnology and genetic engineering
- Explain the role of biotechnology to secure the food of the society
- Explain some of the traditional uses of biotechnology
- Explain the role of microorganisms in making bread, teji, tella, wine and beer
- Describe the application biotechnology in agriculture
- Explain what genetically modified organisms means
- Describe the benefits of creating transgenic crops.
- Describe the application of environmental biotechnology
- explain the process and importance of gene therapy
- explain bioremediation and biopesticides.
- Explain the modern application of biotechnology

UNIT 1: APPLICATION OF BIOLOGY

The applications of biology in our daily to day life are numerous. Biology is the science in charge of studying all living beings. It helps us to understand every living organism, from the smallest bacteria to the blue whales. Biologists often focus on a different subset of living organisms, such as birds, plants, or bacteria, to determine their character.

Biological science is very useful science to determine where some diseases and pests come from, such as infections, animal pathologies, and damage to plants. Biology covers the study of the functions of living organisms, the evolution of species, the factors that produce diseases, as well as the discovery of new drugs.

This discipline allows human beings to explore topics such as genetic engineering, research applications with mother cells, and global warming. It also helps to understand nature and how humans, animals, and plants interact in life.

Biology offers a vision of how living things evolve. Understanding the rates of extinction and how a species depends on and affects the habitat where it lives improves the effectiveness of conservation efforts.

A practical application of biology with which most people are familiar is hand washing, domestication, and traditional fermentation. Regular soap washing removes acquired microbes from the skin and helps control the spread of infectious diseases.

In addition, another application of biology is the set of instructions given to take all the pills of antibiotics in a recipe. Then, to broaden your understanding of this further you will look at the more everyday examples of the application of this science in the next topics.

1.1. Application in Conservation of natural resources

After the successful completion of this section, you will be able to:

- Define natural resources.
- Identify renewable and non-renewable resources.
- Classify natural resources as renewable and non-renewable.
- Describe the role of conservation in protecting and preserving our natural environment.
- Discuss the role of Biology in the conservation of natural resources

Keywords

Natural resources: resources (actual and potential) supplied by nature

Renewable: Capable of being produced indefinitely. It will not be used as there will always be more available.

Non-renewable: There is a definite amount available. Once used, it cannot be easily made or replaced.

Extinct: No longer in existence.

Conservation: the act of preserving, guarding or protecting

Extinict: No longer in existence

Conservation is the careful maintenance and wise use of natural resources to prevent them from disappearing. Natural resources are physical supplies that exist in nature and include soil, water, air, plants, animals, and energy. Conservation biology is a mission-oriented science that focuses on how to protect and restore biodiversity, or the diversity of life on Earth. Ethiopia has many natural resources, such as, gold, platinum, potash, limestone, natural gas, coal, in addition to lakes, volcanic ocean, and water falls (Figure 1.1). Timber, many different crop plants, coffee plantations, many different species of animals and plants, and many different birds, make up rich ecosystems (Figure 1. 2).

Self-questioning

- Do you agree with the conservations of natural resources? Why?
- Explain the concept that human should practice sustainable management of natural resources?
- How the knowledge and application of conservation is critical to the survival of our people?

Natural resources can be classified as renewable or non-renewable. Renewable resources are mainly living things and their products. The main sources of renewable resources are sun, wind, water, the earth's heat (geothermal) and biomass (which relates to living things). When managed carefully, they can be used, reused and replaced. Examples of renewable resources are crop plants, trees, cattle, and chickens. Non-renewable resources are natural substances that are not replaced at the same speed as that are used up. They are finite resources. Examples of non-renewable resources include metals such as gold and iron and fossil fuels such as oil, natural gas, and coal.

.

Hawassa Lake ` Volcanoes with sulfur deposits in Ethiopia (Ertale) Tis Abay

Figure 1.1. Natural landscapes in Ethiopia

Conserve the non-renewable natural resources of Ethiopia is economically advantageous to mine metals and fossil fuels, doing so have a negative impact on the the earth (i.e. climate change) and that by conserving them instead and using renewable energy sources would be more beneficial to the planet. Even renewable

resources can be lost if we do not manage them carefully. What should be done to conserve trees? coppicing and having a continual cycle of planting trees to replace the ones being cut down, bearing in mind that there is a lag time for trees to grow.

Therefore, biologists takes a position in conserving species and saving them from extinction through the preservation of animals and plants in terms of zoos and seed banksand by stopping the destruction of their natural habitats so the populations are able to thrive.

Activity 1.1: Work in group and report what are natural resources of Ethiopia and discuss their economic importance

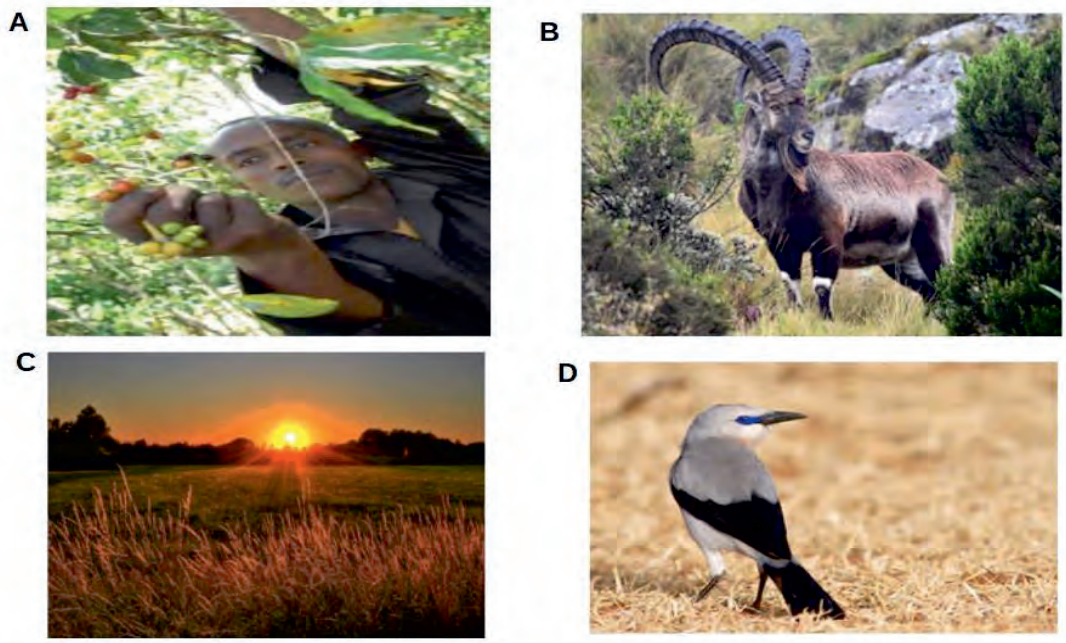


Figure 1.2. Some important renewable resources in Ethiopia: Coffee (a), Walia Ibex (*Capra walie*) (b), Sunlight as a source of energy (c), Ethiopian Bush crow (*Zavattariornis stresemanni*).

To protect our natural resources, both in Ethiopia and around the world, become more aware of the need for the conservation of natural resources, both non-renewable and renewable. Conservation is protecting and preserving our natural environment so that non-renewable resources are used sparingly, and renewable resources should be appropriately managed. Otherwise, they can last for an extended period of time in the future

Activity 1.2: Natural resources of Ethiopia

Have a brainstorming session and think of as many natural resources of Ethiopia as you can. Given that read what the types of renewable and non-renewable resources are, and specify with a specific examples. Forexample name rivers, wind that goes across named plain, and natural gas from named place.

Now divide them up into renewable resources and nonrenewable resources.

Make a poster or collage to show these natural resources. Divide the poster into renewable and non-renewable resources.

1.2. Food and nutrition security

After the successful completion of this section, you will be able to:

- Define food security.
- Practice school-based gardening as one strategy of addressing food security.

Food security, as defined by the United Nations' Committee on World Food Security, is a state in which when all the people have physical, social, and economic access to sufficient, safe, and nutritious food that meets their food preferences and dietary needs for an active and healthy life at all times. However, food insecurity is often rooted in poverty and has long-term impacts on the potential of families, communities, and countries to develop and prosper. Prolonged undernourishment stunts growth (Figures 1.4 & 1.5), slows cognitive development and increases susceptibility to illness.

Key word

Food Security: The state in which all people have physical, social and economic access to sufficient, safe and nutritious food.

Self-questioning

- Which biological application is important for maintaining food security?

Nutrition security is a situation whereby individuals have access to sufficient, safe and nutritious foods, safe water and adequate sanitation, the ability to access health care services, and knowledge of sound household and community practices in child care, food storage and preparation and hygiene.

Therefore, food security is ensured only if: enough food is available for all in a county; when all individuals have the capacity to buy food of acceptable quality, and when there is no barrier to access food. Therefore, Biology plays a key role in producing high-nutrient staple crops and developing new products that can combat malnutrition, and thereby improving food utilization. Biotechnologests design the manufacturing processesand machinery used to produce food and drink. Biologists work to ensure food security within a country, and across the world, through these innovations. This allws ptoducts to have consistant flavor, color, and texture to be produced in large quantities.

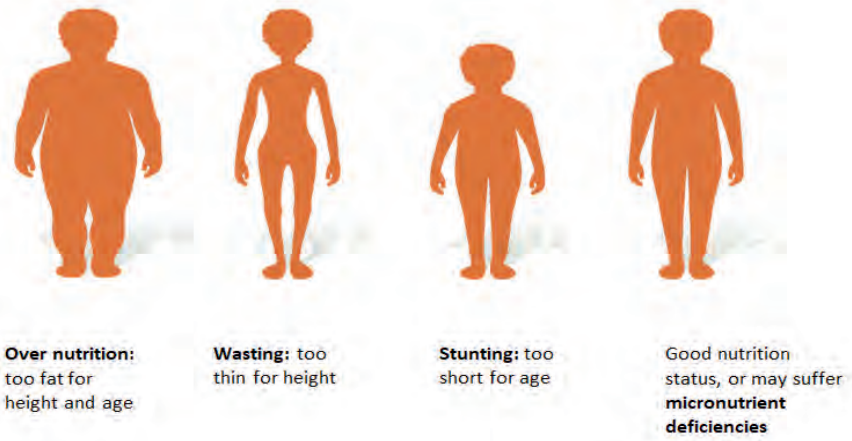


Figure 1.3 Types of mal nutrition (Source: FAD 2015)

Activity 1.3a. Individual work.

Write a short set of notes in order to answer the following question(s). Use the internet, as well as digital and print educational resources.

- What factor influence food availability?
- What is the difference between food security, nutritional security, and food insecurity?
- What are the effects of food and nutritional insecurity?

Activity 1. 3b:

Make a group of 5-10 students to grow different vegetables in the school compound on the ground/in pots to easily demonstrate how one can secure food in a small area.

1.3. Creating conscious citizens and ensuring sustainable development

After the successful completion of this section, you will be able to:

- Define conscious citizen.
- Explain the role of biology in creating a conscious citizen.
- Describe the meaning of a conscious global citizen.
- Explain how sustainable development can be ensured AND/OR Define

Conscious citizen: Person who places value on being fully human while connecting with a higher purpose.

A conscious citizen is one who places value on being fully human while connecting with a higher purpose; one who values human life and a relationship with all living things, and one who takes the responsibility for transforming skills into action through ethical decision making, to ultimately improve life and living on the planet. Biology has a vital role in creating conscious citizen by expanding awareness of the social, global, and environmental conditions.

Biology empowers conscious citizens to assume personal responsibility by engaging in and being committed to initiating positive impact. Nowadays, the interplay of biology and technology (biotechnology) has become vital to facilitating sustainable development initiatives and conscious citizens will use biotechnology applications to improve life and living on the planet. Conscious citizen biologists develop innovative and cost effective bio-based technologies which consume fewer resources, incorporate recycling, reuse components and reduce production of wastes, and use strategies for sustaining a greener earth and improving food production.

Self-questioning

- Why do we need to be conscious global citizens?
- How can we become a conscientious citizen?

Activity 1.3c

In a group of 5-10 students, discuss how students like you can be a conscious global citizen. Then prepare a report and present it to the class..

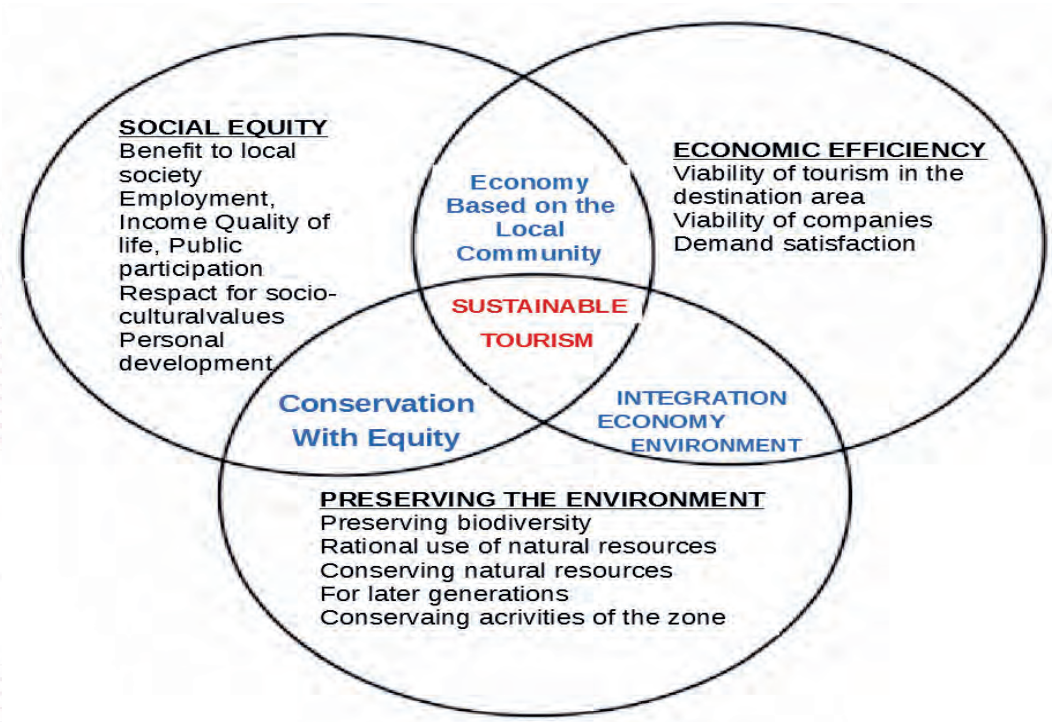


Figure 1.4. A model for sustainable development

According to the World Conservation Union (IUCN, 2006), the three dimensions of sustainability (economic, social and environmental) are represented either as pillars, embedded circles or in the popular Venn diagram (Fig 1.4) of three overlapping circles.

The conscious global citizens (biologists) in Ethiopia involved in sustainable development through protecting, managing and monitoring the existing resources of our land including: analysing soil, water and air for chemical pollution. finding ways to clean up pollution.

identifying, recording and monitoring the plants and animals that share the land we use.

1.4. Applications in biotechnology

Biotechnology is the application of technologies that involve the use of living organisms, or products from living organisms, for the development of products that benefit humans. Genetically modified organisms (GMOs) have received genetic material via recombinant DNA technology. If an organism has received genetic material from a different species, it is called a transgenic organism. A gene from one species that is introduced into another species is called a transgene.

The organisms used may be genetically modifed to make them more suitable. Crops can be modifed to increase yields and to obtain novel products. Biotechnology can be used in the prevention and mitigation of contamination from industrial, agricultural and municipal wastes. Biotechnology can also be used in the diagnosis and treatment of disease.

(i). Application of Biology for Food processing and production

This method involves the increasing of foods productivity using Microorganisms. A technology that shows some promise in increasing world food productivity. Forexample **single-cell protein (SCP)**. This material is produced from waste materials such as molasses from sugar refining, petroleum by-products, and agricultural wastes. In developed countries, an animal feed called Pruteen is produced by mass culture of the bacterium *Methylophilus methylotrophus.* Mycoprotein, a product made from the fungus *Fusarium venenatum,* is also sold there. The filamentous texture of this product makes it a likely candidate for producing meat substitutes for human consumption.

After the successful completion of this section, learner will be able to :

- Define food security.
- Practice/Apply school-based gardening as one strategy of addressing food security.

Health food stores carry bottles of dark green pellets or powder that is a culture of a spiral-shaped cyanobacterium called *Spirulina.* This microbe is harvested from the surface of lakes and ponds, where it grows in great mats. In some parts of Africa, Asia, and Mexico, *Spirulina* has become a viable alternative to green plants as a primary nutrient source. It can be eaten in its natural form or added to other foods and beverages.

Vitamins are also produced using biotechnology. Vitamin C was the first vitamin to be produced during a fermentation process by using **bacteria.** Previously, Vitamin B_{12} or cyanocobalamin and B_2 or Riboflavin were obtained from animal liver extract. But, nowadys the production of vitamin-B_{12} involved fermentation by propionic bacteria. In nature B_2 is found in cereals, vegetables and yeast but the yield of B_2 can be enhanced hundred to three hundred fold by using microbes.

(a). Dairy Products

Microorganisms are used in making a wide variety of dairy products. Cultured buttermilk, popular in the developed countires, is made by adding *Streptococcus cremoris* to pasteurized skim milk and allowing fermentation to occur until the desired consistency, flavor, and acidity are reached. Other organisms—*Streptococcus lactis, S. diacetylactis and Leuconostoc citrovorum, L. cremoris, or L. dextranicum*—make buttermilk with different flavors because of variations in fermentation products. Sour cream is made by adding one of these organisms to cream. Yogurt is made by adding *Streptococcus thermophilus* and *Lactobacillus bulgaricus* to milk. These organisms release still other products, and so yogurt has a different texture and flavor.

Fermented milk have been made for centuries in various countries in the globe, especially Africa, Asia, and Eastern European. The products vary in acidity and alcohol content. Acidophilus milk is made by adding *Lactobacillus acidophilus* to sterile milk. Sterilization prevents uncontrolled fermentation by organisms that might already be present in nonsterilized milk. Bulgarian milk is made by *L. bulgaricus*; it is similar to buttermilk except that it is more acidic and lacks the flavor imparted by the leuconostocs.

(b). Fermented Meats

Microbes such as *Lactobacillus plantarum* and *Pediococcus cerevisiae* add flavor by fermenting meats such as salami, summer sausage, and Lebanon bologna. The heterolactic acid fermentation helps preserve the meat and also gives it a tangy flavor. Fungi such as *Penicillium* and *Aspergillus*, growing naturally on the surfaces of country hams, help to produce their distinctive flavor.

(c). Production of beer, wine, and spirits

Beer and wine are made by fermenting sugary juices; spirits, such as whiskey, gin, and rum, are made by fermenting

juices and distilling the fermented product. Distillation separates alcohol and other volatile substances from solid and nonvolatile substances. Strains of Saccharomyces are the fermenters for all alcoholic beverages. Many different strains have been developed, each having distinctive characteristics. Both the organisms and how they are used are carefully guarded brewers' secrets.

To make beer, cereal grains (usually barley) are malted (partially germinated) to increase the concentration of starch-digesting enzymes that provide the sugar for fermentation. Malted grain is crushed and mixed with hot water (about 65^0C), producing mash. After a few hours, a liquid extract called wort is separated from the mix. Hops (flower cones from the hop plant) are added to the wort for flavoring, and the mixture is boiled to stop enzyme action and precipitate proteins. A strain of Saccharomyces is added. Fermentation produces ethyl alcohol, carbon dioxide, and other substances, including amyl and isoamyl alcohols and acetic and butyric acids, which add to the flavor of the beer. After fermentation, the yeast is removed, and the beer is filtered, pasteurized, and bottled.

Most wine is made from juice extracted from grapes, although it can be made from any fruit and even from nuts or dandelion blossoms. Juice is treated with sulfur dioxide to kill any wild yeasts that may already be present. Sugar and a strain of Saccharomyces are then added, and fermentation proceeds. Although ethyl alcohol is the main product of fermentation, other products similar to those in beer add to the flavor of the wine. In both beer and wine, the particular characteristics of the juice and the yeast strain determine the flavor of the final product. When fermentation is completed, liquid wine is siphoned to separate it from yeast sediment and, if necessary, cleared with agents such as charcoal to remove suspended particles. Finally, it is bottled and aged in a cool place.

Spirits are made from the fermentation of a variety of foods, including malted barley (Scotch whiskey), rye (rye whiskey, gin), corn (bourbon), wine or fruit juice (brandy), potatoes (vodka), and molasses (rum). After fermentation, distilling separates alcohol and other volatile substances that impart flavor from the solid and nonvolatile substances. Because of distillation, the alcohol content of spirits ranges from 40 to 50% much higher than the typical 12% for wine and 6% for beer.

Activity 1.4: Group work

1. Discuss in your groups the ingredients, recipe, and biotechnology of the following traditional fermentation products.

i) Making "*Injera*"

ii) Making "*Areki*"

iii) Making "*Tej*"

iv) Making "*Tella*"

Self-questioning

1. List microorganisms used in each ermentation product and explain why?
2. What type of metabolites or bio-products are produced by those microorganisms?
3. What is back slopping traditional food fermentation? Can you give one example?

(d). Bread making

Microorganisms accomplish three functions in bread making:

1. leavening the flour-based dough,

2. imparting flavor and odor, and

3. conditioning the dough to make it workable.

Leavening is achieved primarily through the release of gas to produce a porous and spongy product. Without leavening, bread dough remains dense, flat, and hard. Although various microbes and leavening agents can be used, the most common ones are various strains of the baker's yeast *Saccharomyces cerevisiae.* Other gas-forming microbes such as coliform bacteria, certain *Clostridium* species, heterofermentative lactic acid bacteria, and wild yeasts can be employed, depending on the type of bread desired.

Yeast metabolism requires a source of fermentable sugar such as maltose or glucose. Because the yeast respires aerobically in bread dough, the chief products of maltose fermentation are carbon dioxide and water rather than alcohol (the main product in beer and wine). Other contributions to bread texture come from kneading, which incorporates air into the dough, and from microbial enzymes, which break down flour proteins (gluten) and give the dough elasticity.

Besides carbon dioxide production, bread fermentation generates other volatile organic acids and alcohols that impart delicate flavors and aromas. These are especially well developed in home-baked bread, which is leavened more slowly

than commercial bread. Yeasts and bacteria can also impart unique flavors, depending upon the culture mixture and baking techniques used. The pungent flavor of rye bread, for example, comes in part from starter cultures of lactic acid bacteria such as *Lactobacillus plantarum, L. brevis, L. bulgaricus, Leuconostoc mesenteroides,* and *Streptococcus thermophilus.* Sourdough bread gets its unique tang from *Lactobacillus sanfrancisc.*

Figure 1.5. sourdough bread

1.5. Genetic Engineering

Genetic engineering is the process of transferring DNA from one organism into another that results in a genetic modification; the production of a transgenic organism. Specific applications of genetic engineering are abundant and increasing rapidly in number. Genetic engineering is being used in the production of pharmaceuticals, gene therapy, and the development of transgenic plants and animals.

a. Animal breeding and transgenic animals, and plants and disease, and pest management

Animal breeding addresses the genetic value of livestock. Selecting for breeding animals with superior traits in growth rate, egg, meat, milk, or wool production, or with other desirable traits has revolutionized the livestock and plant production throughout the entire world (Figure 1.2). Animals can also be genetically modified (transgenic animals) for valuable traits. There are many potential applications of transgenic methodology in developing new and improved strains of livestock. Practical applications of transgenic technology in livestock production include enhancing the prolificacy and reproductive performance, increasing feed utilization and growth rate, improving carcass composition, improving milk production and/or composition, modification of hair or fiber, and increasing disease resistance in animals. The development of transgenic farm animals will allow more flexibility in the direct genetic manipulation of livestock. Gene transfer is a relatively rapid way - of altering the genome of domestic livestock

Figure 1.6. Biotechnology improves the quality of breeds

b. Tissue culture

Plants can be propagated quickly and in large quantity by **tissue culture** technique. Tissue culture, a method of biological research in which fragments of tissue from an animal or plant are transferred to an artificial environment in which they can

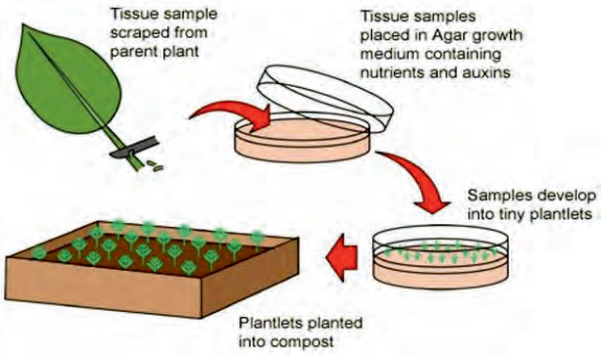


Figure 1.7. Tissue calture

continue to survive and function. The cultured tissue may consist of a single cell, a population of cells, or a whole or part of an organ. Plants produced in large amount using this technique include palm trees, orchids, bananas, and carrots. Using this technology large quantity of food with desired quality can be produced in reasonably little area. Therefore, tissue culture is seen as an important biotechnology for developing countries for the production of disease-free, high quality planting material and the rapid production of many uniform plants. (Figure 1.7).

1.6. Health and wellbeings

Human drugs such as insulin for diabetics, growth hormone for individuals with pituitary dwarfism, and tissue plasminogen activator for heart attack victims, as well as animal drugs like the growth hormones, bovine or porcine somatotropin, are being produced by the fermentation of transgenic bacteria that have received the appropriate gene from human, cow, or pig.

a. The manufacture of antibiotics

When micro-organisms are used for the production of antibiotics, it is not their fermentation products that are wanted, but complex organic compounds, called **antibiotics**, that they synthesise. Most of the antibiotics we use come from bacteria or fungi that live in the soil. The function of the antibiotics in this situation is not clear. One theory suggests that the chemicals help to suppress competition for limited food resources, but the evidence does not support this theory.One of the most prolific sources of antibiotics is *Actinomycetes*. These are filamentous bacteria that resemble microscopic mould fungi. The actinomycete *Streptomyces* produces the antibiotic **streptomycin**.

Perhaps the best known antibiotic is **penicillin**, which is produced by the mould fungus *Penicillium* and was discovered by Sir Alexander Fleming in 1928. Penicillin is still an important antibiotic but it is produced by mutant forms of a different species of *Penicillium* from that studied by Fleming. The different mutant forms of the fungus produce different types of penicillin. The penicillin types are chemically altered in the laboratory to make them more effective and to 'tailor' them for use with different diseases. 'Ampicillin', 'methicillin' and 'oxacillin' are examples.

Antibiotics attack bacteria in a variety of ways. Some of them disrupt the production of the cell wall and so prevent the bacteria from reproducing, or even cause them to burst open; some interfere with protein synthesis and thus arrest bacterial growth. Those that stop bacteria from reproducing are said to be bacteriostatic; those that kill the bacteria are bacteriocidal. Animal cells do not have cell walls, and the cell structures involved in protein production are different. Consequently, antibiotics do not damage human cells although they may produce some side-effects such as allergic reactions

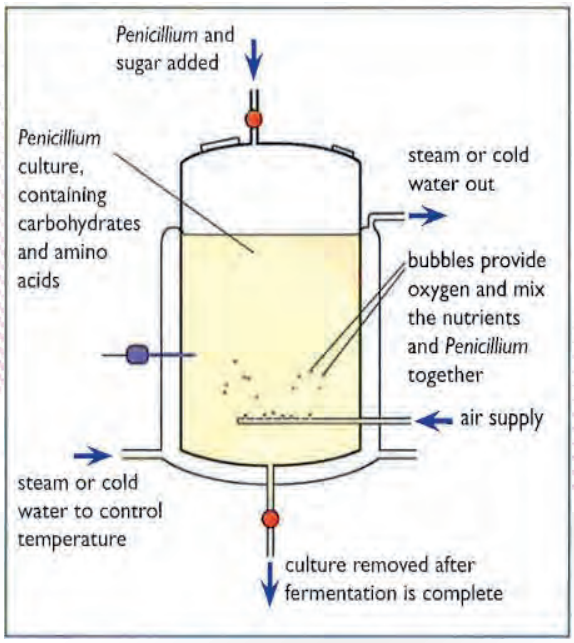


Figure 1.8. A fermenter used for producing penicilli

Activity 1.5.

- What are biosensors?
- Where do we apply biosensors?

Some vaccines are also adaptable to mass production through fermentation. Vaccines for *Bordetella pertussis, Salmonella typhi, Vibrio cholerae,* and *Mycobacterium tuberculosis* are produced in large batch cultures. *Corynebacterium diphtheriae* and *Clostridium tetani* are propagated for the synthesis of their toxins, from which toxoids for the DT vaccines are prepared.

b. Biosensors

A rapidly developing area of biotechnology, arousing intense international scientific interest, is that of **biosensor** production. In this field of bioelectronics, living microorganisms (or their enzymes or organelles) are linked with electrodes, and biological reactions are converted into electrical currents by these biosensors. Biosensors are being developed to measure specific components in beer, to monitor pollutants, and to detect flavor compounds in food. It is possible to measure the concentration of substances from many different environments. Applications include the detection of glucose, acetic acid, glutamic acid, ethanol, and biochemical oxygen

Activity 1.6.

- Where do we apply forensic science?
- Why is forensic science more reliable than other types of investigative approaches?
- What tools are used in forensic science?

demand. In addition, the application of biosensors to measure cephalosporin, nicotinic acid, and several B vitamins has been described. Recently biosensors have been developed using immunochemical-based detection system. These new biosensors will detect pathogens, herbicides, toxins, proteins, and DNA. Many of these biosensors are based on the use of a streptavidin-biotin recognition system.

One of the most interesting recent developments using these approaches is a handheld aflatoxin detection system for use in monitoring food quality. This automated unit, based on a new column-based immunoaffinity fluorometric procedure, can be used for 100 measurements before being recharged. The unit can detect from 0.1 to 50 ppb of aflatoxins in a 1.0 ml sample in less than 2 min. Aflatoxins. Rapid advances are being

made in all areas of biosensor technology. These include major improvements in the stability and durability of these units, which are being made more portable and sensitive. Microorganisms and metabolites such as glucose can be measured, thus meeting critical needs in modern medicine.

c). Forensic Science

Forensic biologists inspect crime scenes to examine potential sources of evidence using blood, saliva, and hair, and then they analyze the specimens in a laboratory, focusing on DNA analysis (Figure1.5). Additionally, fingerprints are also important tools to investigate crime and determine the paternity case of a child. This is because each individual has unique fingerprints that do not change throughout life. Based on their investigations, forensic biologists write up their findings in technical reports and are called upon to testify in court. Finally, this data is used to investigate the related transgression, and then these facts are put forward in the court that‘s quite helpful iin order to castigate the criminal. These days 'bioinformatics is widely acceptable in the field of forensic science because, with the help of computational tools, it has become quite easer and reliable to gather evidence regarding a particular crime scene.

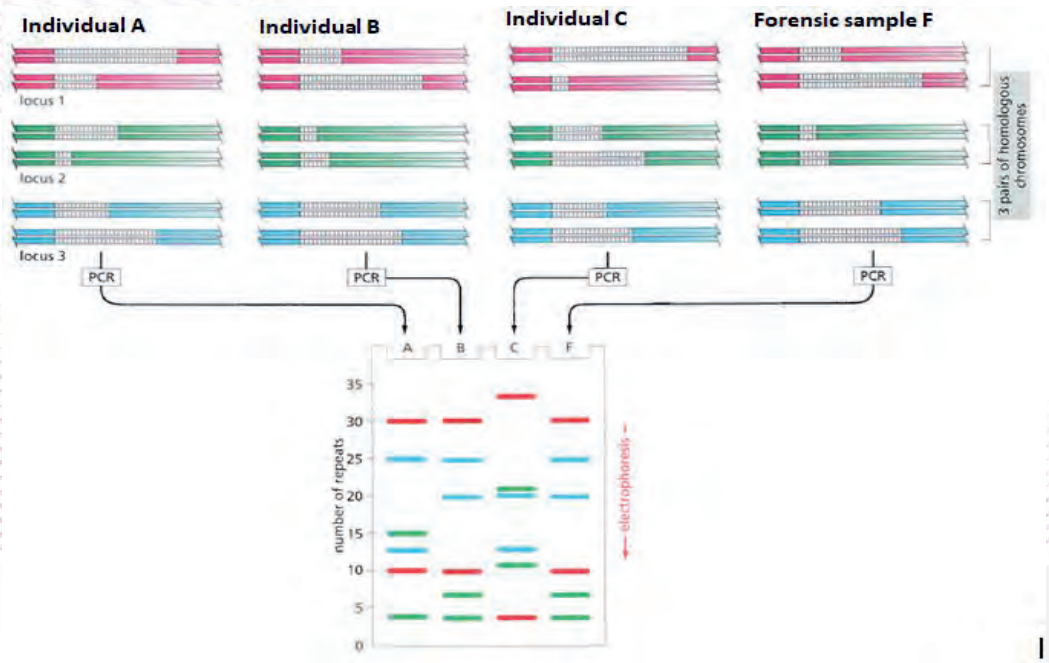


Figure 1.9. PCR as forensic science tools

Activity 1.7. Group work

Make a group and discuss the role of biological knowledge in forensic science.

When examining the variability at 5-10 different Variable Number of Tandem Repeat (VNTR) loci, the odds that two random individuals would share the same genetic pattern by chance can be approximatel1y 1 in 10 billion. In the case shown here, individuals A and C can be eliminated from further enquiries whereas individual B remains a clear suspect for committing the crime. A similar approach is now routinely used for paternity testing. The gel electrophoresis results of two-suspected paternity tests.

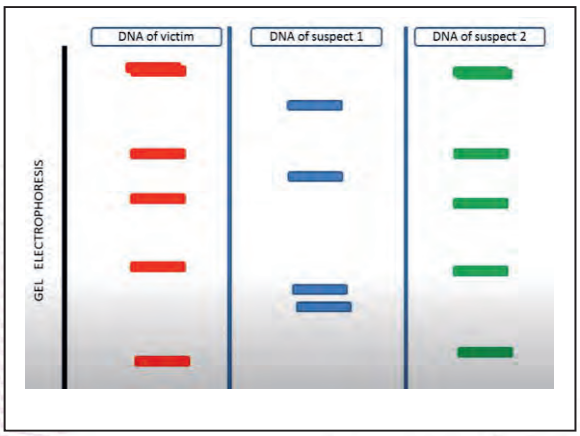


Figure 1.10. PCR in forensic application. Therefore, the suspected person (DNA of suspected 2) is the father of the baby (victim).

V). Applications in biomining

1). Microbiological mining

As the availability of mineral-rich ores decreases, methods are needed to extract minerals from less concentrated sources. This need spawned the new discipline known as biohydrometallurgy, the use of microbes to extract metals from ores. Copper and other metals originally were thought to be leached from the wastes of ore crushing as a result of an inorganic chemical reaction such as those reactions used to extract metals from ores.

It was then discovered that this leaching is due to the action of *Thiobacillus ferrooxidans*.

1.8. Activity

1. Describe how biofuel is produced?
2. Distinguish between biodegradable and non-biodegrdable substances.
3. Explain the importance of microorganisms in wastewater

This chemolithotrophic acidophilic bacterium lives by oxidizing the sulfur that binds copper, zinc, lead, and uranium into their respective sulfide minerals, with a resultant release of the pure metal. Copper in low-grade ores is often present as copper sulfide. When acidic water is sprayed on such ore, *T. ferrooxidans*

obtains energy as it uses oxygen from the atmosphere to oxidiz

the sulfur atoms in sulfide ores to sulfate. The bacterium doesn't use the copper; it merely converts it to a water-soluble form that can be retrieved and used by humans. Other minerals also can be degraded by microbes, *T. ferrooxidans* releases iron from iron sulfide by the same process. Combinations of *T. ferrooxidans* and a similar organism, *T. thiooxidans*, degrade some copper and iron ores more rapidly than either one does alone. Another combination of organisms, *Leptospirillum ferrooxidans* and *T. organoparus,* degrades pyrite (FeS2) and chalcopyrite (CuFeS2), although neither organism can degrade the minerals alone. Other bacteria can be used to mine uranium, and bacteria may eventually be used to remove arsenic, lead, zinc, cobalt, and gold. However, of late, fewer mining companies are actually using microbes in their processing. **(Vi).**

Vi. Application in the Environment

A). Solid waste treatment: composting and landfill

Most of it ends up in landfill sites- huge holes in the ground where refuse is deposited to prevent it being a hazard. The non-biodegradable components (metals, plastics, rubble, etc.) remain there more or less indefinitely; however, over a period of time biodegradable material (food waste, textiles, paper, etc.) undergoes a decomposition process. The rate at which this happens is dependent on the nature of the waste and the conditions of the landfill, but can take several decades. Aerobic processes give way to anaerobic ones and a significant result of the latter is the generation of methane. Modern landfill sites incorporate systems that remove this to prevent it being a fire/explosion hazard, and may put it to good use as a fuel source. Many householders separate organic waste items such as vegetable peelings and grass cuttings and use them to make *compost*. This practice, apart from providing a useful gardening supplement, also substantially reduces the volume of material that has to be disposed of by other means.

Self-questioning

- What are the three stages of wastewater treatment?
- Why are they important?

Fungi and bacteria, particularly actinomycetes, break down the organic

matter to produce CO2, water and humus, a relatively stable organic end product. Compost is not really a fertiliser, since its nitrogen content is not high, but it nevertheless provides nutrients to a soil and generally helps to improve its condition. Composting is carried out on a large scale by local authorities using the waste generated in municipal parks and gardens.

B). Wastewater treatment

The aim of wastewater treatment is the removal of undesirable substances and hazardous microorganisms in order that the water may safely enter a watercourse such as a river or stream. Further purification procedures are required before it can be used as drinking water. Wastewater treatment is fundamental to any developed society, and greatly reduces the incidence of waterborne diseases such as cholera. Wastewater may come from domestic or commercial sources; highly toxic industrial effluents may require pretreatment before entering a water treatment system. *Sewage* is the term used to describe liquid wastes that contain faecal matter (human or animal).

The effectiveness of the treatment process is judged chiefly by the reduction of the wastewater's *biochemical oxygen demand* (BOD). This is a measure of the amount of oxygen needed by microorganisms to oxidise its organic content. A high BOD leads to the removal of oxygen from water, a certain indicator of pollution.

Wastewater treatment usually occurs in stages, the first of which (primary treatment) is purely physical, and involves the removal of floating objects followed by sedimentation, a process that removes up to one-third of the BOD value.

Secondary treatment involves microbial oxidation, leading to a substantial further reduction in BOD. This may take one of two forms, both of which are aerobic, the traditional *trickling filter* and the more recent *activated sludge* process (Figure 1.7.). In the former, the wastewater is passed slowly over beds of stones or pieces of moulded plastic. These develop a biofilm comprising bacteria, protozoans, fungi and algae, and the resulting treated water has its BOD reduced by some 80–85%. Activated sludge facilities achieve an even higher degree of BOD reduction. Here the wastewater is aerated in tanks that have been seeded with a mixed microbial sludge. The main component of this is the bacterium *Zoogloea,* which secretes slime, forming aggregates called *flocs*, around which other microorganisms such as protozoans attach. Some of the

water's organic content is not immediately oxidised, but becomes incorporated into the flocs. After a few hours' residence in the tank, the sludge is allowed to settle out, and the treated water passes out of the system. Before being discharged to a watercourse, it is treated with chlorine to remove any pathogenic microorganisms that may remain. The principal operating problem encountered with activated sludge is that of *bulking*. This is caused by filamentous bacteria such as *Sphaerotilus natans*; it results in the sludge not settling properly and consequently passing out with the treated water.

Both secondary treatment processes result in some surplus sludge, which undergoes anaerobic digestion, resulting in the production of methane and CO2. The methane can be used as a fuel to power the plant, and any remaining sludge is dewatered and used as a soil conditioner. Care must be taken in this context, however, that the sludge does not contain toxic heavy metals.

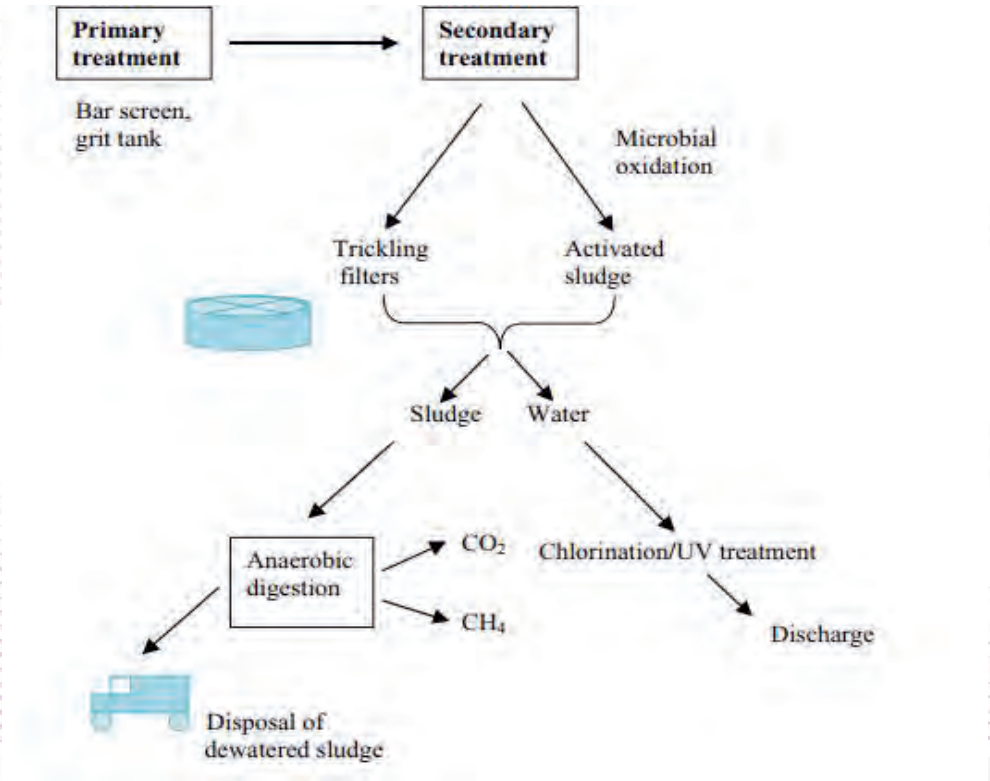


Figure 1.11. The role of microoorganisms in wastewater treatment

C). Bioremediation

Bioremediation is the use of living organisms or their products for the detoxification and degradation of environmental pollutants. Today many

pollutants are degraded with the help of saprophytic microbes; this process is also known as **biodegradation.** Genetically engineered bacterias are used to clean up pollutants from the environment. The engineered bacteria metabolically breakdown toxic pollutants into harmless compounds. Mercury resistant bacteria process metallic mercury (which damages the nervous system) into a nontoxic compound.

During bioremediation via microorganisms, enzymes produced by a microorganism modify a toxic pollutant by altering or transforming its structure. This event is called **biotransformation.** In many cases, biotransformation results in **biodegradation,** in which the toxic pollutant is degraded, were yielding less complex, nontoxic metabolites. Alternatively, biotransformation without biodegradation can also occur. For example, toxic heavy metals can often be rendered less toxic by oxidation or reduction reactions carried out by microorganisms.

D). Biofuels

The need to become independent of fossil fuels is driven by both political and environmental concerns. This has accelerated interest in and use of biofuels - fuel (chiefly ethanol) that is obtained by the fermentation of plant material. While corn is currently the substrate of choice, the use of crop residues could significantly boost biofuel yields. Crop residues are the plant material that is usually left in the field after harvest, and it consists of cellulose and hemicellulose. These polysaccharides are polymers of five different hexoses and pentose: glucose, xylose, mannose, galactose, and arabinose. While no microorganism naturally ferments all five sugars, a *Saccharomyces cerevisiae* strain has been engineered to ferment xylose and an *E. coli* strain that expresses *Zymomonas mobilis* genes is able to ferment all these sugars.

Another area of research focuses on degrading the cellulose and hemicellulose to release these monomers. This is commonly done by heating the plant material and treating it with acid, which is both expensive and corrosive. Work to harvest cellulase- and hemicellulase-producing fungi as well as bioprospecting for enzymes from thermoacidophiles are ongoing in an effort to replace the harsh thermochemical approach with a biological treatment.

E). Biogas production

Biogas is produced by bacteria and archaeans from organic matter in fermenters. Biogas is a combustible gas produced from the anaerobic breakdown of organic matter such as manure, waste plant matter from crops and household organic waste by the activities of the microorganisms. Depending on the construction of the fermenter, biogas

is mostly methane with some carbon dioxide, though other gases may be present.

Three different communities of anaerobic microbes are required. The first group converts the raw organic waste into a mixture of organic acids, alcohol, hydrogen and carbon dioxide. The second group use the organic acids and alcohol from the first stage to produce acetate, carbon dioxide and hydrogen. These first two communities are Eubacteria. The last group are Archaea called methanogens. The methanogens produce methane by one of the following two reactions:

$CO_2 + 4H_2 \rightarrow CH_4 + 2H_2O$
(reduction of carbon dioxide to methane)

$CH_3COOH \rightarrow CH_4 + CO_2$
(splitting ethanoic acid to form methane and carbon dioxide)

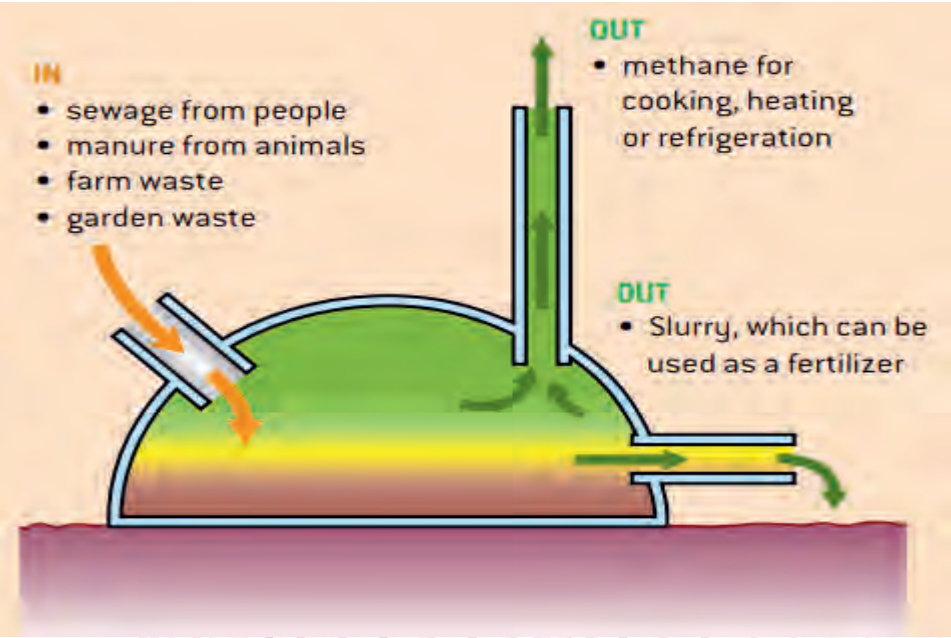


Figure 1.12. Methane generator. Conditions inside must be anaerobic

Advantages of biogas

1. Biogas is a fuel used to cook food, and light lamps.
2. Slurry left after biogas production forms a soil conditioner (manure).
3. Biogas is much cheaper than Liquefied Petroleum Gas for home use.

C) Composting

Composting is an aerobic microbial driven process that converts solid organic wastes into a stable, sanitary, humus-like material that has been considerably reduced in bulk and can be safely returned to the environment. It is, in effect, a low-moisture, solid substrate fermentation process as previously discussed. To be totally effective, it should only use as substrates readily decomposable solid

organic waste. In large-scale operations using largely domestic solid organic wastes, the final product is mostly used for soil improvement, but in more specialised operations using specific organic raw substrates (straw, animal manures, etc.), the final product can become the substrate for the worldwide commercial production of the mushroom *Agaricus bisporus*.

Composting has only recently become a serious waste management technology, and both theoretical and practical development of the technology is still in its infancy. The primary aim of a composting operation is to obtain, in a limited time within limited compost, final compost with a desired product quality. A composting plant must function under environmentally safe conditions.

Composting is carried out in a packed bed of solid organic particles in which the indigenous microbes will grow and reproduce. Free access to air is an essential requirement. The starting materials are arranged in static piles (windrows), aerated piles or covered tunnels, or in rotating bioreactors (drums or cylinders). Some form of pre-treatment of the waste may be required, such as particle size reduction by shredding or grinding. The basic biological reaction of the composting process is the oxidation of the mixed organic substrates with oxygen to produce carbon dioxide, water and other organic by-products (Figure 1.10). After the composting process is completed, the final product most often needs to be left for variable time periods to stabilize.

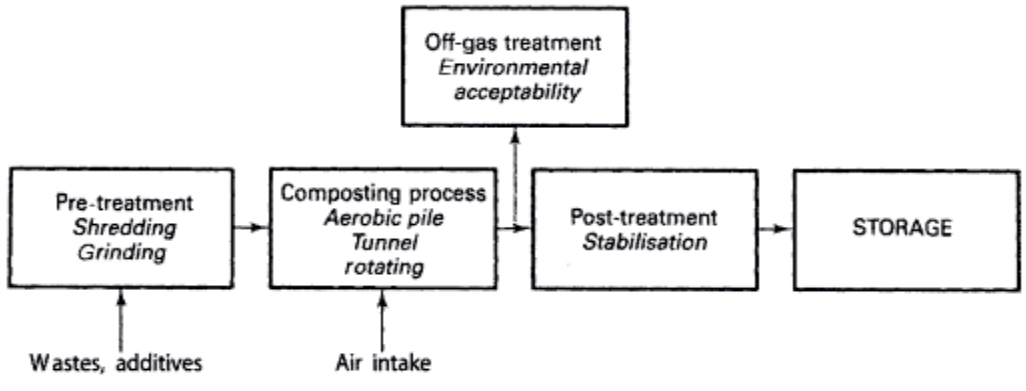


Figure 1.13. flow chart compositing plants processes

Vii. Application in Industry

A). Enzymes

Enzymes can be produced by commercial fermentation using readily available feed stocks such as corn-steep liquor or molasses. Fungi (e.g. *Aspergillus*) or bacteria (e.g. *Bacillus*) are two of the commonest organisms used to produce the enzymes. These organisms are selected because they are non-pathogenic and do not produce antibiotics. The fermentation process is similar to that described for penicillin. If the enzymes are extracellular then the liquid feedstock is filtered from the organism and the enzyme is extracted. If the enzymes are intracellular, the micro-organisms have to be filtered from the feedstock. They are then crushed and the enzymes extracted with water or other solvents.

Some commercial uses of enzymes are listed below.

- **Proteases**: In washing powders for dissolving stains from, e.g. egg, milk and blood; removing hair from animal hides; cheese manufacture; tenderising meat.
- **Lipases**: Flavors enhancer in cheese; in washing powders for removal of fatty stains.
- **Pectinases**: Clarification of fruit juices; maximizing juice extraction.
- **Amylases**: Production of glucose from starch.

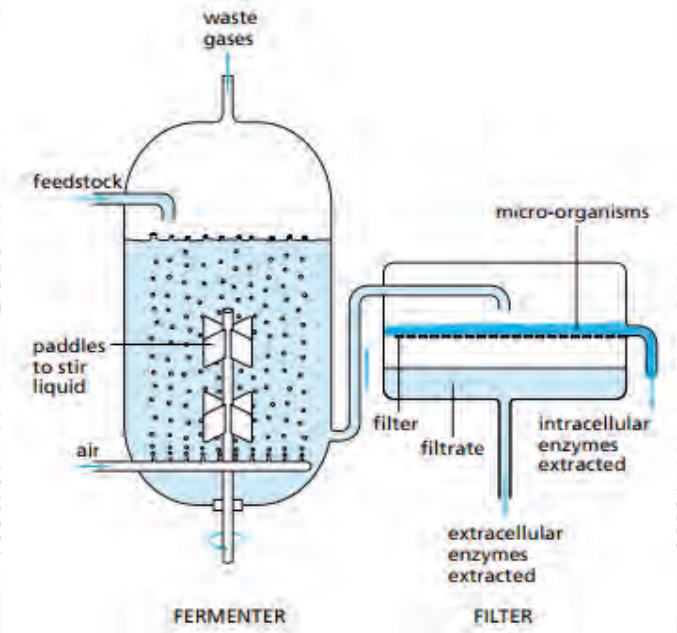


Figure 1.14. Principles of enzyme production from micro-organisms

B). Biological washing powders

The majority of commercial enzyme production involves protein-digesting

enzymes (proteases) and fat-digesting enzymes (lipases) for use in the food and textile industries. When combined in washing powders they are effective in removing stains in clothes caused by proteins, e.g. blood, egg and gravy, and fats, e.g. grease. Protein and fat molecules tend to be large and insoluble.

When they have been digested the products are small, soluble molecules, which can pass out of the cloth. Biological washing powders save energy because they can be used to wash clothes at lower temperatures, so there is no need to boil water. However, if they are put in water at higher temperatures the enzymes become denatured and they lose their effectiveness.

Viii. Applications in agriculture

A). Biopesticides

There has been a long-term interest in the use of bacteria, fungi, and viruses as **bioinsecticides** and **biopesticides**. These are defined as biological agents, such as bacteria, fungi, viruses, or their components, which can be used to kill a susceptible insect.

Bacteria: *Bacillus thuringiensis* and *Bacillus popilliae* are the two major bacteria of interest. *Bacillus thuringiensis* is used on a wide variety of vegetable and field crops, fruits, shade trees, and ornamentals. *B. popilliae* is used primarily against Japanese beetle larvae. Both bacteria are considered harmless to humans. *Pseudomonas fluorescens,* which contains the toxin-producing gene from *B. thuringiensis,* is used on maize to suppress black cutworms.

Viruses: Three major virus groups that do not appear to replicate in warm-blooded animals are used: nuclear polyhedrosis virus (NPV), granulosis virus (GV), and cytoplasmic polyhedrosis virus (CPV). These occluded viruses are more protected in the environment.

Fungi : Over 500 different fungi are associated with insects. Infection and disease occur primarily through the insect cuticle. Four major genera have been used. *Beauveria bassiana* and *Metarhizium anisopliae* are used for control of the Colorado potato beetle and the froghopper in sugarcane plantations, respectively. *Verticillium lecanii* and *Entomophthora* spp., have been associated with control of aphids in greenhouse and field environments.

B). The use of Ti plasmid as a vector

Use of tumour-inducing (Ti) plasmid of *Agrobacterium tumefaciens* to introduce glyphosate resistance into soybean crops. One way to introduce transgenes into plants is to

use *Agrobacterium tumefaciens*. This is a species of bacteria that has a plasmid, called the Ti plasmid, that causes tumours in the plants it infects.

The glyphosate resistance gene is inserted into the Ti plasmid along with an antibiotic resistance gene. The construct is then re-inserted into an *A. tumefaciens* bacterium. Plant cells are then exposed to the transgenic bacterium and cultured on a plate containing antibiotic. The only plant cells that grow are those that have taken up the plasmid. The others are killed by antibiotic.

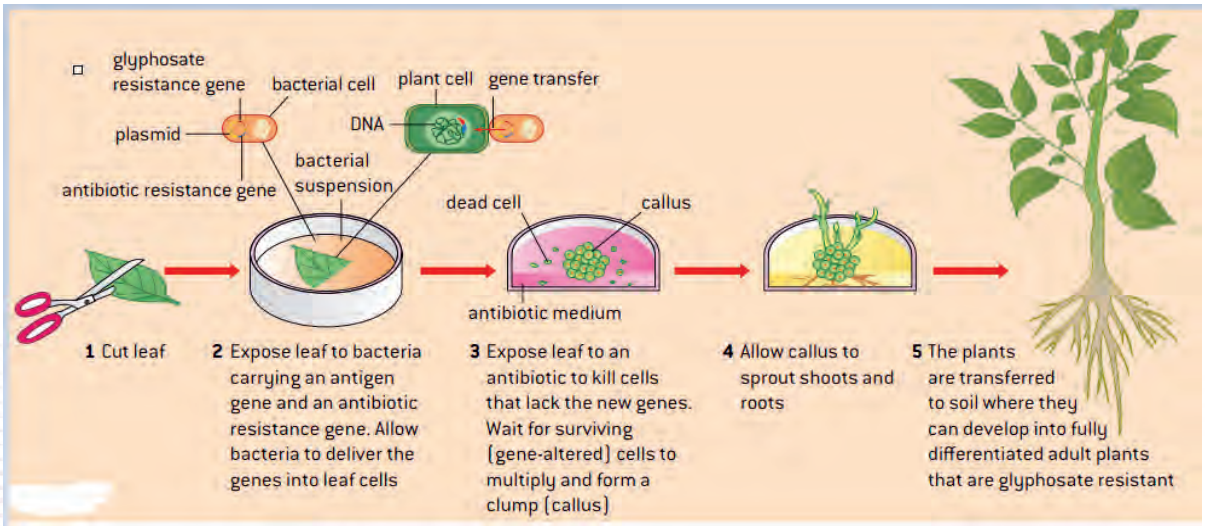


Figure 1.15. The use of Ti palmids as vector

C). Insect-resistant crops

Another important agricultural development is that of genetically modifed plants protected against attack by insect pests. Maize is protected against the corn borer, which eats the leaves of the plants and then burrows into the stalk, eating its way upwards until the plant cannot support the ear. Cotton is protected against pests such as the boll weevil (Figure 1.13). In both plants, yield is improved.

Insect-resistant tobacco also exists, and is protected against the tobacco bud worm, but as yet it has not been grown commercially. The most likely detrimental effects on the environment of growing an insect-resistant crop are:

- The evolution of resistance by the insect pests
- a damaging erect on other species of insects
- the transfer of the added gene to other species of plant.

However, less pesticide is used, reducing the risk of spray carrying to and erecting non-target species of insects in other areas. Remember also that only insects that actually

eat the crop are a erected.

Figure 1.16. a). Corn borer, b). Boll weevil

D). Pest Resistant Crops

Pest attack is one of the very common problems in a number of different crops all around the globe, these crops may include fodder crops or other crops for the purpose of getting food. One the example of such crops is BT-Cotton. The genes of *Bacillus thuringiensis* (Bt), a very common, are inserted in cotton crop in order for development of certain protein in it. The protein is very toxic to a number of different insects. With this aid of biotechnology, the developed BT-Cotton leads to a less pest attack ultimately leading to a significant more production.

E). Transgenic Animals

Although several recombinant proteins used in medicine are successfully produced in

bacteria, some proteins require a eukaryotic animal host for proper processing. For this reason, the desired genes are cloned and expressed in animals, such as sheep, goats, chickens, and mice. Animals that have been modified to express recombinant DNA are called transgenic animals. Several human proteins are expressed in the milk of transgenic sheep and goats, and some are expressed in the eggs of chickens. Mice have been used extensively for expressing and studying the effects of recombinant genes and mutations.

F). Transgenic Plants

Manipulating the DNA of plants (i.e., creating GMOs) has helped to create desirable traits, such as disease resistance, herbicide and pesticide resistance, better nutritional value, and better shelflife (Figure 1.14). Plants are the most important source of food for the human population. Farmers developed ways to select for plant varieties with desirable traits long before modern-day biotechnology practices were established.

Attention: Transgenic crops are being created that resist disease, are tolerant of herbicides and drought, and have improved nutritional quality. Plants are also being used to produce pharmaceuticals, and domesticated animals are being genetically modified to produce biologically active compounds. Plants that have received recombinant DNA from other species are called transgenic plants. Because they are not natural, transgenic plants and other GMOs are closely monitored by government agencies to ensure that they are fit for human consumption and do not endanger other plant and animal life.

Figure 1.17. Corn

Because foreign genes can spread to other species in the environment, extensive testing is required to ensure ecological stability. Staples like corn, potatoes, and tomatoes were theirst crop plants to be genetically engineered.

F). Pest resistance

Self-questioning

- What is biological warfare?
- Why the use of biological warfare is not allowed?

The bacterium, *Bacillus thuringiensis*, produces a toxin that kills caterpillars and other insect larvae. The toxin has been in use for some years as an insecticide. The gene for the toxin has been successfully introduced into some plant species using a bacterial vector. The plants produce the toxin and show increased resistance to attack by insect larvae.

G). Herbicide resistance

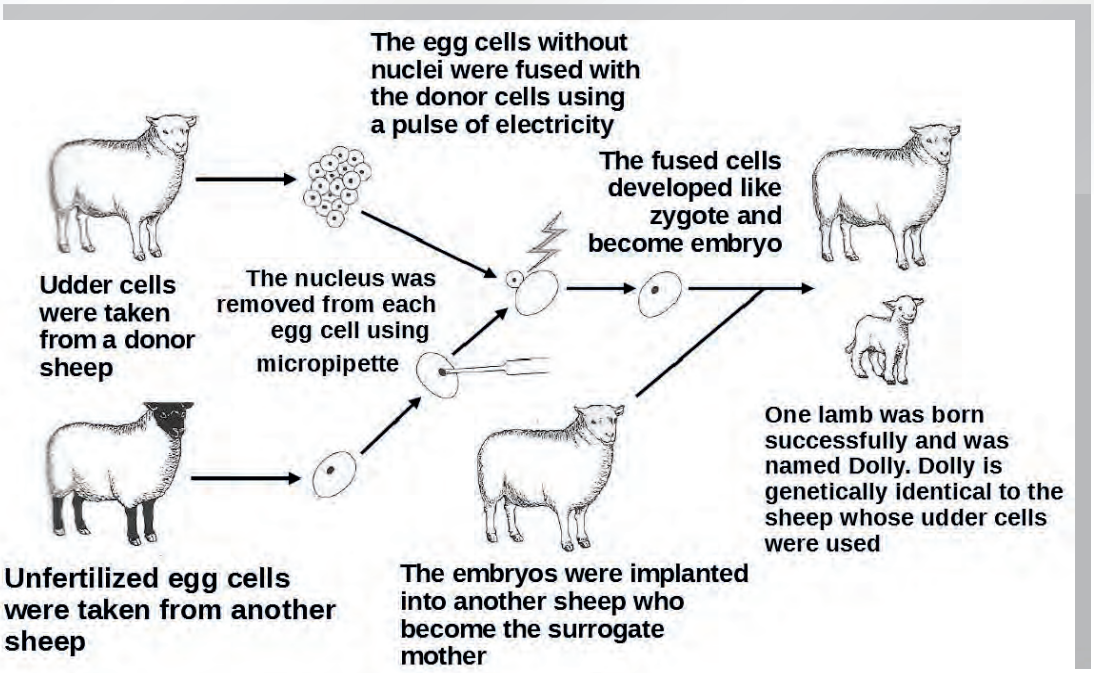
Some of the safest and most effective herbicides are those, such as glyphosate, which kill any green plant but become harmless as soon as they reach the soil. These herbicides cannot be used on crops because they kill the crop plants as well as the weeds. A gene for an enzyme that breaks down glyphosate can be introduced into a plant cell culture. This should lead to a reduced use of herbicides.

Cloning

Cloning is a method of producing identical copies of genes, cells, or organisms. The products of cloning are called a **clone**. A clone is a group of genetically identical organisms or cells produced either by asexual reproduction or artificially by cloning techniques. The main advantage of these techniques is that they can make large numbers of plants or animals, which are exact copies of a parent with desirable characteristics. Sometimes cloning is used to produce skin or other tissues needed to treat a patient.

H). Animal cloning

Animals cannot be cloned in the same way from parts of their bodies. If animal embryos are divided up at an early stage into several pieces, each piece can develop into a separate animal. However, it is hard to predict which embryos can develop into animals with desirable characteristics and should therefore be cloned. The first successful reproductive cloning of an adult with known characteristics is **Dolly** the **sheep.** Study how Dolly was cloned which is illustrated in the diagram below



Fi gure 1.18. Biotechnologically producing identical copies sheep from udder call

X. Biological warfare

Biological warfare (BW) also known as germ warfare is the use of biological toxins or infectious agents such as bacteria, viruses, and fungi with the intent to kill or incapacitate humans, animals or plants as an act of war.

Biological weapons include any microorganism (such as bacteria, viruses, or fungi) or toxin (poisonous compounds produced by microorganisms) found in nature that can be used to kill or injure people. The act of bioterrorism can range from a simple hoax to the actual use of these biological weapons, also referred to as agents. A number of nations have or are seeking to acquire biological warfare agents, and there are concerns that terrorist groups or individuals may acquire the technologies and expertise to use these destructive agents. Biological agents may be used for an isolated assassination, to cause incapacitation or death to thousands. If the environment is contaminated, a long-term threat to the population could be created.

UNIT SUMMARY

- Biotechnology is the application of scientific knowledge by industries that produce biological products like food supplements, enzymes, and drugs.
- Yeasts (Fungi), moulds (Fungi) and bacteria are important microorganisms used in industries.
- Fermentation is a process by which sugar is converted into alcohol and CO_2 by yeast.
- Fermentation by the yeast Saccharomyces yields beer and that by Lactobacillus, yields butter milk.
- Tissue culture, a method of biological research in which fragments of tissue from an animal or plant are transferred to an artificial environment in which they can continue to survive and function. The cultured tissue may consist of a single cell, a population of cells, or a whole or part of an organ.
- Genetic engineering involves the transfer of genes from one organism to an unrelated species. Genetic engineering is being used in the production of pharmaceuticals, gene therapy, and the development of transgenic plants and animals.
- Wastewater may come from domestic or commercial sources; highly toxic industrial effluents may require pretreatment before entering a water treatment system. *Sewage* is the term used to describe liquid wastes that contain faecal matter (human or animal).
- Biogas is made by the action of methanogenic bacteria on waste matter such as the faeces of humans or of cattle.
- Transgenic plants may be obtained by using the plasmid of the bacterium Agrobacterium tumefaciens.
- Transgenic animals are produced by microinjection of foreign DNA into fertilised eggs or by using retrovirus for introducing foreign DNA into early embryonic stages.
- Bioremediation refers to the use of living organisms or their products to decrease pollutants in the environment. Genetically engineered bacteria can clean up pollutants from the environment. The transformed bacteria metabolically break down toxic pollutants into harmless compounds.
- Biological warfare is the use of biological toxins or infectious agents such as bacteria, viruses, and fungi

with the intent to kill or incapacitate humans, animals or plants as an act of war.

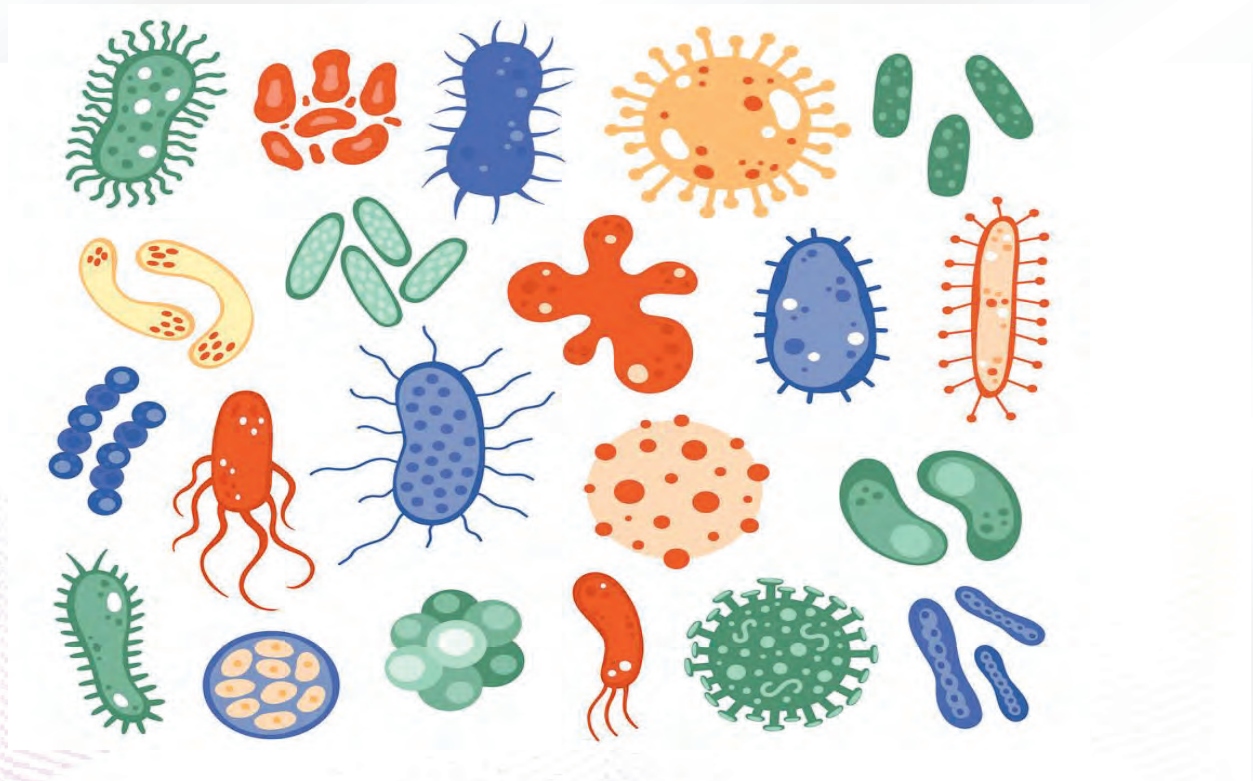
- Cloning is a technique scientists use to make exact genetic copies of living things. Genes, cells, tissues, and even whole animals can all be cloned. Some clones already exist in nature. Single-celled organisms like bacteria make exact copies of themselves each time they reproduce.

III. Review Questions

Answer the following questions

1. Define genetic engineering.
2. Differentiate between gene cloning and gene therapy.
3. List at least three different properties that have been acquired by crop plants via genetic engineering.
4. What is single-cell protein?
5. What are the different communities of anaerobic microbes involved in biogas Production?
6. Explain some medical applications of DNA technology?
7. Describe how bioremediation is helpful to humans?
8. Explain how the practical application of DNA- based biotechnology affect the human lives in many ways.
9. Why has the production of human insulin by recombinant DNA methods had significant medical advantages for diabetics?
10. How are the two fusion proteins used to make human insulin?
11. Write the benefits of biofuels.
12. Explain how can plant biotechnology be used for enhancing the nutritive qualities of Foods.
13. Explain the strategies adopted for creating transgenic plants with the following traits: (a) Disease resistance (b) Herbicide tolerance (c) Insect resistance.
14. What are the social and ethical problems associated with plant genetic engineering? How can these be addressed?
15. Explain the social, economical, and environmental implications of genetic engineering techniques.

UNIT 2: Microorganisms



Unit learning outcome

By the end of this unit, you will be able to:

- State the characteristics of microorganisms
- Describe the prevention and treatment mechanisms of diseases caused by microorganisms
- Explain different groups of microorganisms
- Explain diseases caused by microorganisms
- Discuss the importance of microorganisms in industry and the environment
- Explain diseases caused by microorganisms and viruses
- Describe prevention and treatment mechanisms of diseases caused by microorganisms

UNIT 2: MICROORGANISMS

At the end of this lesson, student will be able to:

- Categorize microorganisms based on their morphology, mode of nutrition, molecular and biochemical analysis.
- Explain the the ways of transmission and prevention of pathogenic microorganisms
- Explain the siginificances of Koch's postulates for the advancement of microbiological techniques.
- Apply the principles of microbiological techniques.
- Discuss the economic importance of microorganisms.
- Appreciate the various uses or application of microbiology in daily life.

Self-Assessment Questions;

- How do you define microorganism?
- Why is microscopic life vital for all life on earth?
- Where do microorganisms live?
- Why is it important to study microbiology?

Self-questioning

- List the types of microorganisms?
- What are the bases for microbial classifications?

Overview

Microorganisms are organisms too small to be seen clearly by the unaided eyes. *Micro* means very small—anything so small that it must be viewed with a **microscope** (an optical instrument used to observe very small objects). Therefore, **microbiology** can be defined as the study of microbes. Individual microbes can be observed only with the use of various types of microscopes. They are very small life forms so small that individual microorganisms cannot be seen without magnification. They include fungi, bacteria, algae, protozoa and viruses. Some microorganisms however, like the eukaryotic microorganisms are visible without magnification. Thus, microbiology is concerned with the study of microorganisms which include: bacteria, viruses, fungi, protozoa, algae, and helminthes (parasitic worms).

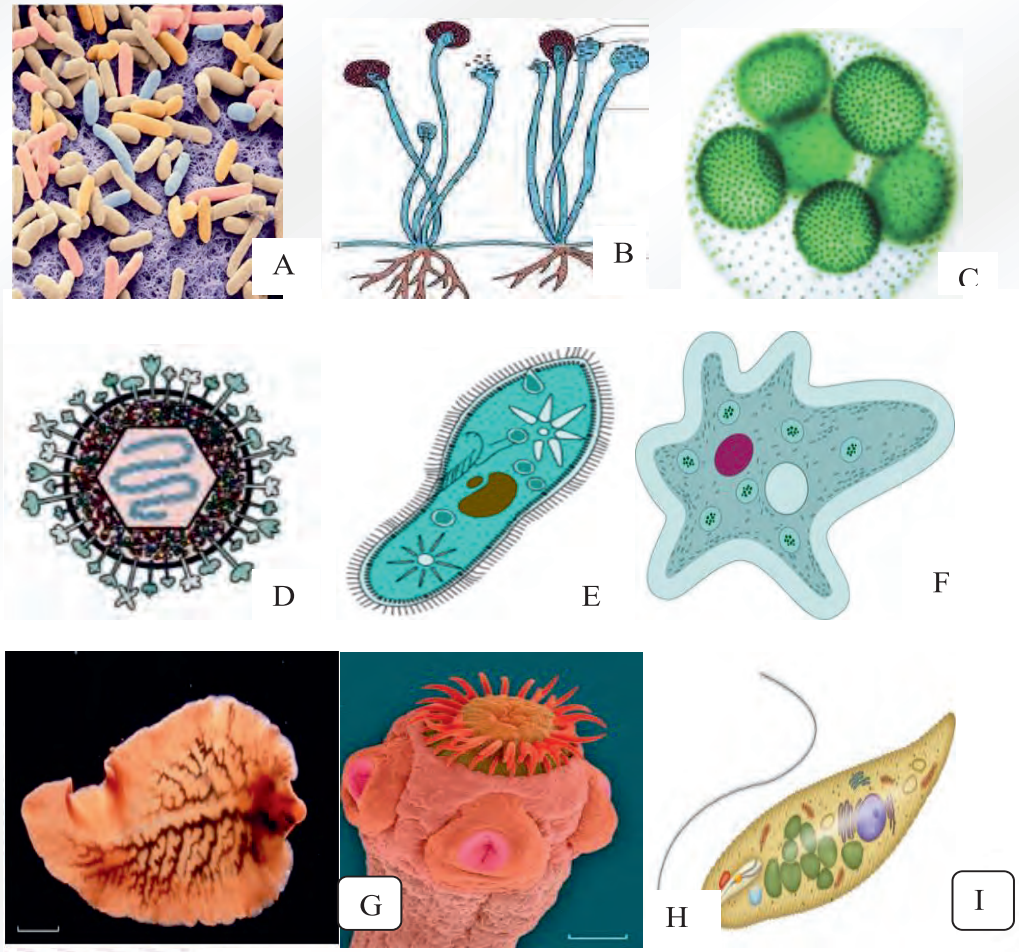


Figure 2.2. Microbial community structure: **Prokaryotes** [Bacteria (A)]; **Eukaryotes** [Fungi (B), Algae (C), Amoeba (E), Paramecium (F), Fluke (G), Tapeworm (H), Flagellates (I)] and **Acellular** [Viruses (D)]

Based on evolutionary lines, organisms are grouped into three domains: these are bacteria, archaea and eukarya (Figure 2.1).

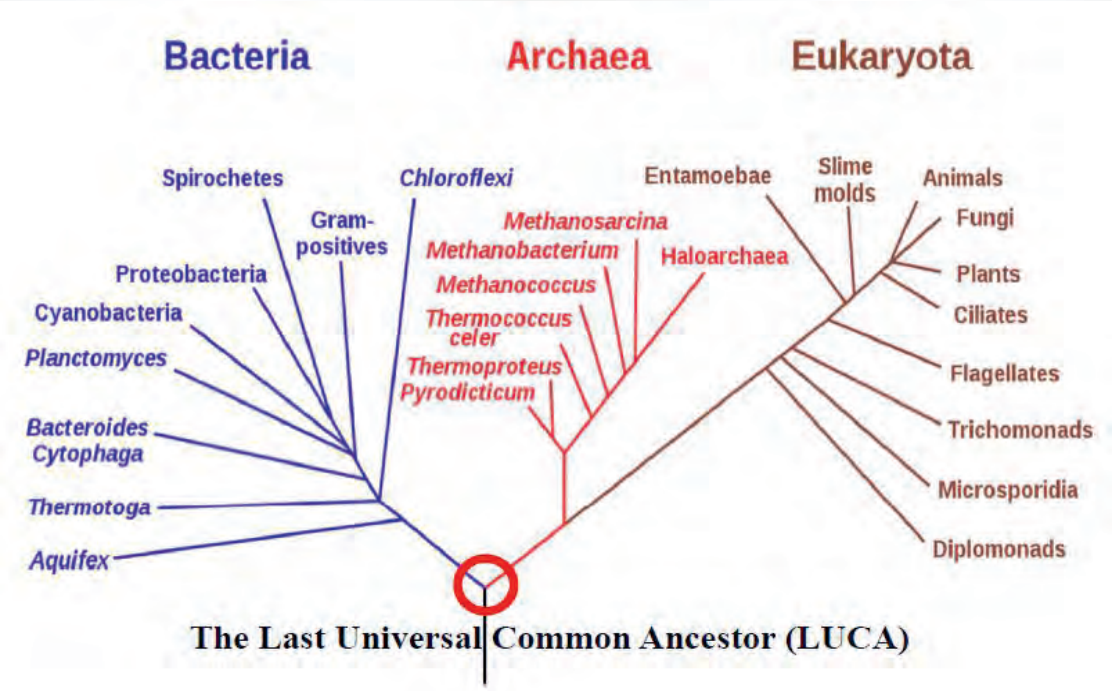


Figure 2. 1. Universal phylogenetic tree life constructed from small subunit of rRNA (SSU= **small subunit**, rRNA) gene sequence analysis

2.1. Eubacteria

At the end of this sub-section, the student will be able to:-

- describe the general features of bacteria.
- Describe the mode of nutrition in bacteria.
- explain the mechanism
- categorize bacteria based on their shapes.
- Distinguish Gram's positive and Gram's negative bacteria.
- mention some of the human diseases caused by bacteria.

Self-questioning

- What are Eubacteria and what do they do?
- Where can we get them?
- What is the difference between eukaryotes and prokaryotes?

Eubacteria (biology = definition): Literally means "**true bacteria**". Bacteria are relatively simple in structure. They are prokaryotic unicellular organisms with no nuclear membrane, mitochondria, Golgi bodies, or endoplasmic reticulum that reproduce by asexual division.

General Characteristics

- They are omnipresent i.e. present in soil, air and water.
- They are unicellular, prokaryotic microorganism.
- The cell bears a thick rigid cell wall (Peptidoglycan) outside the plasma membrane.
- They have great variation in the mode of nutrition i.e. may be autotrophic and heterotrophic. In heterotrophism mode of nutrition they may be
- parasite saprophyte or symbiotic in nature.
- They lack true chlorophyll but few photosynthetic bacteria have a special type of chlorophyll called bacteriochlorophyll.
- Lack true nucleus (lacking nuclear membrane and nucleolus).
- They lack mitochondria, Golgi apparatus, plastid and endoplasmic reticulum.
- Both DNA and RNA are present in the bacterial cell.

Key words

- **Prokaryotes:** are a unicellular organism that lacks a nuclear membrane-enclosed nucleus.
- **Eukaryotes:** are organisms whose cells have a nucleus enclosed within a nuclear envelope.
- **Unicellular:** a unicellular organism has just one cell
- **Multicellular:** a multicellular an organism has more than one cell

Self-questioning

- What are the Characteristics Common to all of Bacteria?

Activity 2.1

Work in small groups on the structure and function of flagella and pili in bacteria. Make a table comparing the two different

Structure of Bacterial Cell

As bacterial cells are very minute, they are studied under electron microscope in which it reveals various structures. Some of these are external to the cell wall while other are internal to the cell wall. The brief descriptions of the readily evident structures of bacteria is as follows:

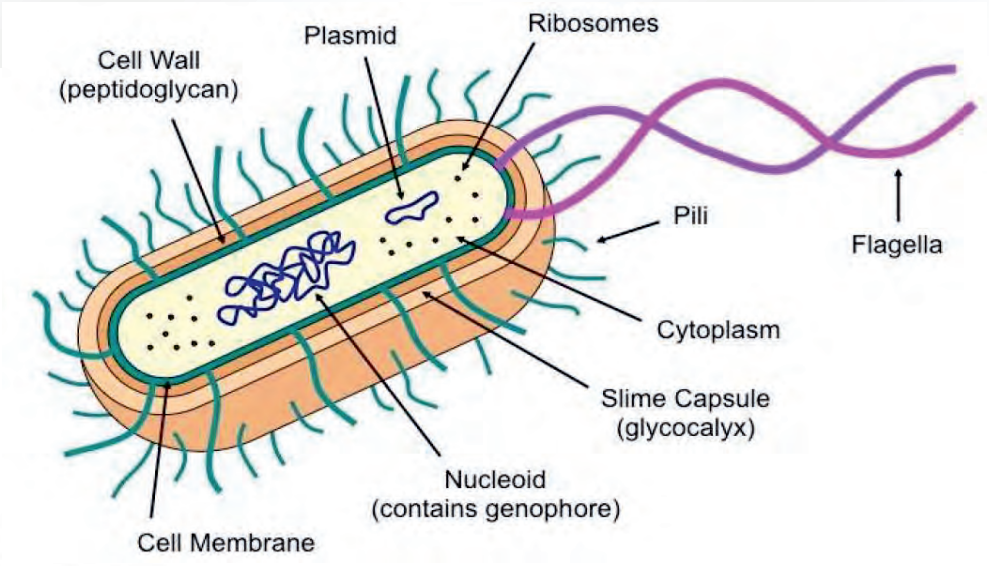
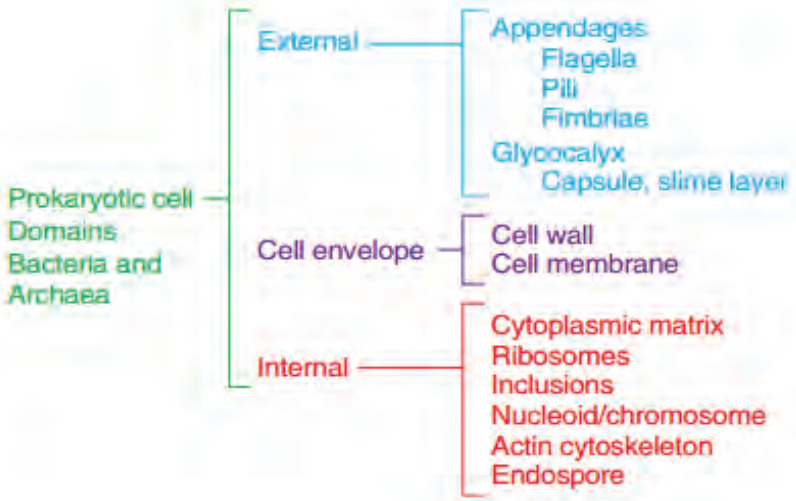


Figure 2.2. The major features of prokaryotic cell, Bacteria

The general structural plan of a prokaryotic cell can be represented with this Flowchart:



Eukaryotic microbes are an extraordinarily diverse group, including species with a wide range of life cycles, morphological specializations, and nutritional needs. Although more diseases are caused by viruses and bacteria than by microscopic eukaryotes, these eukaryotes are responsible

for some diseases of great public health importance.

Eukaryotic microbes have eucaryotic cells that have a variety of complex membranous organelles in the cytoplasmic matrix and the majority of their genetic material within membrane-delimited nuclei. Each organelle has a distinctive structure directly related to specific functions.

? Whate are the function of bacterial structure

A cytoskeleton composed of microtubules, microfilaments, and intermediate filaments helps give eukaryotic cells shape; the cytoskeleton is also involved in cell movements, intracellular transport, and reproduction. When eucaryotes reproduce, genetic material is distributed between cells by the highly organized, complex processes called mitosis and meiosis.

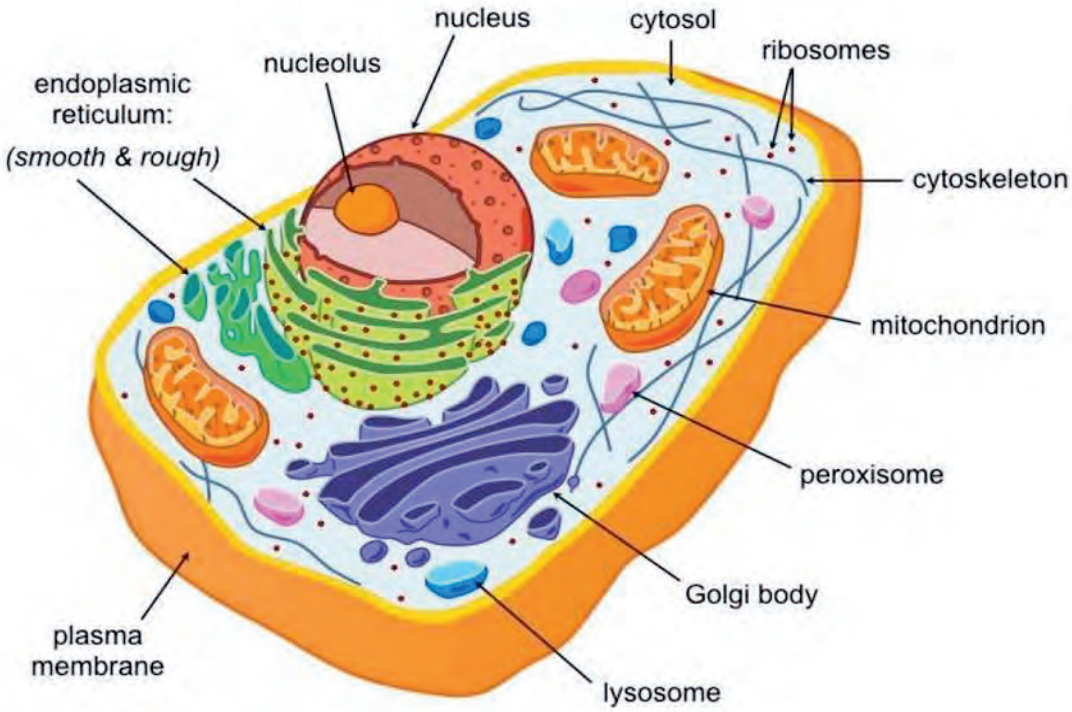


Figure 2.3. Major features of a eukaryotic cell. (Source: BioNinja)

Activity 2.2

Work in small groups and brainstorm all the differences you can think of between eukaryotic and prokaryotic cells. Make a table comparing the two different types of cells and share your ideas with the rest of the whole class.

2.1.1. Bacterial Shapes

Self-questioning

- What are the functions of bacterial structure?
- What is the difference between bacteria and archaea.
- Explain the structural difference between prokaryote and eukaryote cells?
- Can you draw the structures of prokaryote and eukaryote cells?
- What are the major differences among bacteria, archaea, and Eukaryia?

In fact, morphologically bacteria are classified based on numerous features. For example, cell shape (Figure 2.5), nature of multi cell aggregates, motility, formation of spores, and reaction to the gram stain are some of the most important features that are used to classify bacteria.

Self- questioning

- Do all bacteria have the same shapes?
- What are the three basic shapes of bacteria?
- How do the bacterial cells cluster together?

Bacterial cells can be grouped into the following three main shapes:'

- **Cocci** (singular, coccus) – spherical bacteria
- **Bacilli** (singular, bacillus) – rod-shaped bacteria
- **Spirochaetes** – spiral or corkscrew-shaped bacteria
- **Comma**-Vibrio cholera

Self-questioning

- What are the external structures of bacteria?

Activity 2.3

Streptococcus and Lactobacillus are the two well-researched types of bacteria. Carry out a library search to find out what shapes these two bacteria have, and why they are so important to us. Also, try to find two diseases caused by spiral-shaped bacteria.

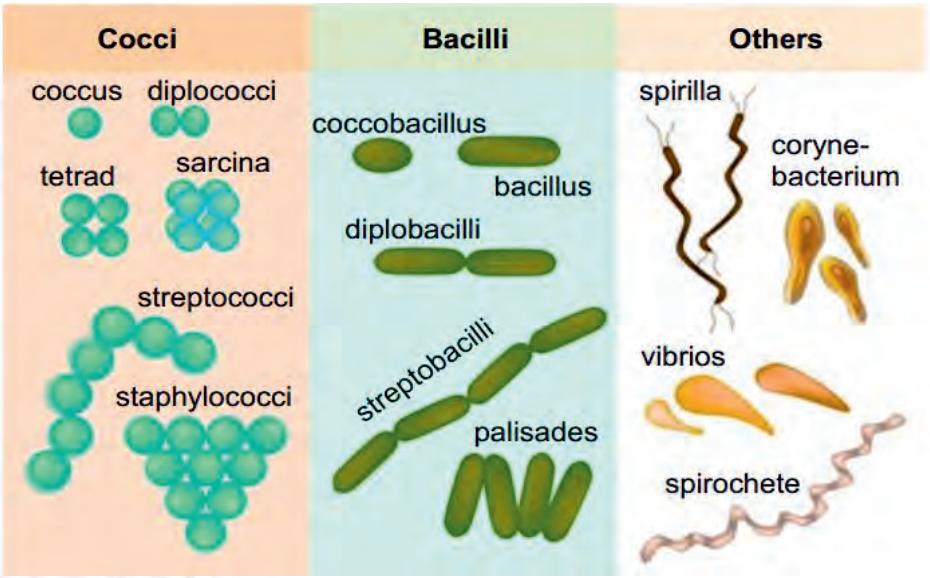


Figure 2.4. Basic bacterial shape and their clustering

Self- questioning

- Describe the following : outer membrane, periplasmic space, envelope, teichoic acid, adhesion site, lipopolysaccharide, and porin protein.
- Describe in detail the structure and composition of peptidoglycan of, gram-positive and gram-negative cell walls. Include labeled diagrams in your answer.
- Explain the role of a cell wall in protecting against lysis and illustrate how this role may be experimentally demonstrated.
- Distinguish the difference between protoplasts and spheroplasts.
- Distinguish the difference between protoplasts and spheroplasts.

Bacterial cell wall

Besides their shape bacteria can be classified based on their cell wall composition. One of these ways is whether they are retaining dyes during Gram's stain (Figure 2.6a & b). In this case, bacteria can be categorized into two:

- **Gram-positive** – Gram-positive bacteria have a distinctive purple appearance when observed under a light microscope following **Gram staining**. This is due to retention of the purple crystal violet stain in the thick peptidoglycan layer of the cell wall (Figure 2.5). Examples of Gram-positive bacteria include all staphylococci, all **streptococci** and some **listeria** species.

Gram-negative – Gram-negative bacteria **lose the crystal violet** stain (and take the color of the red counterstain) in Gram's method of staining. This is characteristic of bacteria that have a cell wall composed of a thin layer of a particular substance (called **peptidoglycan**).

Table 2.1. Comparison of Gram-Positive and Gram-Negative Cell Walls

Characteristic	Gram-Positive	Gram-Negative
Number of major layers	1	2
Chemical composition	Peptidoglycan Teichoic acid Lipoteichoic acid	Lipopolysaccharide Lipoprotein Peptidoglycan
Overall thickness	Thicker (20–80 nm)	Thinner (8–11 nm)
Outer membrane	No	Yes
Periplasmic space	Narrow	Extensive
Porin proteins	No	Yes
Permeability to molecules	More penetrable	Less penetrable

Self- questioning

- What is the purpose Gram's stain?

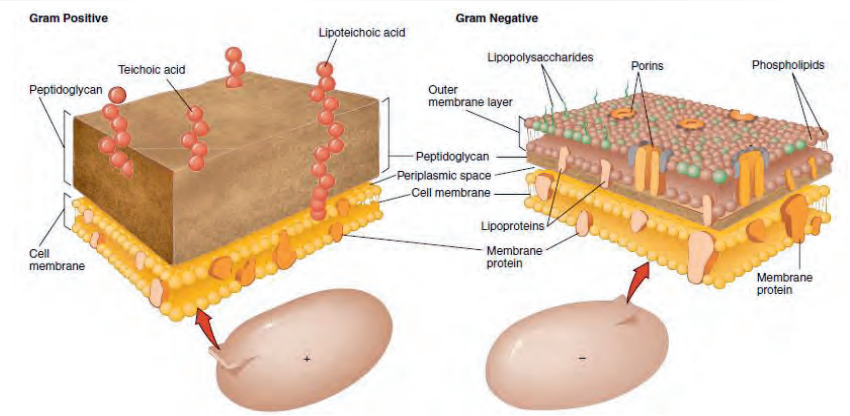


Figure 2.5. Grame positive and gram negative bacterial cell wall

The Gram staining process includes four basic steps, including:

(1) Heat fix/ attach the bacteria to the silde,

(2) . Applying a primary stain (crystal violet),

(3) . Adding a mordant (Gram's iodine),

(4) . Rapid decolorization with ethanol, acetone or a mixture of both, &

(5) . Counterstaining with safranin. Then observe the color change at each levels.

- The staining technique includes simple staining (use single dye) or differencial staining (use different dyies) (Figure 2.6)

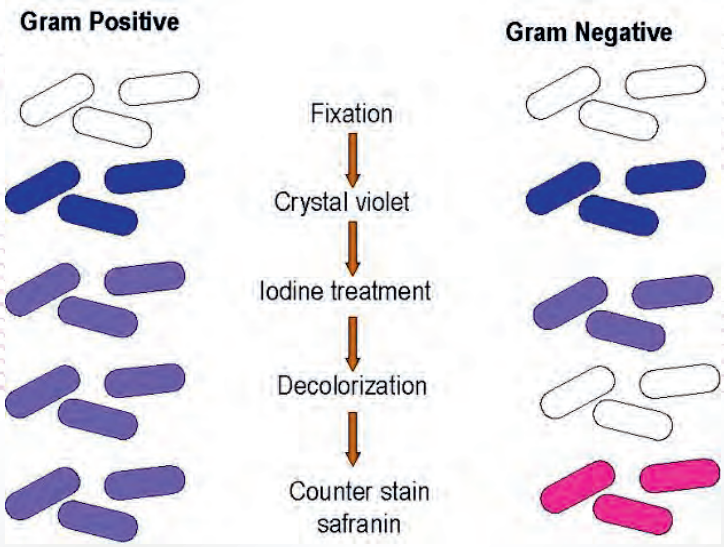


Figure 2.6. Gram staining

Key Words

Gram's staining: a test for distinguishing bacteria (named after Hans Christian Gram, who developed the technique in 1884)

Differential staining: is a staining procedure that distinguishes organisms based on their staining properties

Peptidoglycan: The rigid layer of the cell walls of Bacteria, a thin sheet composed of N-acetylglucosamine, N-acetylmuramic acid, and a few amino acids.

Endotoxin: The lipopolysaccharide portion of the cell envelope of certain gram-negative bacteria, which is toxin to human when solubilized.

Activity 2.4: Looking at bacteria in yoghurt

Per Student

- Clean glass slides and cover glasses
- Compound microscope with ocular micrometer
- Cytological stains (e.g., methylene blue, I_2KI)
- Methyl cellulose
- Immersion oil
- Cotton swabs
- Lens paper
- Lens cleaning solution
- Yoghurt
- Water
- A dropping pipettes Method
- Take a small sample of yoghurt using the pipette, and place it on a slide.
- If the sample seems too thick, dilute it with a drop of water.
- Lower a coverslip onto the yoghurt, taking care not to produce air bubbles.
- First, observe the bacteria at low power (×100) to find a good place to start looking. The diaphragm setting should be very low (small) because these bacteria are almost transparent.
- Switch to the highest power to identify the bacteria according to the shape of

cells and arrangement of the cells (pairs, clusters, chains, etc.).

- Make a drawing of the different bacteria you can see.

Question

- Why gram-positive bacterial retain crystal violate colour at the end of staining?
- Why gram-negative bacteria red at the end of staining?
- How gram-negative bacteria differ from gram positive? Discuss on their difference.

2.1.2. Nutritional types of bacteria

Bacteria have evolved many mechanisms to acquire the energy and nutrients they need for growth and reproduction. Many are autotrophs, organisms that obtain their carbon from inorganic CO_2. Autotrophs that obtain their energy from sunlight are called **photoautotrophs,** while those that harvest energy from inorganic chemicals are called **chemoautotrophs.** Other bacteria are heterotrophs, organisms that obtain at least some of their carbon from organic molecules like glucose. Heterotrophs that obtain their energy from sunlight are called **photoheterotrophs,** while those that harvest energy from organic molecules are called **chemoheterotrophs**. Sources of energy includes;

(1) light energy, and

(2) the energy derived from oxidizing organic or inorganic molecules.

Phototrophs use light as their energy source; **chemotrophs** obtain energy from the oxidation of chemical compounds (either organic or inorganic). Bacteria also have only two sources for electrons. **Lithotrophs** (i.e., "rock-eaters") use reduced inorganic substances as their electron source, whereas **organotrophs** extract electrons from reduced organic compounds.

Despite the great metabolic diversity seen in bacteria, most may be placed in one of five nutritional classes based on their primary sources of carbon, energy, and electrons. The majority of bacteria thus far studied are either photolithoautotrophic or chemoorganoheterotrophic.

Photolithoautotrophs (often called simply **photoautotrophs**) use light energy and have

CO_2 as their carbon source. Photosynthetic bacteria and cyanobacteria employ water as the electron donor and release oxygen. Other photolithoautotrophs, such as the purple and green sulfur bacteria, cannot oxidize water but extract electrons from inorganic donors such as hydrogen, hydrogen sulfde, and elemental sulfur.

Chemoorganoheterotrophs (sometimes called **chemoheterotrophs** or chemoorganotrophs) use organic compounds as sources of energy, hydrogen, electrons, and carbon.

Frequently the same organic nutrient will satisfy all these requirements. Nearly all pathogenic microorganisms are chemoorganoheterotrophs. The other nutritional types have fewer known microorganisms but are very important ecologically. Some photosynthetic bacteria (purple and green bacteria) use organic matter as their electron donor and carbon source. These **Photoorganoheterotrophs** are common inhabitants of polluted lakes and streams. Some of these bacteria also can grow as photolithoautotrophs with molecular hydrogen as an electron donor.

Chemolithoautotrophs oxidize reduced inorganic compounds such as iron, nitrogen, or sulfur molecules to derive both energy and electrons for biosynthesis. Carbon dioxide is the carbon source.

Chemolithoheterotrophs use reduced inorganic molecules as their energy and electron source but derive their carbon from organic sources. Chemolithotrophs contribute greatly to the chemical transformations of elements (e.g., the conversion
of ammonia to nitrate or sulfur to sulfate) that continually occur in ecosystems.

Table 2.2. Major Nutritional Types of Microorganisms

Nutritional Type	Carbon Source	Energy Source	Electron Source	Representative Microorganisms
Photolithoautotroph	CO_2	Light	Inorganic e^- donor	Purple and green sulfur bacteria, cyanobacteria
Photoorganoheterotroph	Organic carbon	Light	Organic e^- donor	Purple nonsulfur bacteria, green nonsulfur bacteria
Chemolithoautotroph	CO_2	Inorganic chemicals	Inorganic e^- donor	Sulfur-oxidizing bacteria, hydrogen-oxidizing bacteria, methanogens, nitrifying bacteria, iron-oxidizing bacteria
Chemolithoheterotroph	Organic carbon	Inorganic chemicals	Inorganic e^- donor	Some sulfur-oxidizing bacteria (e.g., *Beggiatoa*)
Chemoorganoheterotroph	Organic carbon	Organic chemicals often same as C source	Organic e^- donor, often same as C source	Most nonphotosynthetic microbes, including most pathogens, fungi, and many protists and archaea

2.1.3. Reproduction of bacteria

2.1.3.1. A sexual reproduction

Most bacterial organisms reproduce by an asexual process called **binary fission**, which usually occurs after a period of growth in which the cell doubles in mass. At this time, the chromosome (DNA) replicates and the two DNA molecules separate (Figure 2.7). Chromosome segregation is not well understood. Unlike eukaryotic cells,bacterial cells lack a mitotic spindle to separate replicated chromosomes. The segregation process does involve specialized chromosomal-associated proteins but there is no clear picture describing how most of these proteins work to ensure accurate chromosome segregation. In any event, cell fission at midcell involves the synthesis of a partition, or septum, that separates the mother cell into two genetically identical daughter cells.

2.1.3.2. Asexual reproduction in bacteria

In conjugation, two cells of different mating types come together, and genetic material is transferred from one to the other. In contrast to transformation and transduction, conjugation involves contact between two cells.

Conjugation has been most extensively studied in the bacterium *E. coli.* In the *E. coli* population there are donor cells, or F cells, that have DNA that can be transmitted to recipient cells, or F cells (Figure 2.8). F cells have a DNA sequence known as the F factor (F stands for fertility) that is necessary for a bacterium to serve as a donor during conjugation. The F factor, which consists of about 20 genes, can be in the form of a plasmid or it can be part of the DNA in the bacterial chromosome.

genes encode enzymes essential for transferring DNA. Certain F genes encode sex pili, long, hairlike extensions that project from the cell surface. The sex pilus recognizes and binds to the surface of an F cell, forming a cytoplasmic conjugation

bridge between the two cells. The F plasmid replicates itself, and DNA is transferred from donor to recipient bacterium through the conjugation bridge. F plasmids may also have other types of genes, including those that determine resistance to antibiotics.

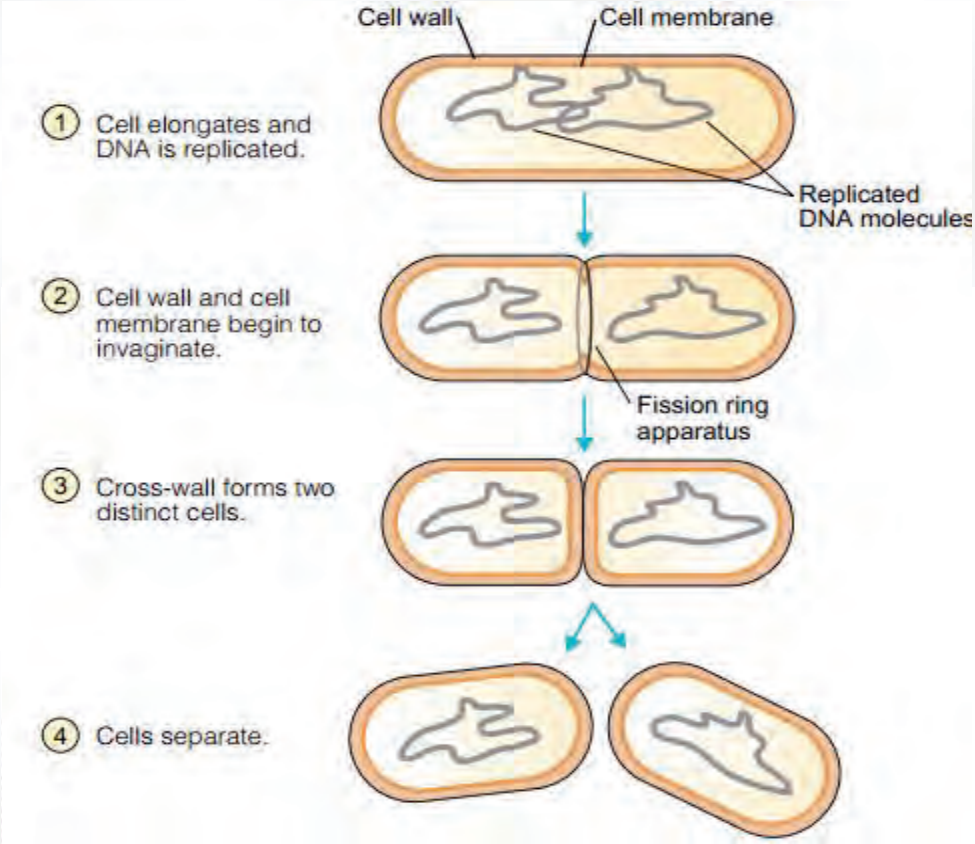


Figure 2.7. The Process of Binary Fission

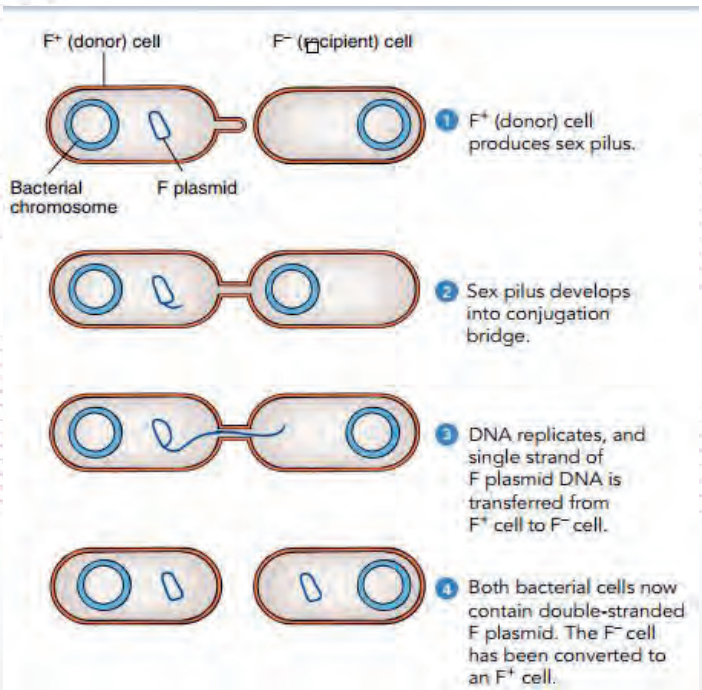


Fig 2.8. Bacterial conjugation

Table 2.3. Common bacterial diseases

Diseas	**Causative agent**	**Description of agent**	**Organs affected**	**Transmission/ vector**
Pertussis	Bordetella pertussis	Gram (-) rod	Upper resp. tract	air
Meningococcal Meningitis	Neisseria meningtidis	Gram (-) diplococcus	Upper resp. tract, meninges	air
Tuberculosis	Mycobacterium tuberculosis	Acid-fast	Lungs, bones, other organs	air
Typhoid Fever	Salmonella typhi	Gm(-) rod	Intestine, blood, gall bladder	Food, water
Cholera	Vibro cholerae	Gm(-) rod	intestine	Food, water
Tetanus	Clostridium tetani	Gm(+) spore-forming anaerobic rod	Nerves at synapse	soil
Syphilis	Treponema pallidum	Spirochete	Skin, cardiovascular organs	sexual
Gonorrhea	Neisseria gonorrhoeae	Gm(-) diplococcus	Urethra, cervix, fallopian tubes, epididymis, eyes, pharynx	sexual
Leprosy	Mycobacterium leprae	Acid-fast	Skin, bones, periphereal nerves	contact

*explain the causative agent, mode of transmission and prevention methods of chancroid, diphtheria, pneumonia, anthrax and shigellosis

2.2. Archaea

At the end of this lesson, student will be able to:

- give definition of archaea.
- describe the general features archaea.
- categorize archaea based on physiological characteristics.
- explain the beneficial aspects of archaea.

Self- questioning

Where are archaea bacteria found?

Why are archaea-bacterial and other bacterial

groups placed together?

List the general characteristics of archaea-

bacteria.

What do you think archaea-bacteria

ARCHAEA: Similar to bacteria, species in the Domain Archaea known as archaeans, are unicellular, microscopic organisms that live as producers or decomposers.

Characterisctics of Archaea

- They are prokartyotic.
- They single celled organisms.
- They lack memebrane bound nucleus and membrane bounded organelles.
- Archaea lack true peptidoglycan in their cell walls.
- Their cell membrane lipids have branched hydrocarbon chains.
- Many are found in extreme environments.

Generally three major groups of archaea are recognized: methanogens (they generate methane), extreme halophiles, and extreme thermophiles (Figure 2.9 & 2.10). These groupings are based on physiological characteristics of the organisms and therefore cannot be considered phylogenetic, or evolutionary, classifications.

The methanogens are strictly anaerobic organisms, having been isolated from such divergent anaerobic environments as waterlogged soils, lake sediments, marshes, marine sediments, and the gastrointestinal tracts of animals, including humans. As members of the anaerobic food chain, they degrade organic molecules to methane.

2. Extreme halophiles grow in highly saline environments such as the Great Salt Lake, the Dead Sea, salt evaporation ponds, and the surfaces of salt-preserved foods. Unlike the methanogens, extreme halophiles are generally obligate aerobes.

3. Extreme Thermophiles:The extreme thermophiles (hyperthermophiles) are found near volcanic vents and fissures that release sulfurous gases and other hot vapors. With optimum temperatures usually in excess of 80^0C, they may be either obligate aerobes, facultative aerobes, or obligate anaerobes.

4. Thermophilic Extreme Acidophiles: Members of two genera, *Thermoplasma* and *Picrophilus,* are notable for growing in extremely acidic, hot environments.

Keywords

- Thermophiles live at high temperatures
- Hyperthermophiles live at really high temperatures (present record is 121°C!)
- Psychrophiles (also called cryophiles) like it cold (one in the Antarctic grows best at 4°C)
- Halophiles live in very saline environments (like the Dead Sea)
- Acidophiles live at low pH (as low as pH 1 and who die at pH 7!)

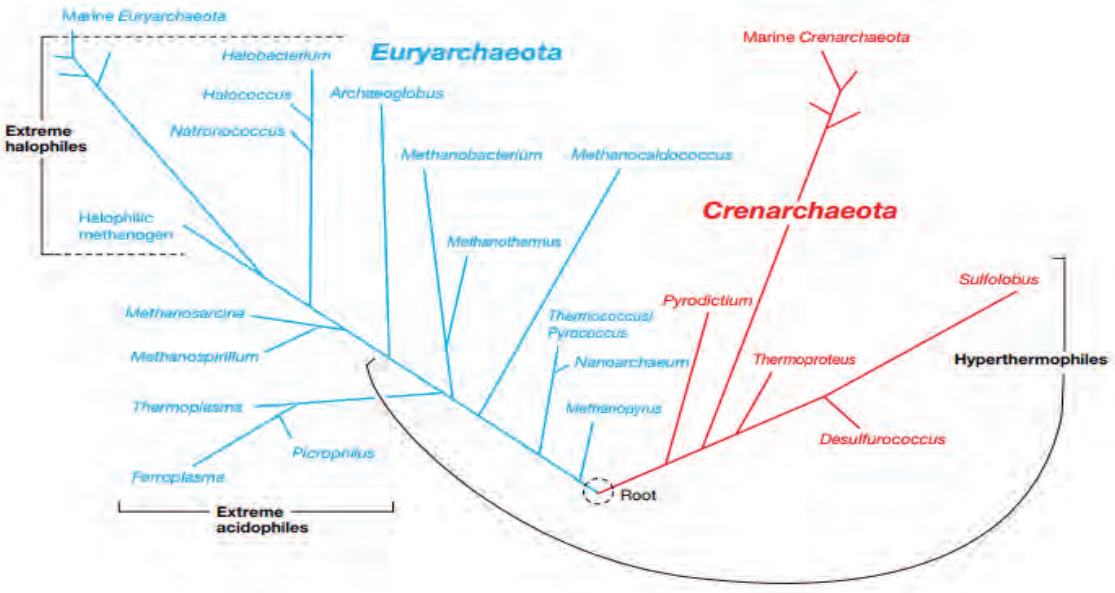


Figure 2.9. Phylogenetic tree of archaea. The tree, based on sequences of 16S ribosomal RNA genes, reveals a major evolutionary split of Archaea into two phyla, the Crenarchaeota and the Euryarchaeota.

Scientists originally identified archaea as a distinct type of prokaryotes on the basis of unique rRNA sequences. Archaea also share other common features that distinguish them from bacteria:

- Archaea lack true peptidoglycan in their cell walls.
- Their cell membrane lipids have branched hydrocarbon chains.
- The initial amino acid in their polypeptide chains, coded by the AUG start codon, is methionine (as in eukaryotes and in contrast to the N-fomylmethionine used by bacteria).

Archaea multiply by binary fission, budding, fragmentation or other mechanisms. Nutritionally they are either aerobic, facultative anaerobic or strictly anaerobic, chemolithoautotroph to organotrophs.

Figure 2.10. Diversity of Archaea

2.2.1. Beneficial aspects of Archaea

Because of their tolerance to high temperatures and relatively extreme environments, some members of the domain have already been exploited for a wide variety of commercial uses.

These Archaea become the source of enzymes that

1) are usually added to detergents in order to help it maintain its activity even at higher temperature and pH.
2) Proteases and lipases derived from alkaliphilic bacteria are being used as detergent additives to increase their stain removal ability
3) Some Archaea also bear the potential for bioremediation or help in cleaning contaminated sites.
4) The thermophilic Archaea, Thermus aquaticus, is an essential part of the development of molecular biology as a science. As a result, Archean has become the source of the enzyme harnessed as the basis for the amplification of the DNA in a technique called *Polymerase Chain Reaction* (**PCR**).

2.2.2. Physical factors that affecting microbial growths

The environments in which some microorganisms grow would kill most other microorganisms, the major physical factors which affect microbial growth are solutes and water activity, pH, temperature, oxygen level, pressure and radiation (Figure 2.11)

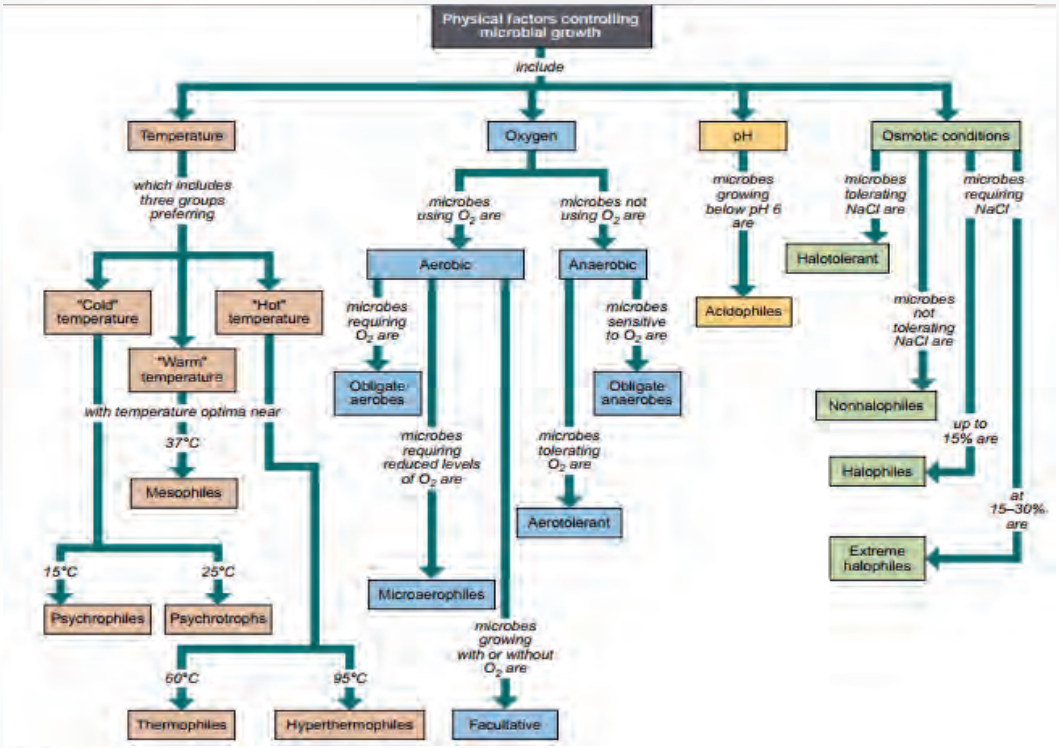


Figure 2.11. Classes of Microbes Based on Physical Factors.

After completing this section, the student will be able to:

- describe the features of fungi.
- label the structures a typical fungus.
- explain the mode of reproducuction in fungi.
- explain how fungi obtain nutrients.
- explain the economic importance of fungi.
- describe the major dvisions of fungi.
- justify why yeasts are useful to humans.
- explain the mode of transmission of fungal diseases of humans.

2.3. Fungi

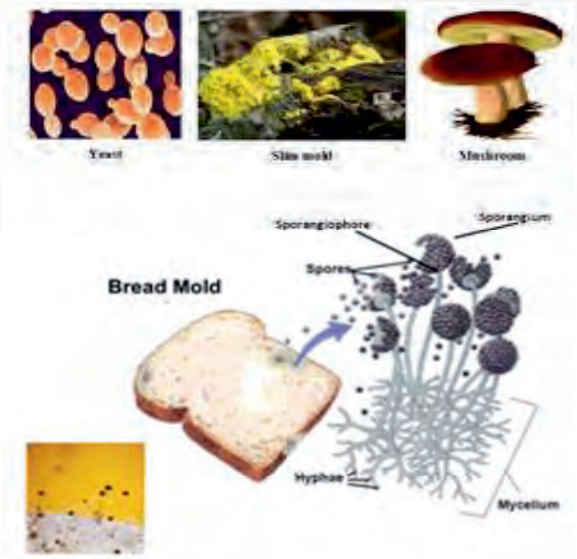


Figure 2.13. Different forms of Fungi

Microbiologists use the term **fungus** [pl., fungi; Latin *fungus,* mushroom] to describe eucaryotic organisms that are spore-bearing, have absorptive nutrition, lack chlorophyll, and reproduce sexually and asexually. Scientists who study fungi are **mycologists** [Greek *mykes,* mushroom, and *logos,* science], and the scientific discipline devoted to fungi is called **mycology.** The study of fungal toxins and their effects is called **mycotoxicology,** and the diseases caused by fungi in animals are known as **mycoses** (s., mycosis).

Self-questioning

- ? What are the general characteristics of true fungi?
- ? Do fungi have a benefit for humans? If yes, list their importance.
- ? Where do different fungal groups grow in nature? How do different fungi get their nutrients?
- ? What features of fungal species may be interpreted as adaptations to life as a saprobe, a pathogen, or a mutualistic/ symbiont? Which fungal features permit some fungal species and not others to be pathogenic on humans?
- ? What are the ecological roles of different kinds of fungi?
- ? How do different kinds of fungi, including lichenized fungi, reproduce asexually? sexually?
- ? Why are there two names for the same fungus or fungal group? Provide specific examples.

2.3.1. General Characteristics of True Fungi

1. All are eukaryotic

- Possess membrane-bound nuclei (containing chromosomes) and a range of membrane bound cytoplasmic organelles (e.g. mitochondria, vacuoles, endoplasmic reticulum).

2. Most are filamentous (Figure 2.12, & 2.13)

- Composed of individual microscopic filaments called hyphae, which

exhibit apical growth and which branch to form a network of hyphae called a mycelium (Figure 2.8).some have septate hyphae, and the other have nonsepatate (coenocytic hyphae)

3. Some are unicellular • e.g. yeasts. (Figure 2.12)

4. Protoplasm of a hypha or cell is surrounded by a rigid wall

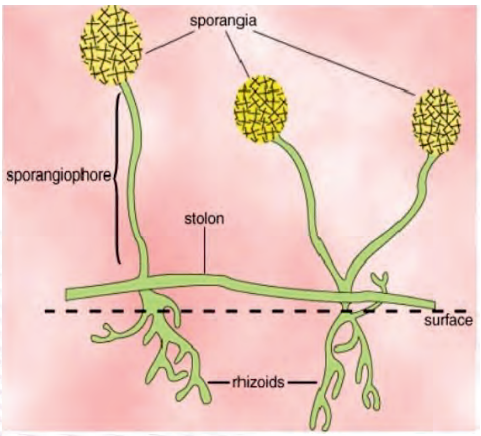


Figure 2.13. mycelium

- Composed primarily of chitin and glucans, although the walls of some species contain cellulose.

5. Many reproduce both sexually and asexually

• Both sexual and asexual reproduction often result in the production of spores.

6. Their nuclei are typically haploid and hyphal compartments are often multinucleate

• Although the Oomycota and some yeasts possess diploid nuclei.

7. All are achlorophyllous

• They lack chlorophyll pigments and are incapable of photosynthesis.

8. All are chemoheterotrophic (chemo-organotrophic)

• They utilize pre-existing organic sources of carbon in their environment and the energy from chemical reactions to synthesize the organic compounds they require for growth and energy.

9. Possess characteristic range of storage compounds

Self-questioning

? What are traditional fermented foods?
? List the Ethiopian traditional fermented foods, beverages, and condiments.
? What do you think is the role of fungi in the production of traditional fermented products?
? What is starter culture?
? How do local people preserve starter culture for a long-time use?
? Which fungal groups are commonly involved in food fermentation?

• e.g. trehalose, glycogen, sugar alcohols and lipids.

10. Nutritionally categorized into three (saprophytic, parasitic, and symbiotic).

*what are the differences between fungi and bacteria

2.3.2. Ecology of Fungi

Fungi have colonized nearly all environments on Earth, but are frequently found in cool, dark, moist places with a supply of decaying material. Fungi are saprobes that decompose organic matter. Many successful mutualistic relationships involve a fungus and another organism. Many fungi establish complex mycorrhizal associations with the roots of plants. Some ants farm fungi as a supply of food.

Lichens are a symbiotic relationship between a fungus and a photosynthetic organism, usually an algaor cyanobacterium. The photosynthetic organism provides energy derived from light and carbohydrates,while the fungus supplies minerals and protection.

- Zygomycota. Sporangial fungi, E.g. rhizopus and mucor
- Ascomycota. Ascospore producing fungi. E.g. *Saccharomyces cerviciae*
- Basidomycota. Basidia producing fungi. E.g. rusts & smuts

2.3.4. Reproduction in fungi

Sporulation is the process of spore formation. It usually occurs in structures called **fruiting bodies**, which represent the part of a fungus in which spores are formed and from which they are released. These structures may be asexual and invisible to the naked eye, or sexual structures, such as the macroscopic mushrooms..

Self-questioning

- Which funguses are common in food spoilage?
- What does a fungal disease mean?
- List at least five diseases caused by fungi?
- What are the main human diseases caused by fungi?

Some animals that consume fungi help disseminate spores over long distances.

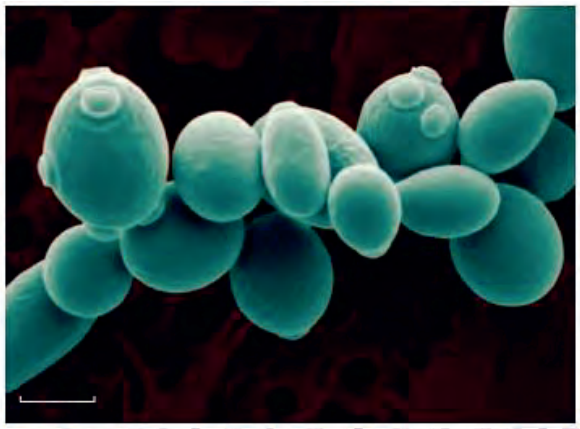
2.3.3. Classification of Fungi

Many mycologists currently recognize five major groups of fungi, although recent genomic evidence indicates that the chytrids and zygomycetes are paraphyletic.

In a group of 5-10, search on Google and discuss the fungal specie that are common to the Ethiopian traditional fermented Enjera, Kocho, Bulla, Tej, Shameta, and different condiments.

The current classifications of fungi includes:

- Chytridomycota. Zoospore producing fungi. E.g. allomyces & water molds
- Glomeromycota. E.g. mycorrhizal fungi.



A) Asexual reproduction

Asexual reproductive structures develop at the ends of specialized hyphae. As a result of mitotic divisions, thousands of spores are produced, all genetically identical.

Many asexual spores develop within sacs or vessels called sporangia (sing., sporangium; *angio* = "vessel"). Appropriately, the spores are called sporangiospores. Other fungi produce spores on supportive structures called conidiophores. These unprotected, dust-like spores are known as conidia (sing., conidium; *conidio* = "dust") (Figure 2.14). Fungal spores are extremely light and are blown about in huge numbers by wind currents. In yet other fungi, spores may form simply by fragmentation of the hyphae yielding arthrospores (*arthro* = "joint"). The fungi that cause athlete's foot multiply in this manner.

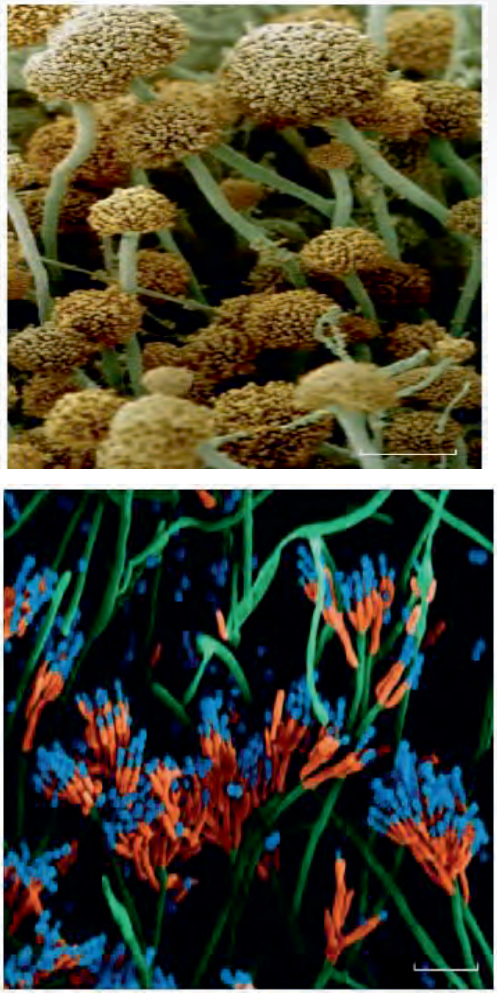


Figure 2. 14. Fungal fruiting bodies

A). Sporangia of the common bread mold *Rhizopus (B)* The conidiophores and conidia

Many yeasts reproduce asexually by **budding (Figure 2.15)**. In this process, the cell becomes swollen at one edge, and a new cell called a **blastospore**, (*blasto* = "bud") develops (buds) from the parent cell. Eventually, the spore breaks free to live independently. The parent cell can continue to produce additional blastospores.

Figure 2.15. Budding in yeast

B) Sexual reproduction

Many fungi also produce spores by sexual reproduction. In this process, opposite mating types come together and fuse (Figure 2.16). Because the nuclei are genetically different in each mating type, the fusion cell represents a heterokaryon (*hetero* = "different"; *karyo* = "nucleus"); that is, a cell with genetically dissimilar nuclei existing for some length of time in a common dikaryotic cytoplasm. Eventually the nuclei fuse and a diploid cell is formed. The chromosome number soon is halved by meiosis, returning the cell or organism to a haploid condition.

A visible fruiting body often results during sexual reproduction and it is the location of the haploid spores. Perhaps the most recognized fruiting body from which spores are produced is the mushroom.

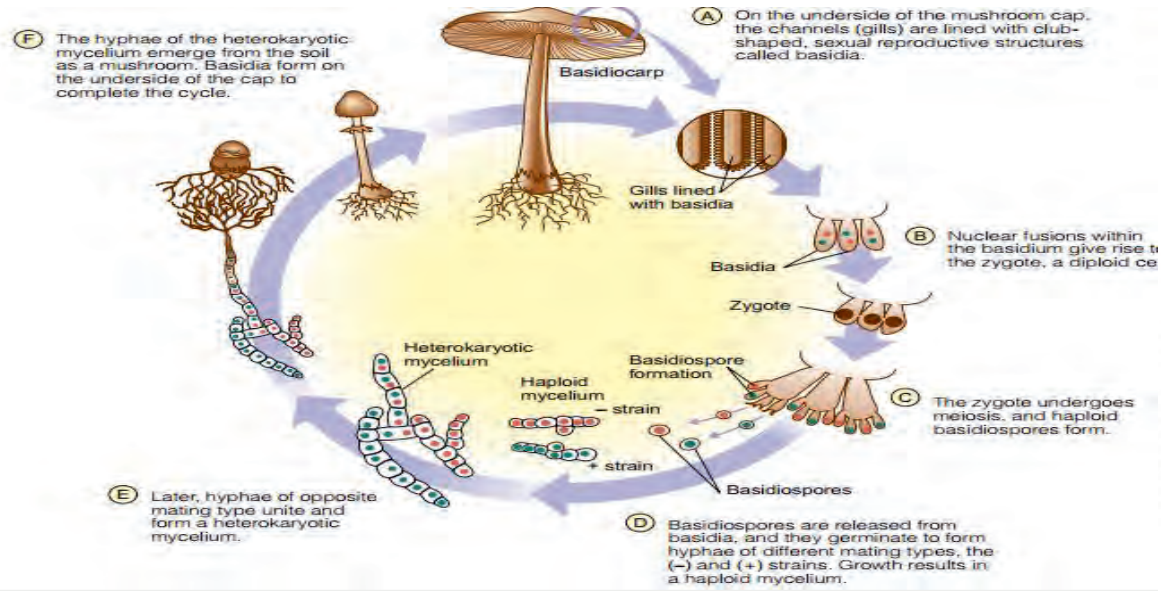


Figure 2 .16 The Sexual Reproduction Cycle of a Typical Basidiomycete.

2.3.5. Economic importance of fungi

Fungi are among the economic important microrganisms. They have both beneficial and harnmful aspects.

Beneficial aspects of fungi

1. Fungi exist either as *saprobes* or *parasites*. Their preeminent ability to break down complex organic substrates of almost every type is an important and essential activity in the recycling of carbon and other elements in the cycle of life.

2. Edible wild or domesticated varieties of mushrooms (*Basidiomycetes*) are important as food sources.

3. Fungi, especially the yeasts, are essential to many industrial processes involving fermentation. Examples include the making of bread, wine, and beer.

4. Fungi also play a major role in the preparation of some cheeses, soy sauce, and

sufu, Enjera, Tela(Farsoo), Tej(Dadhii), bulla, ETC. (Figure 2.17); and in the commercial production of many organic acids (citric, gallic) and certain drugs (ergometrine, cortisone).

5. Molds (such as *Aspergillus* species) are used in the production of citric, oxalic, gluconic and itaconic acid.

Products of fermentation yield industrial alcohol, fats and proteins. A mold (*Fusarium*) can produce within 48 hours, 12-15 grams of fat from a litre of 50% glucose solution.

6. Fungi play a major role in the manufacture of many antibiotics (penicillin, griseofulvin) and the immunosuppressive drug cyclosporin.

7. Actinomycetes and fungi are important sources of antibiotics such as penicillin, amphotericin B, adriamycin and bleomycin, etc.

8. Fungi are useful tools for studying complex eukaryotic events, such as cancer and aging within a simple cell.

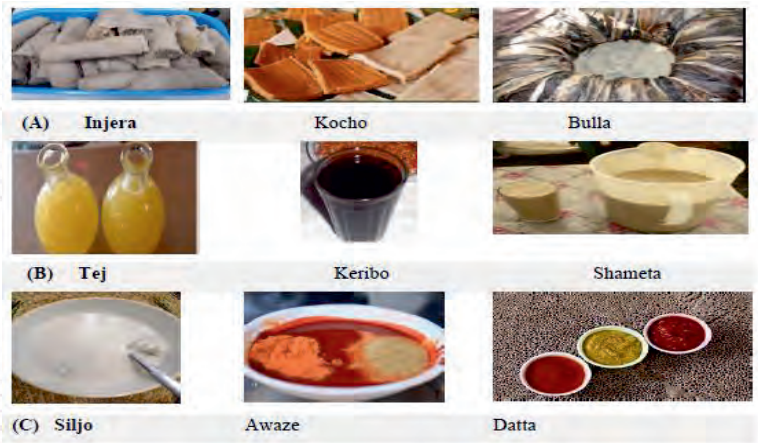


Figure 2.17. Some Ethiopian traditional fermented products prepared by using yeast for fermentations (A= staple foods; B, beverages; and C, condiments)

Harmful aspects of Fungi

1. Fungi are the major cause of plant diseases. Plants are particularly vulnerable to fungal diseases because fungi can invade leaves through their stomates. Over 5,000 species attack economically valuable crops, garden plants, and many wild plants. Fungi also cause many diseases of animals and humans.

2. Molds can cause deterioration of fabrics, leather, electrical insulation and other

manufactured goods. Extensive losses may follow failure to protect material from ravages of fungi in warm humid climates.

3. Fungi can spoil the agricultural produce, if improperly stored. These also destroy vegetables, fruits and cereals.

4. Mycotoxicoses (ingestion of toxins of fungal origin) and mycetismus (mushroom poisoning through ingestion of fungal elements).

A. Aflatoxis: Two closely related fungi, *Aspergillus flavus* and *A. parasiticus,* produce mycotoxins called **aflatoxins**. The molds are found primarily in warm, humid climates, where they contaminate agricultural products such as peanuts, grains, cereals, sweet potatoes, corn, rice, and animal feed. Aflatoxins are deposited in these foods and ingested by humans where they are thought to be carcinogenic, especially in the liver. Contaminated meat and dairy products are also sources of the toxins.

B. Ergotism is caused by *Claviceps purpurea,* an ascomycete fungus producing a powerful toxin. *C. purpurea* grows as hyphae on kernels of rye, wheat, and barley. As hyphae penetrate the plant, the fungal cells gradually consume the substance of the grain, and the dense tissue hardens into a purple body called a **sclerotium**. A group of peptide derivatives called alkaloids are produced by the sclerotium and deposited in the grain as a substance called **ergot**. Products such as bread made from rye grain may cause ergot rye disease, or ergotism.

C. **Mushroom poisoning**, or **mycetism,** can occur from mushrooms that produce mycotoxins that affect the human body.

5. Superficial Fungal infections

Superficial mycoses are fungal infections of the outermost areas of the human body: hair, fingernails, toenails, and the dead, outermost layers of the skin (the epidermis). This a complex of diseases caused by any of several species of taxonomically related flamentous fungi in the genera *Trichophyton, Epidermophyton,* and *Microsporum*.

The various forms of dermatophytosis are referred to as tineas or "ringworm." Clinically, the tineas are classifed according to the anatomic site or structure affected:

1. *Tinea corporis* (ringworm): Microsporum canis and Trichophyton mentagrophytes. Affects hairless skin.
2. *Tinea pedis* (athlete's foot): *T. rubrum*, T. *mentagrophytes*, and *Epidermophyton floccosum*. Affects mainly the lower legs.
3. *Tinea capitis: T. tonsurans* and *M. canis*. the scalp, eyebrows, and eyelashes
4. *Tinea barbae: T. rubrum and T. mentagrophytes*. Beard ringworm.

5. *Tinea unguium* (also known as **onychomycosis**).: *T. rubrum, T. mentagrophytes*, and *E. floccosum*. Affect the nails

Mode of transmission: Superficial mycoses (Dermatomycoses) are infections that are transmitted directly by human contact, animal-human contact or indirectly on inanimate objects (clothes, carpets, moisture, and dust in showers, swimming pools, wardrobes, gyms). The localization of the primary foci corresponds to the contact site. Thus feet, uncovered skin (hair, head, facial skin) are affected most frequently (Figure 2.18).

Prevention: Regular disinfection of showers and wardrobes can contribute to prevention of athlete's foot, a very frequent infection

Onychomycosis caused by *Trichophyton rubrum*

Tinea capitis caused by *Microsporum canis*

Figure 2.18. Fungal disease affecting human and plants

Table 2. 4. Summry of some fungal diseases

Diseases	Causative agent	Signs and symptoms	Transmission	Prevention and control
Dermatophytosis (tineas)	Epidermophyton Microsporum Trichophyton	Blister-like lesions	Fragments of skin on skin floors or surfaces	Keeping skin dry not sharing personal items
Candidiasis	Candida albicans	Itching, burning pains, “cheesy” discharge	sexually	Avoiding baths hot and tubs , Avoiding douches
Thrush	Candida albicans	White flecks on mucous membranes	Passage through the birth canal	Practicing good oral hygiene, limiting sugar intake
Aspergillosis	Aspergillus fumigatus	Bloody cough, chest pain, Wheezing Shortness of breath	Air borne spores	Staying away from sources of mold

2.4. Protozoa

The term protozoa [s., protozoan; Greek protos, first, and zoon, animal] has traditionally referred to chemoorganotrophic protists, and protozoology generally refers to the study of protozoa.

Characteristics of protozoa

- Unicellular microorganisms that lacks cell wall.
- They are may be free living or parasitic.
- Aerobic.
- They have true nucleus.
- They are eukaryotic cells.
- Mostly microscopic, although some are large enough to be seen with the unaided eye (Figure 2.19).
- Locomotion by pseudopodia, flagella, cilia, and direct cell movements; some sessile.
- Nutrition of all types: autotrophic (manufacturing own nutrients by photosynthesis), heterotrophic (depending on other plants or animals for food), saprozoic (using nutrients dissolved in the surrounding medium)
- Aquatic or terrestrial habitat; free-living or symbiotic mode of life
- Reproduction asexually by fission, budding, and cysts and sexually by conjugation or by syngamy (union of

After completing this section, the student will be able to:

- describe the general features of protozoa.
- describe the structures of ameaba, euglena, paramecium.
- explain how protozoans get nutrition.
- describe the modes of reproduction in protozoa.
- describe the process of conjugation in protozoa.
- list the common diseases caused by protozoans.
- explain the ways of transmission of protozoan diseases.
- explain the economic importance of on human protozoans.
- analyze the health impacts of different protozoan diseases.

Self-questioning

- Can protozoan reproduce?
- How are Amoeba, Euglena, and Paramecium reproduced?
- Draw and label the structure of Amoeba, Euglena, and Paramecium.

male and female gametes to form a zygote).

Table 2.5. Classification of protozoa

CATEGORY	MEANS OF MOVEMENT	METHOD OF ASEXUAL REPRODUCTION	METHOD OF SEXUAL REPRODUCTION	REPRESENTATIVES
Ciliates	Cilia	Transverse fission	Conjugation	*Balantidium coli, Paramecium, Stentor, Tetrahymena, Vorticella*
Amebae (amebas)	Pseudopodia (false feet)	Binary fission	When present, involves flagellated sex cells	*Amoeba, Naegleria, Entamoeba histolytica*
Flagellates	Flagella	Binary fission	None	*Chlamydomonas, Giardia lamblia, Trichomonas, Trypanosoma*
Sporozoa	Generally nonmotile except for certain sex cells	Multiple fission	Involves flagellated sex cells	*Plasmodium, Toxoplasma gondii, Cryptosporidium*

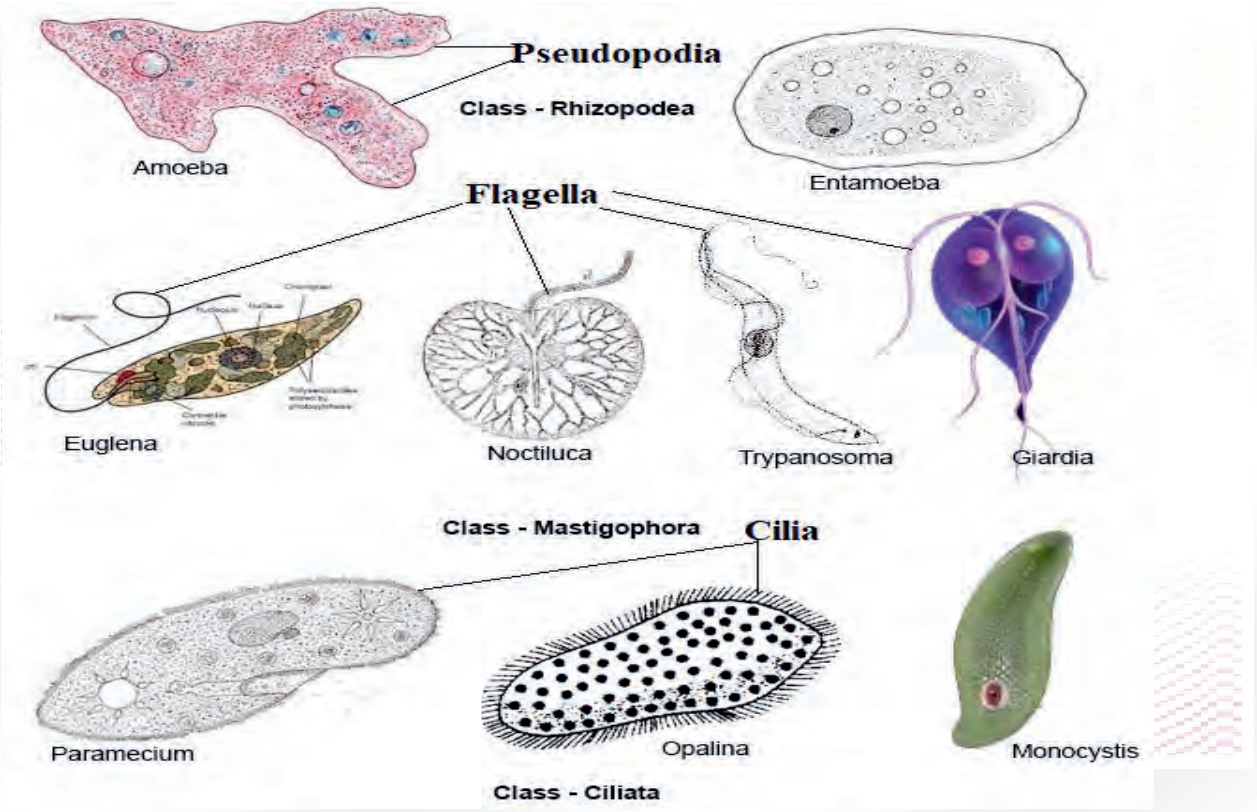


Figure 2.19. Common protozoa

Reproduction in protozoa

Most protozoa are asexual and reproduce in one of three ways. These are:

• *Fission:* Fission occurs when a cell divides evenly to form two new cells (Figure 2.20).

• *Budding:* Budding occurs when a cell divides unevenly.

• *Multiple fission (schizogony):* Multiple fission is when the nucleus of the cell divides multiple times before the rest of the cell divides. Forms around each nucleus when the nucleus divides then each nuclei separates into a daughter cell.

Activity 2.6

Be in a group of 5 students, select one group leader and read your textbook and tell your understanding to the leader and finally discuss together the characteristics and economic importance of some common protozoa. If you have, any difficulties invite your teacher to clarify for you.

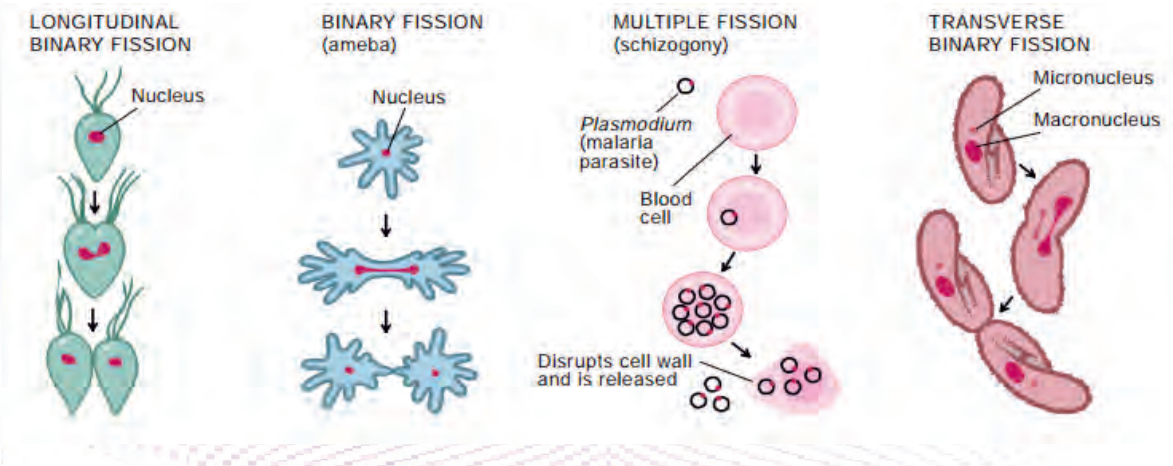


Figure 2.20 Asexual reproduction in protozoa

Sexual reproduction also occurs during the life cycle of most protozoa. A distinctive feature of ciliates is the presence of two types of nuclei: tiny micronuclei and large macronuclei. A cell has one or more nuclei of each type. Genetic variation results from **conjugation (Figure 2.21)**, a sexual process in which two individuals exchange haploid micronuclei but do not reproduce. Ciliates generally reproduce asexually by binary fission, during which the existing macronucleus disintegrates and a new one is formed from the cell's micronuclei. Each macronucleus typically contains multiple copies of the ciliate's genome. Genes in the macronucleus control the everyday functions of the cell, such as feeding, waste removal, and maintaining water balance.

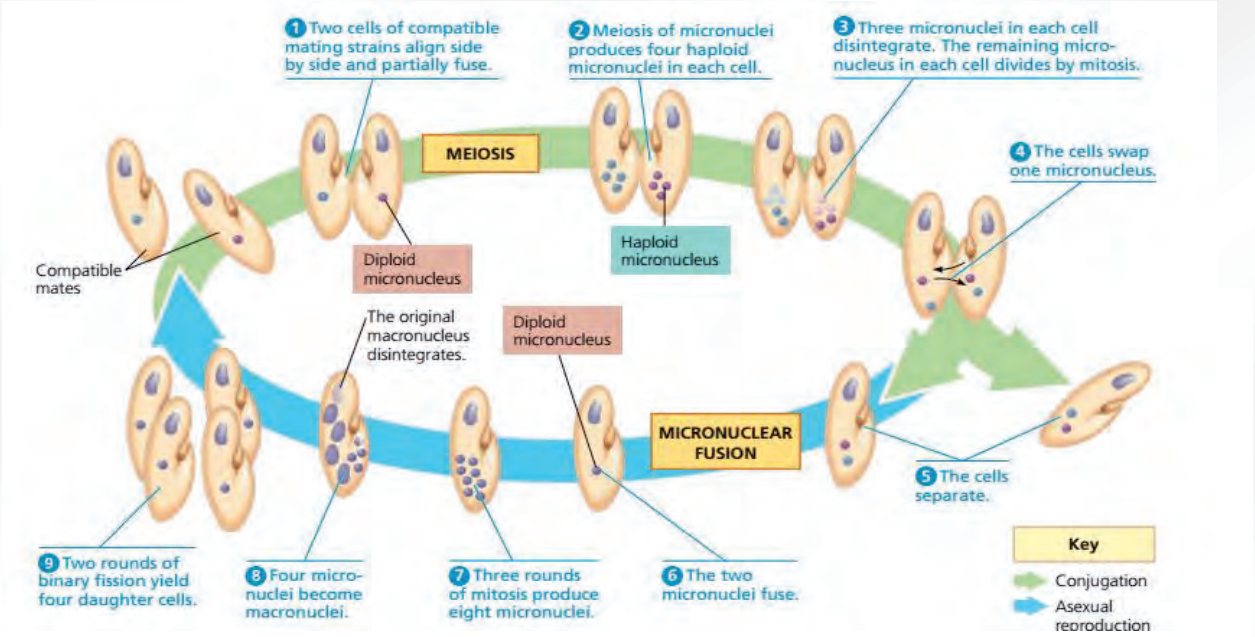


Figure 2.21. Conjugation in Paramecium

Nutrition in Protozoans

Protists receive nutrients by breaking down organic matter (*heterotrophic*) and can grow in both aerobic and anaerobic environments, such as protists that live in the intestine of animals. Some protists, such as *Euglena*, receive nutrients from organic matter and through photosynthesis because they contain chlorophyll. These protists are considered both algae and protozoa. Protists obtain food in one of three ways:

- **Absorption:** Food is absorbed across the protist's plasma membrane.
- **Ingestion:** Cilia outside the protist create a wave-like motion to move food into a mouth-like opening in the protist called a *cytosome*. An example is the paramecium.
- **Engulf:** Pseudopods (meaning "false feet") on the protist engulf food, then pull it into the cell using a process called *phagocytosis*. An example of this type of protist is the amoeba.

Self-questioning

- Can protozoan reproduce?
- How do Amoeba, Euglena, and Paramecium reproduce?
- Draw and label the structure of Amoeba, Euglena, and Paramecium.

Food is digested in the vacuole after the food enters the cell. The *vacuole* is a membrane-bound organelle. Waste products are excreted using a process called exocytosis.

2.4.1.Common diseases caused by protozoa

Most **protozoas** are not harmful but there are a few disease-causing protozoans. Many types of protozoa are even beneficial in the environment because they help make it more productive. They improve the quality of water by eating bacteria and other particles. Some of the human diseases caused by protozoans include malaria, African trypanosomiasis, amoebiasis, giardiasis, chagas disease, leishmaniosis, toxoplasmosis, and cryptosporidiosis (Figure 2.22).

Table 2.6. Major protozoal parasites of human and other organisms

Disease	Causative agent	**Organs affected**	**Transmission /vector**	**Clinical Features**
Amoebiasis	Entamoeba histolytica	Intestine, liver	Ingestion of fecally contaminated food or water; disease associated with poverty, homosexual men, and migrant workers	Multiplication of the organisms and tissue destruction in the intestine and in other body tissues can result in amebic abscesses. The irritating effect of the amebas on the cells lining the intestine causes intestinal cramps and diarrhea. Due to intestinal ulceration, the diarrheal fluid is often bloody, and the condition is referred to as amebic dysentery
Giardiasis	Giardia lamblia	Intestine	Ingestion of fecally contaminated water; person-to-person, in day care centers	Ingested cysts survive stomach passage; trophozoites emerge from the cysts in the small intestine, where some attach to epithelium and others move freely; mucosal function is impaired by adherent protozoa and host immune response.
Trichomon	Trichomonas vaginalis	Urogenital	Sexual contact	Vaginal discharge, odor and edema or erythema

Disease	Causative agent	**Organs affected**	**Transmission /vector**	**Clinical Features**
iasis		organs		
African sickness	Trypanosoma brucei	Blood, brain	Tsetse fly (*Glossina*)	Initial haemolytic phase (fever, joint pains followed by neurological disorder, somnolence)
Leishmaniasis (kala-azar)	Leishmania donovani	White blood cells, skin, intestine	Sand fly (Phlemotomus)	Skin ulcers, mucocutaneous complications and visceral diseases (hepatosplenomegaly)
Malaria	Plasmodium spp.	Liver, red blood cells	Mosquito (Anopheles)	Fever, shivering, cough, respiratory distress, pain in the joints, headache, watery diarrhea, vomiting, convulsions, severe anemia
Toxoplasmosis	Toxoplasma gondii	Blood, eye	Domestic cats, food	Blindness and mental retardation can result in congenitally infected children. Immunosuppressed patients can present more severe symptoms: splenomegaly, polymyositis, dermatomyositis, chorioretinitis, myocarditis, pneumonitis, hepatitis, encephalitis, and multisystem organ failure.

Figure 2.22. Leishmaniasis and trypanosomiasis

By the end of this section, the student will be able to:

- give a brief definition of viruses.
- describe the structures of a virus
- explain the characteristics of viruses.
- label the structures of viruses.
- justify why viruses must find an appropriate host cell to replicate themselves.
- classify viruses as RNA viruses and DNA viruses.
- explain the mechanism of replication viruses.
- Distinguish between lytic and lysogenic cycle of a virus.
- Mention the common viral diseases in Ethiopia.
- Describe the ways of transmission and prevention of HIV/AIDS, Hepatitis B.

Self-questioning

- What are Babesia and Theileria?
- State the common species of plasmodium in Ethiopia.
- Explain the causative agent and vector of Nagana(Gendi).
- Describe the vector and life cycle of leshimania spp.
- State the routes of transmission of Giardia and amoeba.
- Show the life cycle of plasmodium spp. and describe their ecology.

2.5. Viruses

Self-questioning

- Why we studying viruses? Discuss in-group and present to your class.
- How viruses differ from other organisms?
- Write in brief the general features and occurrence of viruses.

Keyword

A virus is a very small, non-cellular parasite of cells. Its genome, which is composed of either DNA or RNA, is enclosed in a protein coat. Viruses are different from other micro-organisms because they have no cellular organelles and so cannot carry out any metabolic processes; they must all enter other cells to reproduce

Viruses are small, obligate, intracellular particles; that is, most can be seen only with the electron microscope (Figure 2.23) and they must infect and take over a host cell in order to replicate. This is because they lack the chemical machinery for generating energy and synthesizing large molecules. Viruses, therefore, must find an appropriate host cell in which they can replicate—and, as a result, often cause disease.

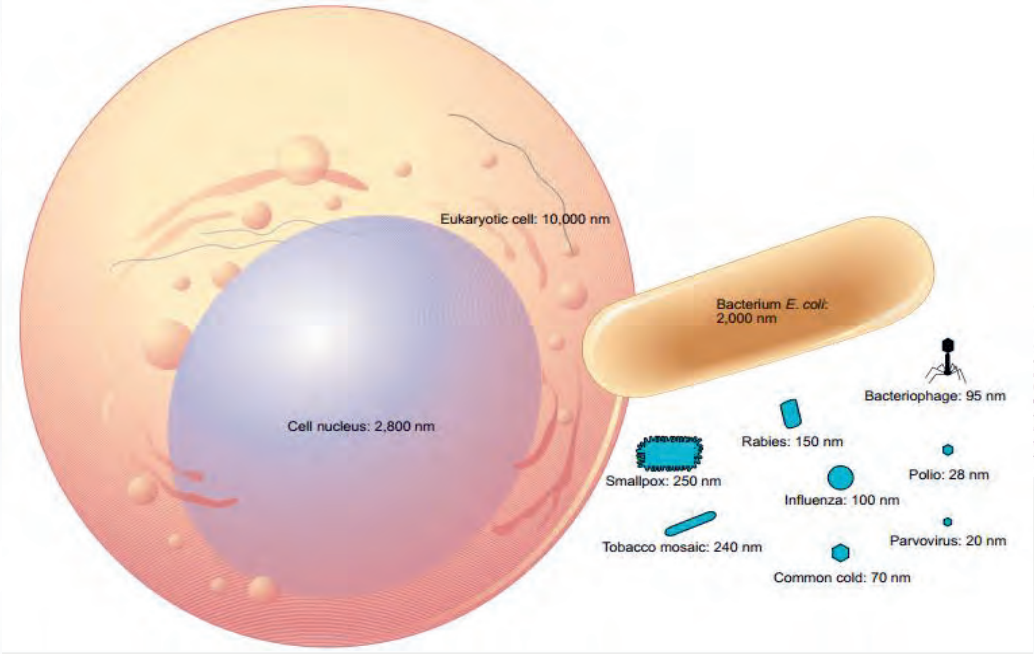


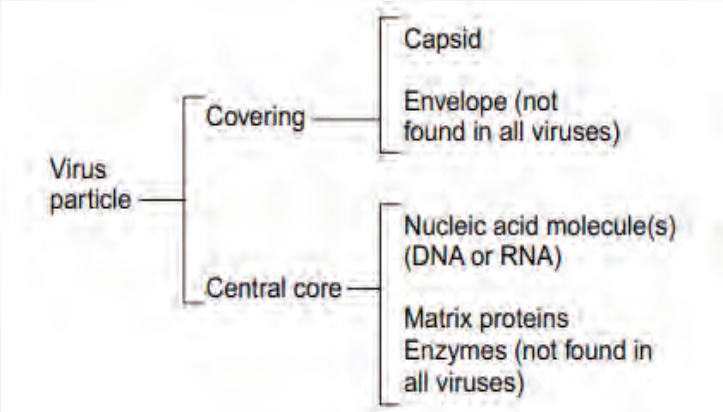
Figure 2.23. Variations in size among eukaryotic cells, prokaryotic cells and viruses

2.5.1.Characteristics of virus

- Viruses have an inner core of nucleic acid surrounded by protein coat known as an envelope.
- They cannot be grown on artifcial cell free media (However, grow in animals, eggs or tissue culture).
- Viruses do not have a cellular organization. They do not have cell wall or cell membrane or cellular organelles including ribosomes.
- They do not occur free in nature but act as obligate intracellular parasite.
- They lack the enzymes necessary for protein and nucleic acid synthesis and are dependent for replication on the synthetic machinery of host cells.
- They are unaffected by antibacterial antibiotics.
- Viruses are inert (nucleoprotein) filterable Agents
- Virus occupy a space in between living and non-living, because they are crystallizable and non-living outside the body of host.
- Obligate intracellular parasites of bacteria, protozoa, fungi, algae, plants, and animals.
- Ultramicroscopic size, ranging from 20 nm up to 450 nm (diameter)
- Do not independently fulfill the characteristics of life
- Inactive macromolecules outside the host cell and active only inside host cells
- Basic structure consists of protein shell (capsid) surrounding nucleic acid core.
- Nucleic acid of the viral genome is either DNA or RNA but not both.
- Nucleic acid can be double-stranded DNA, single-stranded DNA, single-stranded RNA, or double-stranded RNA.
- Molecules on virus surface impart high specif city for attachment to host cell.
- Multiply by taking control of host cell's genetic material and regulating the synthesis and assembly of new viruses
- Lack machinery for synthesizing proteins

Structure of viruses

A basic structure of virus is nucleic acid core (either DNA or RNA but not both) surrounded by protein coat (Figure 2.24). Central core of nucleic acid of a virus is called genome and the protein coat surrounding is called as capsid. In some virus, an envelope made up of glycoprotein and phospholipid bilayer is present outside the capsid.



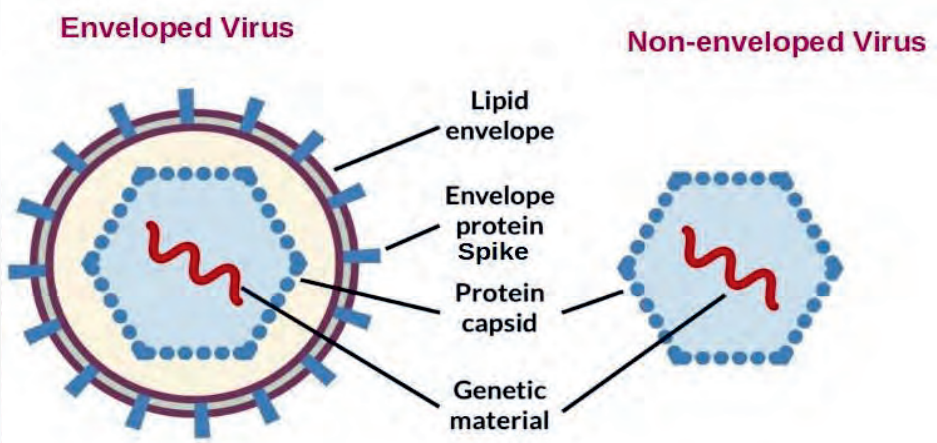


Figure 2.24. Viral Structure:

The basic structural components of a virus are:

Viruses have a very simple structure consists of the following

- Core-the genomic material, either DNA or RNA. DNA or RNA may be single-stranded or double stranded
- Capsid-a protective coat of protein surrounding the core
- Nucleocapsid-the combined structure formed by the core and capsid
- Envelope- a few viruses, such as the HIV and influenza viruses, have an additional lipoprotein layer around the capsid derived from the cell surface membrane of the host cell.
- Capsomeres- capsids are often built up of identical repeating subunits called capsomeres.
- Some virus contains enzymes which play central role during infection process. E.g., Some bacteriophage contains an enzyme

lysozyme, which makes small hole in bacterial cell that allows viral nucleic acid to get in. Some virus contains their own nucleic acid **polymerase** which transcribe the viral genome into mRNA during replication process. E.g., Retrovirus are RNA virus that replicates inside host cell as DNA intermediate. These viruses possess an **RNA dependent DNA polymerase** called **reverse transcriptase**.

2.5.2.Viral symmetry

1. Helical symmetry

There are several viruses found with a **helical** morphology. These viruses consist of identical protein subunits or protomers which assembled in a **helical** structure around the genome (Figure 2.14). This type of protein subunits generally forms a rigid nucleocapsid. Moreover, the helical structure provides flexibility to the filaments. The most common example of a helical virus is the tobacco mosaic virus. Apart from that Sendai virus is also reported to have a helical structure.

1. Icosahedral Symmetry

An icosahedron structure refers to a type of **polyhedron** with 20 equilateral triangular faces and 12 vertices. The rigid structure provides protection to the genome (Figure 2.14). The common examples of viruses reported to have an icosahedral structure are papovavirus, picornavirus, adenovirus, toga virus, etc.

2. Complex Symmetry

These groups of viruses do not come under the above-motioned groups. These viruses consist of complex structural components which made it different from the other two groups (Figure 2.25). A common example of this group of the virus is the pox virus.

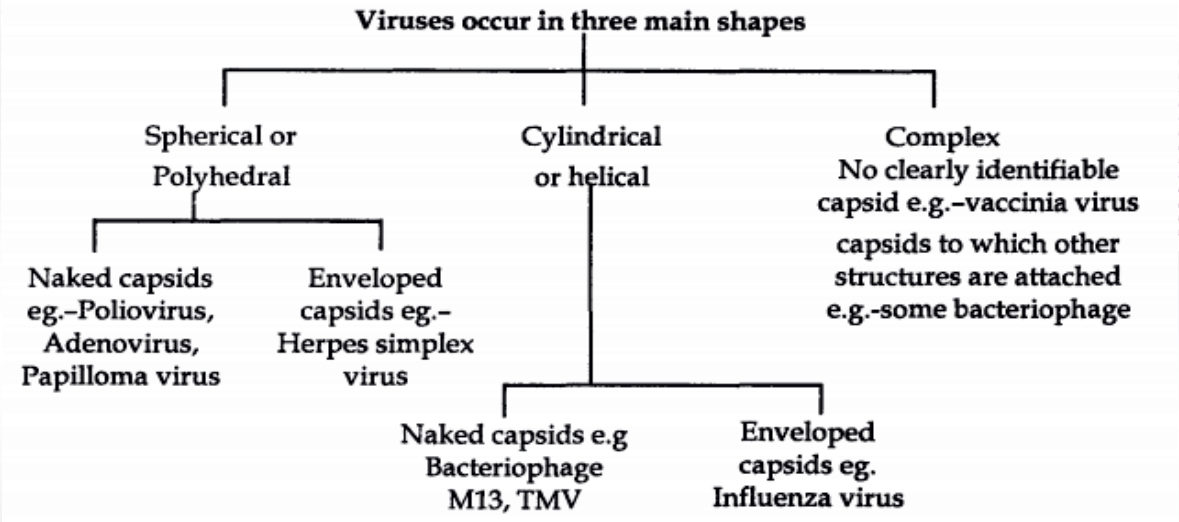


Figure 2.25. Viral classification based ontheir structures

Difference between DNA & RNA Viruses

- The viruses that contain DNA as their genetic material are called the DNA viruses. RNA viruses, on the other hand, contain RNA as their genetic material.
- DNA viruses are mostly double-stranded while RNA viruses are single-stranded.
- RNA mutation rate is higher than the DNA mutation rate.
- DNA viruses replicate in the nucleus while RNA viruses takes place in the cytoplasm.

Activity 2.7

- Be in groups of 5-10, discuss, and give an illustrated explanation on the morphology and chemical structure of viruses and write an essay on multiplication of animal viruses.
- What do you know about bacteriophages? What is it use in biotechnology?

Viruses infect all cellular life forms: eukaryotes (vertebrate animals, invertebrate animals, plants, fungi) and prokaryotes (bacteria and archaea). The viruses that infect prokaryotes are often referred to as bacteriophages, or phages for short.

Self-questioning

- What are the bases for classification of viruses?
- List the different shapes of the virus.
- Do you think viruses are essential for human being? If yes, in what way?
- Can viruses reproduce? What does it mean by the lytic and lysogenic cycle?

? Did you know?

The particle of a virus is called a virion. All virions contain at least two components:

? A protein shell or capsid

? DNA or RNA as the genetic material

? Some also have:

? A membrane made from lipids and proteins outside the capsid

? Other proteins and enzymes inside the capsid

HIV (human immunodeficiency virus) is a virus that attacks the body's immune system. If HIV is not treated, it can lead to AIDS (acquired immunodeficiency syndrome). Learning the basics about HIV can keep you healthy and prevent HIV transmission

KEY WORDS

DNA virus contains genetic information stored in the formof DNA

RNA virus contains genetic information stored in the formof RNA

Retrovirus an RNA virus that converts its genetic information from RNA intoDNA after it has infected ahost

Bacteriophage a virus that uses a bacterium to replicate its genetic information

2.5.3.Classification of Viruses

Today, the primary criteria for delineating the main viral taxa are:

(1) the type and character of the viral genome,

(2) the strategy of viral replication, and

(3) the types of organisms they infect.

Self-questioning

- What are Retroviruses?
- How does HIV reproduce and cause AIDS?
- Can AIDS be treated?
- What is the social and economic impact of AIDS?
- Draw the structure of HIV.
- What is HIV surface spikes made from? What is the importance of HIV spikes?
- What is the reproductive cycle of bacteriophages is?

Viral Replication

A virus invades a living host cell a thousand or more times its size, hijacks the metabolism of the cell to produce copies of itself, and often destroys the host cell when new virions (a completely assembled virus outside its host cell) are released.

Replication has been studied in a wide range of viruses and their host cells. We examine the bacteriophages first and then discuss the animal viruses. One of the best studied processes of replication is that carried out by bacteriophages of the T-even group (T for "type").

In general, viruses go through the following five steps in their **replication cycles** to produce more virions:

1. Adsorption, the attachment of viruses to host cells.

2. Penetration, the entry of virions (or their genome) into host cells.

3. Synthesis, the synthesis of new nucleic acid molecules, capsid proteins, and other viral components within host cells while using the metabolic machinery of those cells.

4. Maturation, the assembly of newly synthesized viral components into complete virions.

5. Release, the departure of new virions from host cells. Release generally, but not always, kills (lyses) host cells.

Bacteriophage

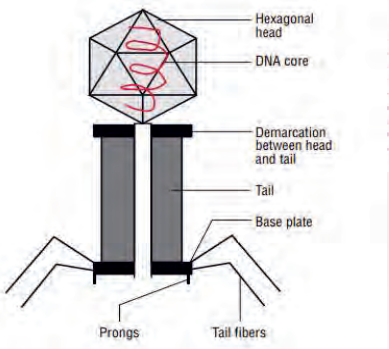


Figure 2.26. bacterioph

Component	Function
Genome	Carries the genetic information necessary for replication of new phage particles
Tail sheath	Retracts so that the genome can move from the head into the host cell's cytoplasm
Plate and tail fibers	Attach phage to specific receptor sites on the cell wall of a susceptible host bacterium

Bacteriophages are bacterial viruses. They are the viruses that infect bacteria. They are obligate intracellular parasites that multiply inside bacteria by making use of some or all of the host biosynthesis machinery. They are also called ***phages***.

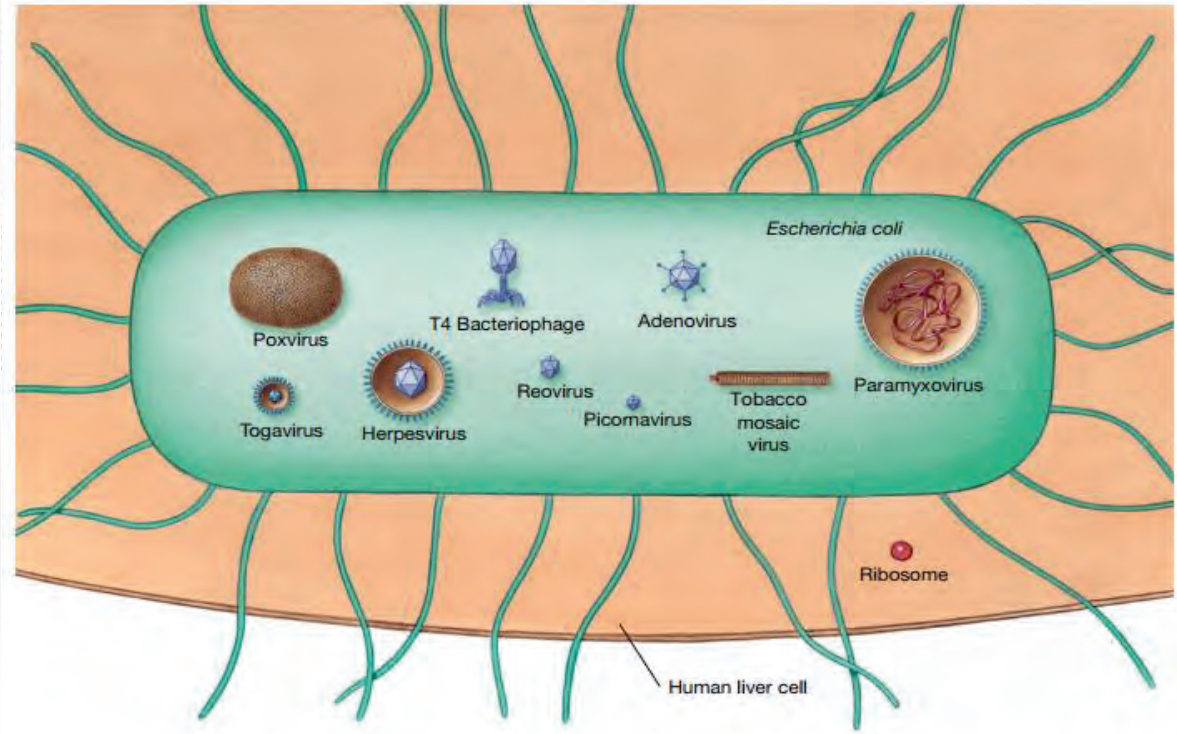


Figure 2.27. Variations in shapes and sizes of viruses compared with a bacterial cell, an animal cell, and a eukaryotic ribosome .

Phages exhibit two different types of life cycle:

Lytic cycle: also known as virulent cycle. In this cycle, intracellular multiplication of the phage results in the lysis of host bacteria, resulting in release of progeny virions (Figure 2.28).

The enzyme lysozyme, which is coded for by a phage gene, breaks down the cell wall, allowing viruses to escape. In the process the bacterial host cell is lysed. Thus, phages such as T4 are called virulent (lytic) phages because they lyse and destroy the bacteria they infect. The released phages can now infect more susceptible bacteria, starting the infection process all over again. Such infections by virulent phages represent a lytic cycle of infection.

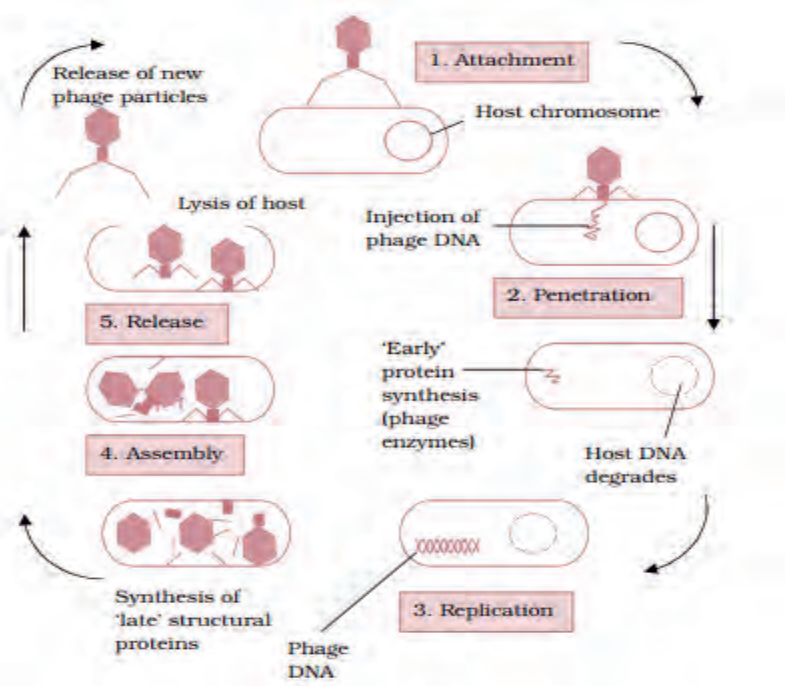


Figure 2.28. The lytic cycle of phage T4

Lytic cycle of bacteriophage (Replication of a virulent bacteriophage). A virulent phage undergoes a lytic cycle to produce new phage particles within a bacterial cell. Cell lysis releases new phage particles that can infect more bacteria).

Lysogenic cycle: Infection with every phage does not result in lysis of the host cells (Figure 2.29). Unlike virulent phages, which cause lysis of the host cell, some phages (such as temperate phages) integrate into the genome of the bacterial chromosome without causing any lysis of the bacteria.

The integrated phage nucleic acid is known as the prophage. The prophage behaves like a segment of the host chromosome and multiplies synchronously with it. This phenomenon is

known as lysogeny. The bacterium that carries a prophage within its genome is called lysogenic bacterium and such phages are called lysogenic or temperate phages.

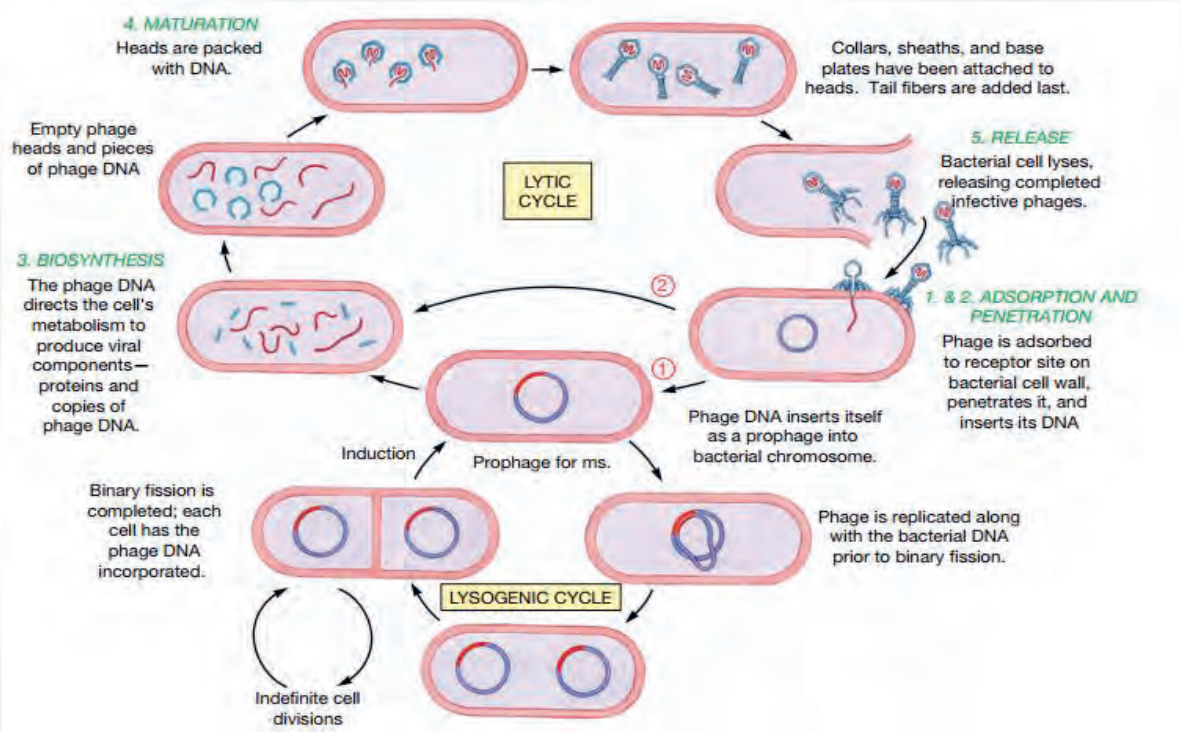


Figure 2.29. Lysogenic bacteriophage cycle (Replication of a temperate bacteriophage).

- What are the differences between lytic and lysogenic cycle

TG. Lytic vs Lysogenic Cycle

Lytic Cycle	**Lysogenic Cycle**
The DNA of the virus doesn't integrate into the host DNA	The DNA of the virus integrates into the host DNA
Host DNA hydrolyzed	Host DNA not hydrolyzed
Absence of prophage stage	Presence of prophage stage
DNA replication of virus takes place independently from the host DNA replication	DNA replication of the virus takes place along with the host DNA replication
Occurs within a short period of time	Takes time

Lytic Cycle	Lysogenic Cycle
Symptoms of viral replication are evident	Symptoms of viral replication not evident
Genetic recombination in the host bacterium not allowed	Genetic recombination in the host bacterium allowed
The cellular mechanism of the host cell is totally undertaken by the viral genome	The cellular mechanism of the host cell is somewhat disturbed by the viral genome

Common viral diseases in Ethiopia

A **viral disease** is any condition that's caused by a **virus**. There are several types of **viral disease**, depending on the underlying **virus**.

Table 2.7. Viral diseases

Disease	Causative agent	Sign and symptoms	Transmission	Prevention and control
Mumps	Mumps virus	Swollen and paifull protid glands .pain on chewing and swallowing	Person to person in infected saliva	vaccine
Measles (rubeola)	Measles virus	Caugh, nasal discharge, eye redness, and high fever,	Droplet contact	vaccine
Rabies	Rabies virus	Tingling, burning, coldness at bite site , Fever, headache, increased muscle, tension, Paralysis and hydrophobia	Bite from rabid animals	Avoiding rabid animals, thoroughly washing bitten area, pre-

Disease	Causative agent	Sign and symptoms	Transmission	Prevention and control
				exposure vaccination when needed
Polio	Polio virus	Often no sign and symptoms	Fecal-oral-route	Polio vaccine, good personal hygien
Common cold ((rhinitis)	Rhinoviruses Adeno viruses & other viruses	Sneezing, sore throat, runny and stuffy nose, hacking cough	Respiratory droplets	Practicing good hygien
Chickenpox ((varicella)	Varicella-zoster virus	Fever, headache, malaise with red, itchy rash on face, scalp, chest, and back	Droplet contact	Hickenpox vaccine

2.6. Normal microbionta

At the end of this lesson, student will be able to:

- Define the normal microbionta.
- Distinguish between resident and transient microbiota.
- Mention three protective roles of the normal microbiota.
- Explain how the composition of the normal microbiota can change over time.
- Describe germ theory of diseases.
- State atleast three postulates formulated by Koch.
- Justify how the Koch's experimental procedure contributed to the advancement microbiological techniques

? What is the role of bacteria and other microorganisms in infectious diseases?

? How do we benefit from the reduction of vitamin K by microbes?

The **normal microbiota** is the population of microorganisms routinely found growing on the body of healthy individuals. Microbes that typically inhabit body sites for extended periods are resident microbiota, whereas temporary occupants are transient microbiota (Figure 2.30).

There are many reasons to acquire knowledge of the normal human microbiota. Three specific examples include:

1. An understanding of the different microorganisms at particular locations provides greater insight into the possible infections that might result from injury to these body sites.

2. A knowledge of the normal microbiota helps the physicianinvestigator understand the causes and consequences of colonization and growth by microorganisms normally absent at a specific body site.

3. An increased awareness of the role that these normal microbiota play in stimulating the host immune response can be gained. This awareness is important because the

immune system provides protection against potential pathogens.

Considering how important this population is to human health, relatively little is known about its members. The normal human microbiontal have protective role from diseases causing microorganisms. One of the most significant contributions of the normal microbiota to health is protection against pathogens. The normal microbiota excludes pathogens by;

(1) covering binding sites that might otherwise be used for attachment,

(2) consuming available nutrients,

(3) producing compounds toxic to other bacteria,

4) to stimulate the adaptive immune system.

When members of the normal microbiota are killed or their growth suppressed, as can happen during antibiotic treatment, pathogens may colonize and cause disease. For example, Oral antibiotics can also inhibit members of the normal intestinal microbiota, allowing the overgrowth of toxin-producing strains of *Clostridium difficile* that cause antibiotic-associated diarrhea and colitis.

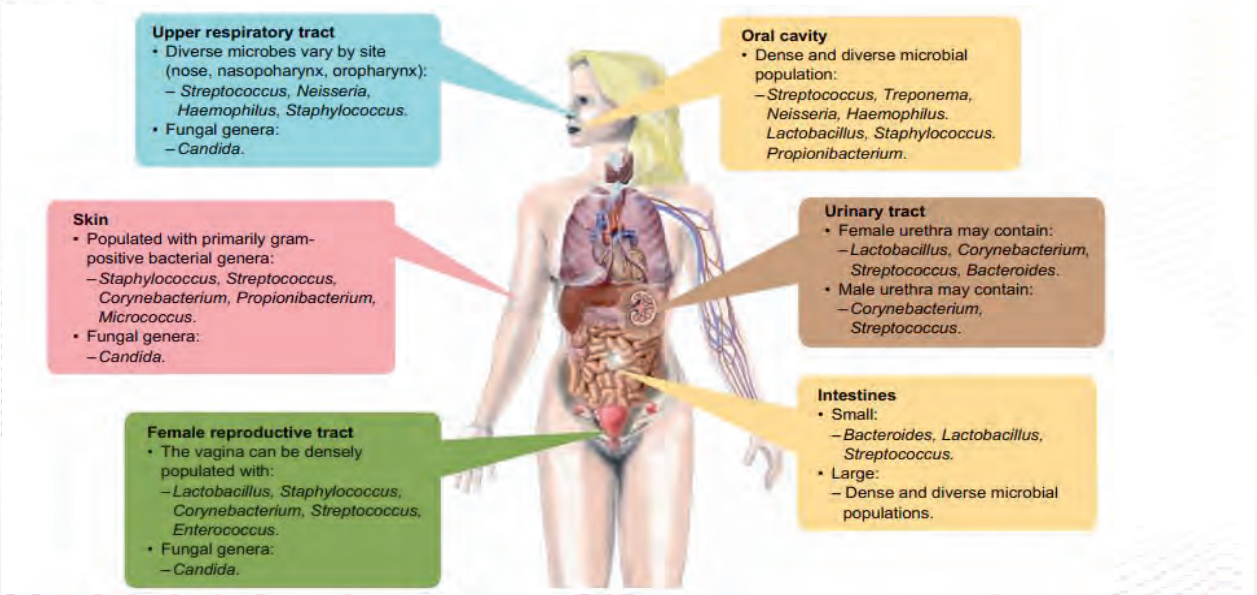


Fig 2. 30. Normal microbionta of the human body

The Germ Theory of Disease and Koch's Postulates

In order to prove whether or not diseases are caused by microorganisms, Koch used mice as experimental animals. Using appropriate controls, Koch demonstrated that when a small amount of blood from a diseased mouse was injected into a healthy mouse, the latter quickly developed anthrax. He took blood from this second animal, injected it

into another, and again observed the characteristic disease symptoms (Figure 2.31). However, Koch carried this experiment a critically important step further. He discovered that the anthrax bacteria could be grown in nutrient fluids outside the host and that even after many transfers in laboratory culture, the bacteria still caused the disease when inoculated into a healthy animal. On the basis of these experiments and others on the causative agent of tuberculosis, Koch formulated a set of rigorous criteria, now known as Koch's postulates, for definitively linking a specific microorganism to a specific disease. Koch's postulates state the following:

1. The disease-causing organism must always be present in animals suffering from the disease but not in healthy animals.

2. The organism must be cultivated in a pure culture away from the animal body.

3. The isolated organism must cause the disease when inoculated into healthy susceptible animals.

4. The organism must be isolated from the newly infected animals and cultured again in the laboratory, after which it should be seen to be the same as the original organism

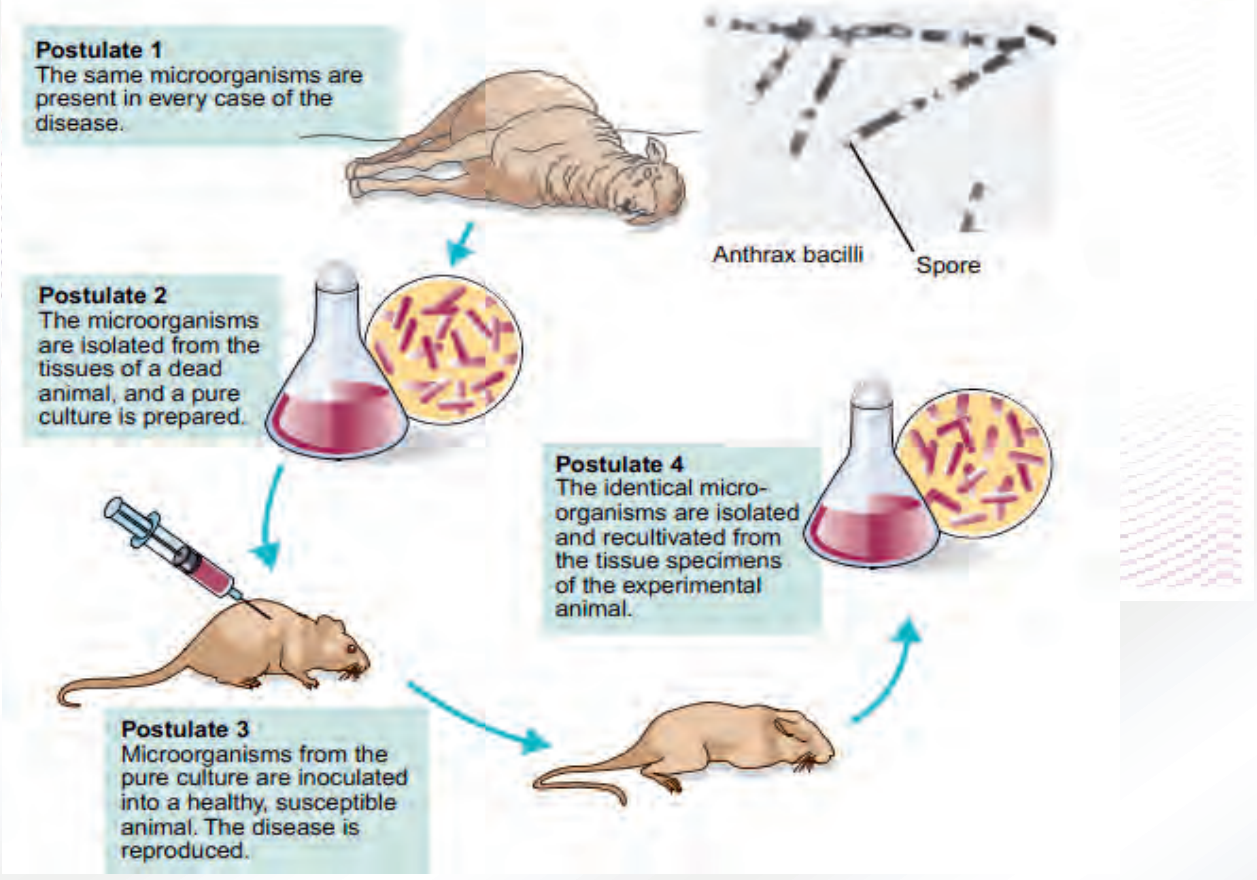


Figure 2.31. Koch's postulate

Self-qestioning

? What is an infectious disease is?

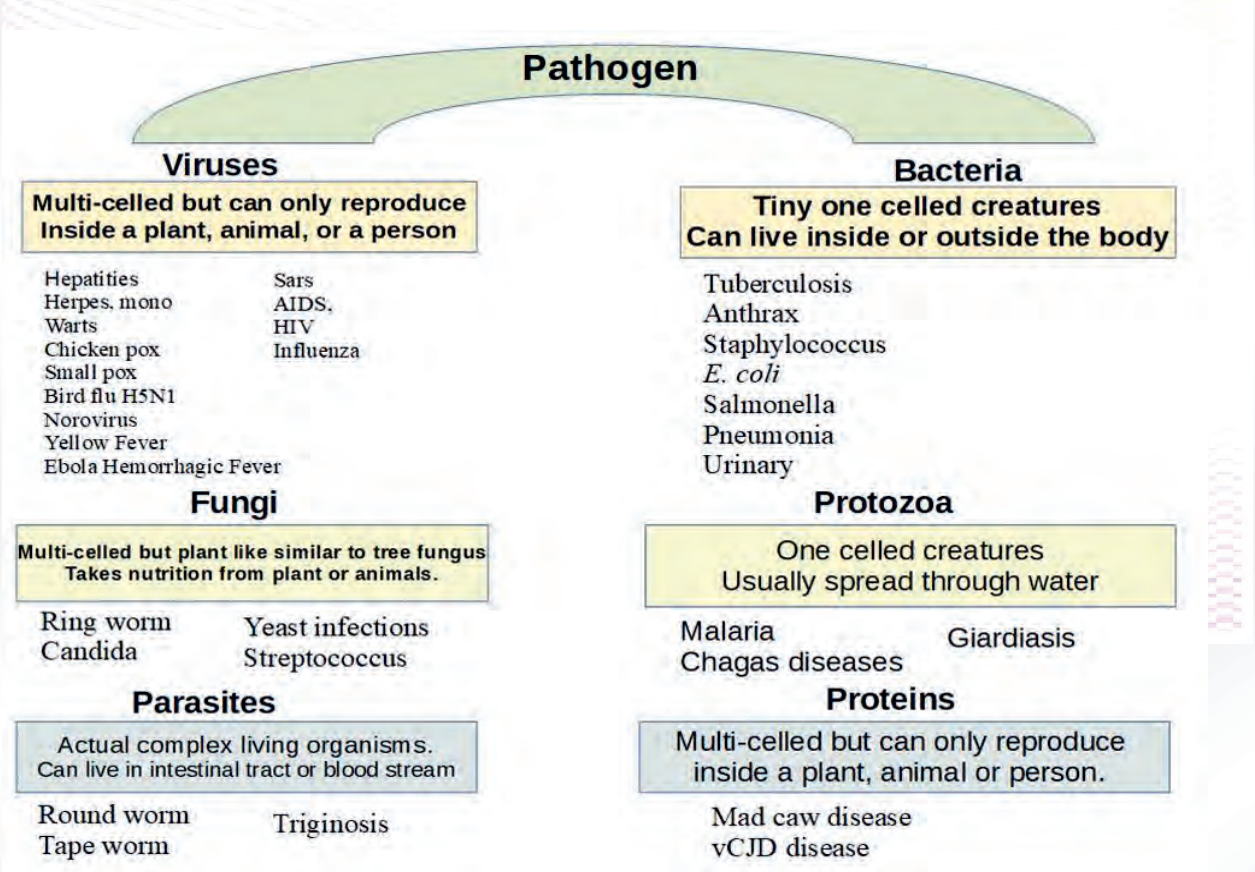
? How will you prove that microorganisms can cause disease?

? How are normal microbiota beneficial?

? Which are the common disease causing microorganisms?

Keywords

- **Normal microbiota or flora**: are the microorganisms that live on another living organism (human or animal) or inanimate object without causing disease.
- **Immune system**: is a complex network of cells, tissues, organs, and the substances they make that helps the body fight infections and other diseases.
- **Inflammatory response**: is a complex biological reaction of the body, which appears when healthy tissues are wounded by physical/chemical stimuli or are invaded by bacteria, viruses, or toxins.



Activity 2.8

Make six groups in a classroom and take one group of organisms among bacteria, fungi, protozoa (malaria, amoeba, and giardia), and virus for each group to study their life cycles and biology using the Jigsaw method. Use your textbook and picture prompt for each group of organisms. Students can also use drama method or role-play (Hint-some students can act as doctor and others as patients) to learn about the cause, symptom, prevention, and treatment method.

2.7. Modes of disease transmission and ways of prevention

Microorganisms are transmitted in health care settings by four main routes:

- Contact
- Droplet
- Airborne
- Common vehicle

At the end of this lesson, student will be able to:

- describe the modes of disease transmission.
- give examples of airborne diseases.
- give examples of food and waterborne diseases.

Table 2.8. Modes of disease transmission

Method of transmission	**How the transmission route works**	**Examples of diseases**
Droplet infection	Many of these diseases are 'respiratory diseases' – diseases affecting the airways of the lungs. The organisms are carried in tiny droplets through the air when an infected person coughs or sneezes. They are inhaled by other people.	Common cold, 'flu, pneumonia
Drinking contaminated water	The micro-organisms transmitted in this way often infect regions of the gut. When unclean water containing the organisms is drunk, they colonise a suitable area of the gut and reproduce. They are passed out with faeces and find their way back into the water.	Cholera, typhoid fever
Eating contaminated food	Most food poisoning is bacterial, but some viruses are transmitted this way. The organisms initially infect a region of the gut.	Salmonellosis, typhoid fever, listeriosis, botulism
Direct Contact	Many skin infections, such as athlete's foot, are spread by direct contact with an infected person or contact with a surface carryingthe organism.	Athlete's foot, ringworm
Sexual intercourse	organisms infecting the sex organs can be passed from one sexual partner to another during intercourse. Some	Candidiasis, syphilis, AIDS,

Method of transmission	How the transmission route works	Examples of diseases
	are transmitted by direct body contact, such as the fungus that causes candidiasis (thrush). others are transmitted in semen or vaginal secretions, such as the AIDS virus. Some can be transmitted in saliva, such as syphilis.	gonorrhoea
Blood-to-blood contact	Many of the sexually transmitted diseases can also be transmitted by blood-to-blood contact. Drug users sharing an infected needle can transmit AIDS.	AIDS, hepatitis B
Animalvectors	Many diseases are spread through the bites of insects. Mosquitoes spread malaria and tsetse flies spread sleeping sickness. in both cases, the disease-causing organism is transmitted when the insect bites humans in order to suck blood. Flies can carry micro- organisms from faeces onto food.	Malaria, sleeping sickness

Activity 2.9: Identifying sources of infection

For each of the methods of transmission given in table 2.4, identify the reservoir of infection

2.8. Uses of microorganisms

At the end of this lesson, student will be able to:

- explain the various uses or application of microorganisms.
- Explain the medical application .
- Describe how vector borne diseases are transmitted.

Self-questioning

? What happens to the body of an animal when it dies? Why?

? What do you think the economic or beneficial uses of microorganisms?

Through the years, microbiologists have made great steps in discovering how microorganisms work, and application of this knowledge has greatly advanced human health and welfare.

Besides understanding microorganisms as agents of disease, microbiology has made great advances in understanding the important role that microorganisms play in food and agriculture, and microbiologists have been able to exploit microbial activities to produce valuable human products, generate energy, and clean up the environment.

Agriculture

Microorganisms play an important role in agriculture. The microorganisms include bacteria, fungi, algae, protozoa, viruses. Microorganisms help in organic matter decomposition, humus formation.

The important role of microorganisms includes nitrogen fixation, phosphate solubilization, potassium mobilization, antagonism towards pathogens, pests. Hence, the role of microorganisms in agriculture is indispensable (Figure 2.32).

The chemical elements carbon, nitrogen, oxygen, sulfur, and phosphorus are essential for life and abundant, but **not necessarily in forms that organisms can use.** Therefore, microorganisms are primarily responsible for converting these elements into forms that plants and animals can use. Microorganisms, especially bacteria and fungi, return carbon

dioxide to the atmosphere when they decompose organic wastes and dead plants and animals. Algae, cyanobacteria, and higher plants use the carbon dioxide during photosynthesis to produce carbohydrates for animals, fungi, and bacteria. Nitrogen is abundant in the atmosphere but in a form not usable by plants and animals. Only bacteria can naturally convert atmospheric nitrogen to a form available to plants and animals (nitrate) (Figure 2.19).

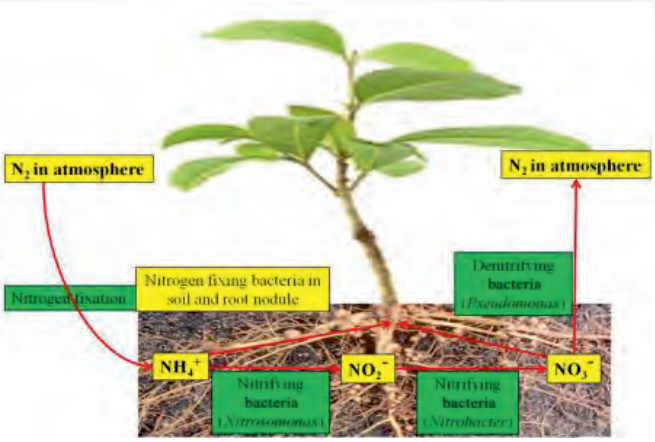


Figure 2.32. The role of micororganisms in agriculture

Sewage treatment

Anaerobic bacteria are used in wastewater treatment on a normal basis. The main role of these bacteria in sewage treatment is to reduce the volume of sludge and produce **methane gas** from it. **Methane gas** can be used as an **alternative energy source**. This is a huge benefit considering the already high wastewater treatment energy consumption levels. **Phosphorus removal** from wastewater is another benefit of anaerobic microbes used in sewage treatment (Figure 2.33).

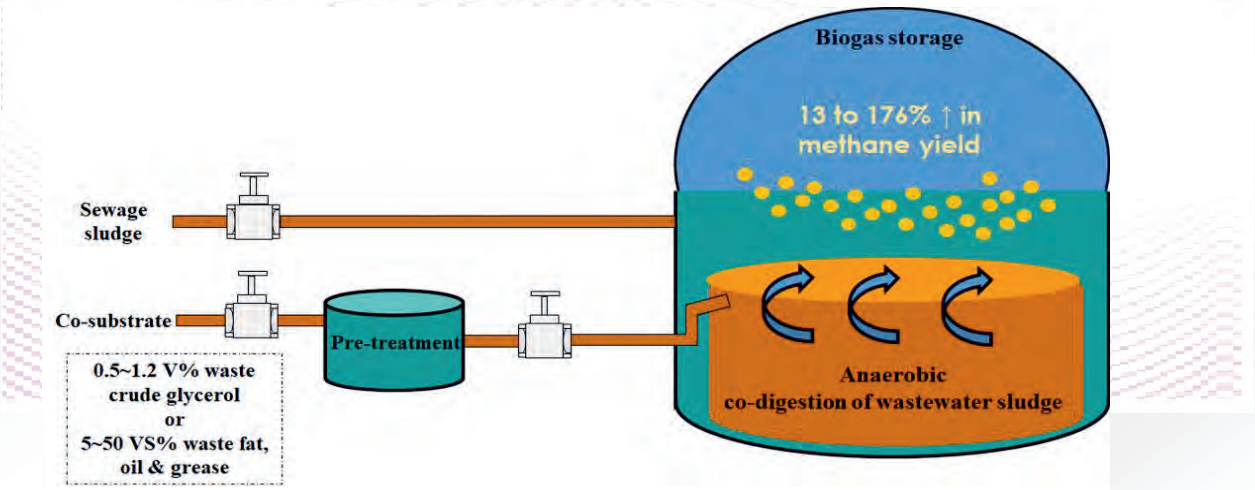


Figure 2.33. Anaerobic co digestion of waste water

Bioremediation

Bioremediation is a natural process that relies on microorganisms and plants and/or their derivatives (enzymes or spent biomass) to degrade or alter environmental contaminants as these organisms carry out their normal life functions. As defined, microbial bioremediation makes use of microorganisms and/or their derivatives (enzymes or spent biomass) to clean-up environmental contaminants.

It is important to note that microorganisms are everywhere and as such pollutants in the different environmental compartments always come into contact with microorganisms. Microbes break down/transform pollutants via their inherent metabolic processes with or without slight pathway modifications to allow the pollutant to be channeled into the normal microbial metabolic pathway for degradation/and biotransformation.

Applied bioremediation methods therefore focus on tapping the naturally occurring microbial catabolic capabilities to degrade, transform or accumulate most of the synthetic compounds such as hydrocarbons (e.g., oil), polychlorinated biphenyls (PCBs), polyaromatic hydrocarbons (PAHs), radionuclides and metals. The natural existence of a large diversity of microbial species expands the variety of chemical pollutants that are degraded or detoxified.

Bioremediation is a natural process, it takes a little time, as an acceptable waste treatment process for contaminated material such as soil. Microbes able to degrade the contaminant and increase in numbers when the contaminant is present. When the contaminant is degraded, the biodegradative population declines. The residues for the treatment are usually harmless products including water, carbon dioxide and cell biomass.

Food production and processing

The tart taste of yogurt, pickles, sharp cheeses, and some sausages is due to the production of lactic acid by one or more members of a group of bacteria known as the lactic acid bacteria. These bacteria - including species (Figure 2.34) of *Lactobacillus, Lactococcus, Streptococcus, Leuconostoc,* and *Pediococcus* are obligate fermenters that characteristically produce lactic acid as an end product of their metabolism. Some also produce flavorful and aromatic compounds that contribute to the overall quality of fermented foods.

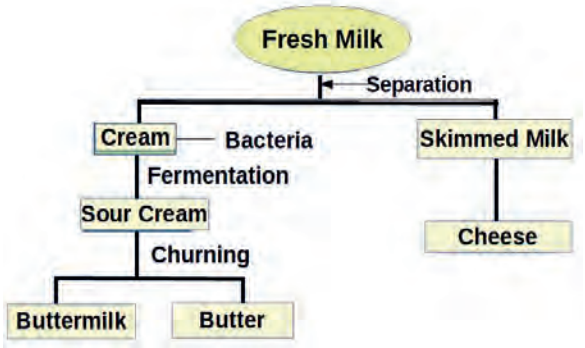


Figure 2. 34. The role of Bacteria in Butter production

Medicine

It is very difficult to decode the human genome if any disorders occur in it as humans are the eukaryotic organisms. It means their body consists of various types pf cells and they are all differentiated into different tissues and organs. Microorganisms have made it possible to make such medicines which when enter the body, target the defected genes and make healthy changes in them and they become functional again. There is a common example of human insulin (Figure 2.35). Insulin is an medicine which is prescribed for the diabetic patients. Now it is possible to synthesize the insulin in microorganisms like bacteria and yeast.

These microorganisms are inserted in the body in the form of vectors and cure the defected genes. Due to the availability of microorganisms in the environment, scientists have made use of them for making many medicines and drugs and also used them for drug delivery.

Health

People will be surprised if they get to know that their body contains ten times more microorganisms than the body cells. These microorganisms are useful for the body and perform various useful functions, for example *E. coli* (specie of bacteria) resides in the intestine and releases such components which help in the digestion of the food. If microorganisms help in performing different body functions, then they also take something from the body that is they take nutrients from the body. One purpose of bacteria in the body is to fight against those harmful bacteria which can cause diseases. For example, there is also a bacterium in the gut which helps in synthesizing the vitamins like biotin, vitamin K and folic acid.

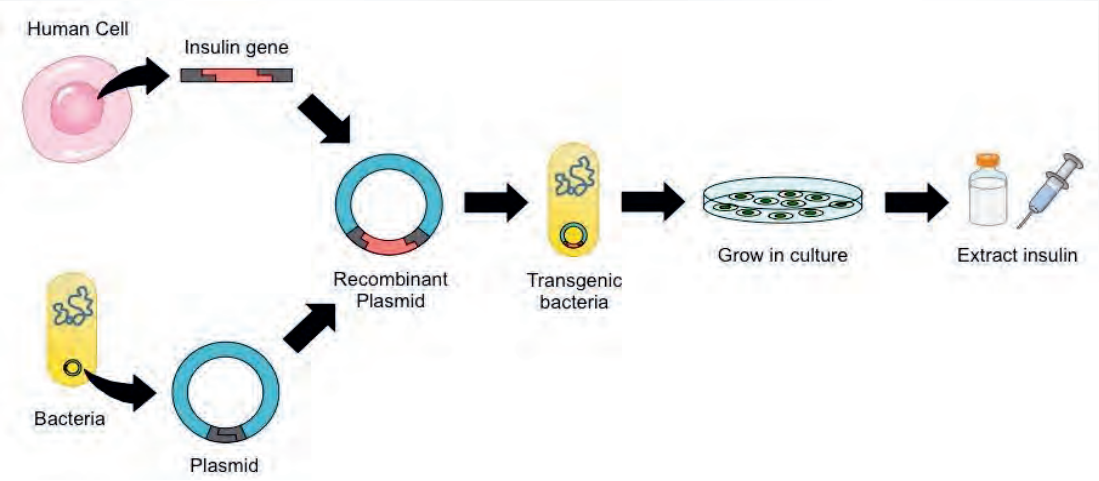


Figure 2.35. Gene transfer to produce human insulin in bacteria. Source. BioNInja

Biotechnology

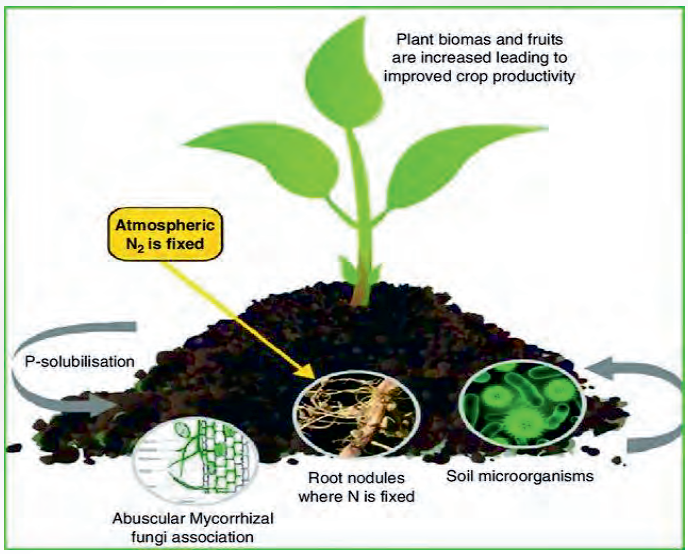
Biotechnology is one field which has made use of microorganisms most. By using the techniques of biotechnology, scientists have succeeded in developing human insulin, growth hormones and other useful components of the body.

Biotechnological processes use microorganisms for the drug delivery in the form of vectors and plasmids. Microorganisms have provided many beneficial things to agriculture as they are responsible for increasing the fertility of the soil. Due to this, the production of the plants increases and economy becomes strong.

Self-Assessment Questions

- List all the Ethiopian traditional fermented food, beverages, and condiments.
- Which are the main types of microorganisms in Ethiopian food fermentation?
- What microorganisms are important in the spoilage of foods?
- What are the industrial roles of microorganisms?

Generally, microorganisms (MOS) have a



big role in, suppression of soil-borne pathogens, recycling and increased availability of plant nutrients, degradation of toxicants including pesticides, Production of antibiotics and other bioactive compounds, Production of simple organic molecules for plant uptake. In addition, MOS are essential in alleviating complexation of heavy metals to limit plant uptake, solubilization of insoluble nutrient sources(Figure 2.23), and Production of polysaccharides to improve soil aggregation

Bacteria are found in every ecosystem they are pretty well everywhere around you and everywhere inside you as well! There are ten bacterial cells inside you for every one of your own cells. Most of these are found in the alimentary canal.

The role of bacteria in recycling minerals through ecosystems

Many bacteria are decomposers. When organisms die, these bacteria break down the complex molecules that are found in the bodies of the dead organisms into much simpler molecules. The bacteria use some of these for their own metabolism, but in the process, they release some minerals (Figure 2. 24), in various forms, into the environment.

Almost, 90% all living organisms are made up of C, O, N and H and these substance are limited in their availability. Thus in order for life to continue the substances should be recycled. This is done by decomposers.

The carbon cycle

All organisms are composed of organic molecules such as proteins, lipids, and carbohydrates. The carbon travels through the food chain as primary producers are eaten by primary consumers, which are then eaten by secondary consumers. Decomposers then use the remains of primary producers and consumers.

Carbon Fixation

A fundamental aspect of the carbon cycle is carbon fixation, the defining characteristic of primary producers. Without primary producers, no other organisms, including humans, could exist.

Respiration and Fermentation When heterotrophs consume organic material, they

break it down using respiration and/or fermentation to release the energy, which is captured to make ATP. The processes usually make CO_2. The type of organic material helps dictate which species degrade it. A wide variety of organisms use sugars, amino acids, and proteins as energy sources, but rapidly multiplying bacteria often the O_2 supply has a strong influence on the carbon cycle. Not only does O_2 allow degradation of certain compounds such as lignin, it also helps determine the types of carbon-containing gases produced. When organic matter is degraded aerobically, a great deal of CO_2 is produced (Figure 2.37). When the O_2 level is low, however, as is the case in marshes, swamps, and manure piles, the degradation is incomplete, generating some CO_2 and a variety of other products.

Methanogenesis and Methane Oxidation

In anaerobic environments, CO_2 is used by methanogens. These archaea obtain energy by oxidizing hydrogen gas, using CO_2 as a terminal electron acceptor, generating methane (CH_4). Methane that enters the atmosphere is oxidized by ultraviolet light and chemical ions, forming carbon monoxide (CO) and CO_2. A group of microorganisms called methylotrophs can use methane as an energy source, oxidizing it to produce CO_2.

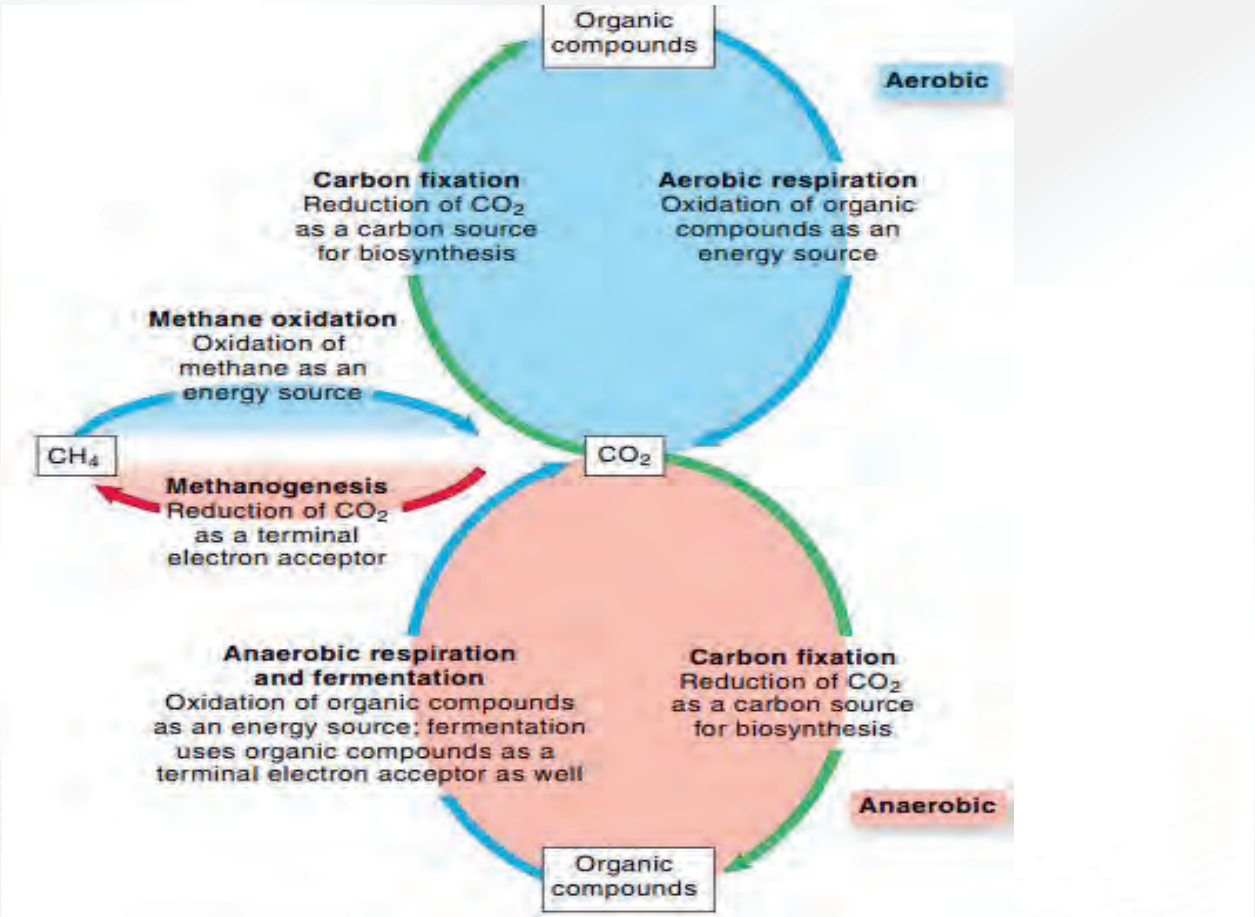


Figure 2.37. The carbon cycle

The Nitrogen cycle

Root nodules are found on the roots of plants, primarily legumes, which form a symbiosis with nitrogen-fixing (Figure 2.38) bacteria. Under nitrogen-limiting conditions, capable plants form a symbiotic relationship with a host-specific strain of bacteria known as rhizobia. Nitrogen fixation in the nodule is very oxygen sensitive.

Key Processes and Prokaryotes in the Nitrogen Cycle

Processes	Example organisms
Nitrification ($NH_4^+ \rightarrow NO_3^-$)	
$NH_4^+ \rightarrow NO_2^-$	*Nitrosomonas*
$NO_2^- \rightarrow NO_3^-$	*Nitrobacter*
Denitrification ($NO_3^- \rightarrow N_2$)	*Bacillus, Paracoccus, Pseudomonas*
N_2 Fixation ($N_2 + 8\,H \rightarrow NH_3 + H_2$)	
Free-living	
Aerobic	*Azotobacter* Cyanobacteria
Anaerobic	*Clostridium*, purple and green phototrophic bacteria *Methanobacterium (Archaea)*
Symbiotic	*Rhizobium Bradyrhizobium Frankia*
Ammonification (organic-N $\rightarrow NH_4^+$)	Many organisms can do this
Anammox ($NO_2^- + NH_3 \rightarrow 2\,N_2$)	*Brocadia*

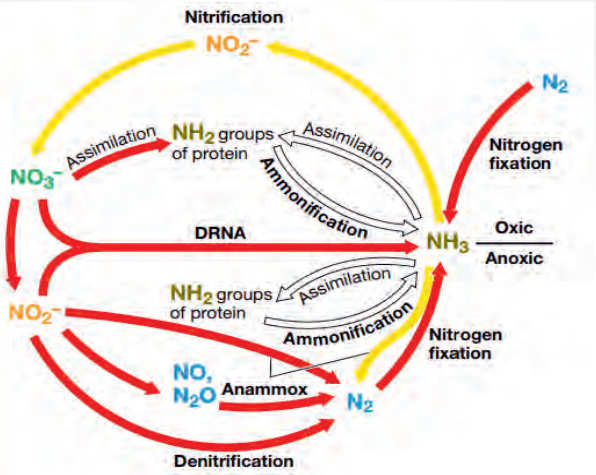


Figure 2.38. The nitrogen cycle

The sulfur cycle

Sulfur is found in fewer types of organic molecules than nitrogen, but it is found in many proteins. Figure 2.39 shows the bacteria involved in the sulfur cycle and the roles they play.

Key Processes and Prokaryotes in the Sulfur Cycle

Process	Organisms
Sulfide/sulfur oxidation ($H_2S \rightarrow S^0 \rightarrow SO_4^{2-}$)	
Aerobic	Sulfur chemolithotrophs (*Thiobacillus, Beggiatoa*, many others)
Anaerobic	Purple and green phototrophic bacteria, some chemolithotrophs
Sulfate reduction (anaerobic) ($SO_4^{2-} \rightarrow H_2S$)	*Desulfovibrio, Desulfobacter Archaeoglobus (Archaea)*
Sulfur reduction (anaerobic) ($S^0 \rightarrow H_2S$)	*Desulfuromonas*, many hyperthermophilic *Archaea*
Sulfur disproportionation ($S_2O_3^{2-} \rightarrow H_2S + SO_4^{2-}$)	*Desulfovibrio*, and others
Organic sulfur compound oxidation or reduction ($CH_3SH \rightarrow CO_2 + H_2S$) (DMSO $\rightarrow$ DMS)	Many organisms can do this
Desulfurylation (organic-S $\rightarrow H_2S$)	Many organisms can do this

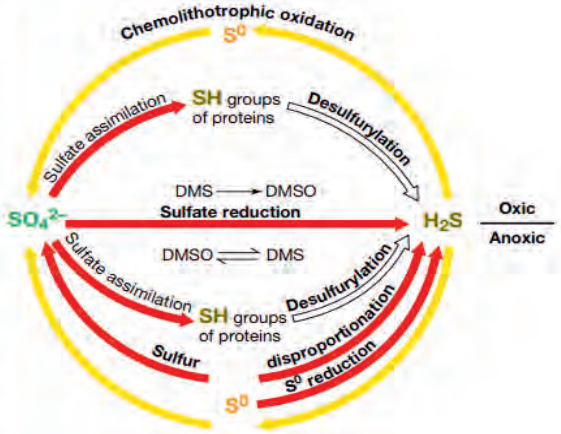


Figure 2.39. The sulfur cycle

The phosphorus cycle

Phosphorus (P) occurs in soils as both organic and inorganic forms (Figure 2.40). Phosphorus can be found dissolved in the soil solution in very low amounts or associated with soil minerals or organic materials. The relative amounts of each form of phosphorus vary greatly among soils, with the total amount of P in a clayey-textured soil being up to ten times greater than in a sandy soil.

Organic P in soils. A large number of compounds make up the organic P in soils, with the majority being of microbial origin. Organic P is held very tightly and is generally not available for plant uptake until the organic materials are decomposed and the phosphorus released via the mineralization process. Mineralization is carried out by microbes, and as with nitrogen, the rate of P release is affected by factors such as soil moisture, composition of the organic material, oxygen concentration and pH.

Inorganic P in soils. The concentration of inorganic P (orthophosphates) in the soil solution at any given time is very small, amounting to less than 1 lb. /A. Phosphorus in the inorganic form occurs mostly as aluminum, iron or calcium compounds.

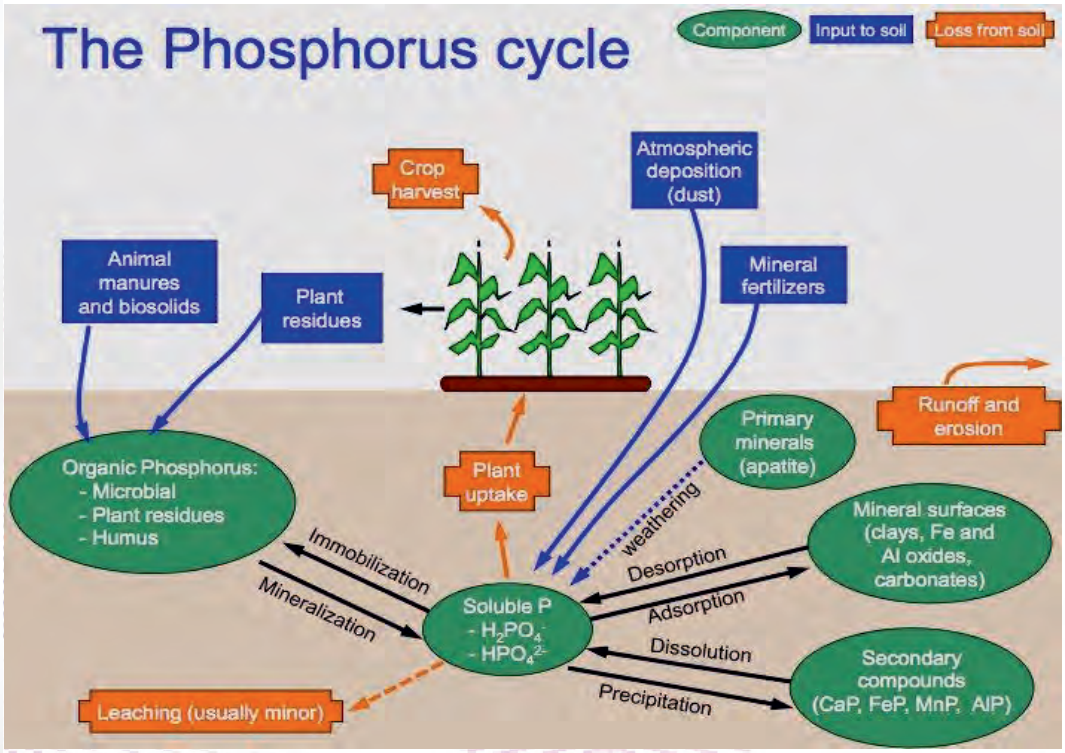


Figure 2.40. The phosphorus cycle

2.9. Controlling microorganisms

Sterilization is defined as a process by which an article, surface, or medium is freed of all living microorganisms either in the vegetative or in the spore state. Any material that has been subjected to this process is said to be *sterile*. These terms should be used only in the absolute sense. An object cannot be slightly sterile or almost sterile; it is either sterile or not sterile. Although most sterilization is performed with a physical agent, such as heat, a few chemicals called *sterilants* can be classified as sterilizing

agents because of their ability to destroy spores.

A *germicide,* also called a *microbicide,* is any chemical agent that kills pathogenic microorganisms. A germicide can be used on inanimate (nonliving) materials or on living tissue, but it ordinarily cannot kill resistant microbial cells. Any physical or chemical agent that kills "germs" is said to have *germicidal* properties.

Disinfection refers to the use of a chemical agent that destroys or removes all pathogenic organisms or organisms capable of giving rise to infection. This process destroys vegetative pathogens but not bacterial endospores. It is important to note that disinfectants are normally used only on inanimate objects because they can be toxic to human and other animal tissue, when used in higher concentrations. Disinfection processes also remove the harmful products of microorganisms (toxins) from materials. Examples of disinfection include

a). applying a solution of 5% bleach to examining table,

(b) Boiling food utensils used by a sick person, and

(c) Immersing thermometers in an isopropyl alcohol solution between uses.

Chemical agents called antiseptics are applied directly to the exposed body surfaces (e.g., skin and mucous membranes), wounds, and surgical incisions to destroy or inhibit vegetative pathogens.

Examples of antisepsis include:

(a) Preparing the skin before surgical incisions with iodine compounds,

(b) Swabbing an open root canal with hydrogen peroxide, and

(c) Ordinary hand washing with a germicidal soap.

Sanitization is any cleansing technique that mechanically removes microorganisms (along with food debris) to reduce the level of contaminants. A sanitizer is a compound (e.g., soap or detergent) that is used to perform this task. Cooking utensils, dishes, bottles, cans, and used clothing that have been washed and dried may not be completely free of microbes, but they are considered safe for normal use. Air sanitization with ultraviolet lamps reduces airborne microbes in hospital rooms, veterinary clinics, and laboratory installations.

Preservation is a general term for measures taken to prevent microbecaused spoilage of susceptible products (pharmaceuticals, foods).

Decontamination is the removal or count reduction of microorganisms contaminating an object. The objective of aseptic measures

and techniques is to prevent microbial contamination of materials or wounds.In antiseptic measures, chemical agents are used to fight pathogens in or on living tissue, for example in a wound.

Physical Methods of Sterilization and Disinfection

Heat

The application of heat is a simple, cheap and effective method of killing pathogens. Methods of heat application vary according to the specific application.

Pasteurization. This is the antimicrobial treatment used for foods in liquid form (milk):

— Low-temperature pasteurization: 61.5 ^{0}C, 30 minutes; 71^{0}C, 15 seconds.

— High-temperature pasteurization: brief (seconds) of exposure to 80–85^{0}C in continuous operation.

— Uperization: heating to 150^{0}C for 2.5 seconds in a pressurized container using steam injection.

Disinfection. Application of temperatures below what would be required for sterilization. Important: boiling medical instruments, needles, syringes, etc. does not constitute sterilization! Many bacterial spores are not killed by this method.

Dry heat sterilization. Sterilization by dry heat includes sterilization by

(a) Flaming : Sterilization of inoculating loop or wire, the tip of forceps, searing spatulas, etc., is carried out by holding them in the flame of the Bunsen burner till they become red hot.

(b) Incineration: Incineration is an excellent method for safely destroying infective materials by burning them to ashes. It has many uses:

■ Incinerators are used to carry out this process and are regularly employed in hospitals and research labs to destroy hospital and laboratory wastes.

■The method is used for complete destruction and disposal of infectious material, such as syringes, needles, culture material, dressings, bandages, bedding, animal carcasses, and pathology samples.

■This method is fast and effective for most hospital wastes, but not for metals and heat-resistant glass materials.

(c) Hot air oven: Sterilization by hot-air oven requires exposure to 160–180°C for 2 hours and 30 minutes, which ensures thorough heating of the objects and destruction of spores.

Moist heat sterilization. Autoclaves charged with saturated, pressurized steam are used for this purpose:

— 121^{0}C, 15 minutes, one atmosphere of pressure (total: 202 kPa).

— 134^{0}C, three minutes, two atmospheres of pressure (total: 303 kPa).

In practical operation, the heating and equalibriating heat up and equalizing times must be added to these, i.e., the time required for the temperature in the most inaccessible part of the item(s) to be sterilized to reach sterilization level. When sterilizing liquids, a cooling time is also required to avoid boiling point retardation and soon."

Intermittent sterilization

Certain heat-labile substances (e.g., serum, sugar, egg, etc.) that cannot withstand the high temperature of the autoclave can be sterilized by a process of intermittent sterilization, known as tyndallization.

Tyndallization is carried out over a period of 3 days and requires a chamber to hold the materials and a reservoir for boiling water. Items to be sterilized are kept in the chamber and are exposed to free-flowing steam at 100°C for 20 minutes, for each of the three consecutive days.

2.10. Bacterial Isolation techniques

Microorganisms (bateria or fungi) can be isolated from food, soil, water or from other materials. For bacterial/fungal isolation, the soil (food) samples are collected from the desired sites (Figure 2.41). Microorganism are separated on artificial media by serial dilution method. Each of the isolates are purified on new media and experimented for the morphological characteristic like shape, gram nature and arrangement of cells, motility etc. Enzymatic activities were tested by biochemical characterization. Finally, molecular techniques are used for further identifications.

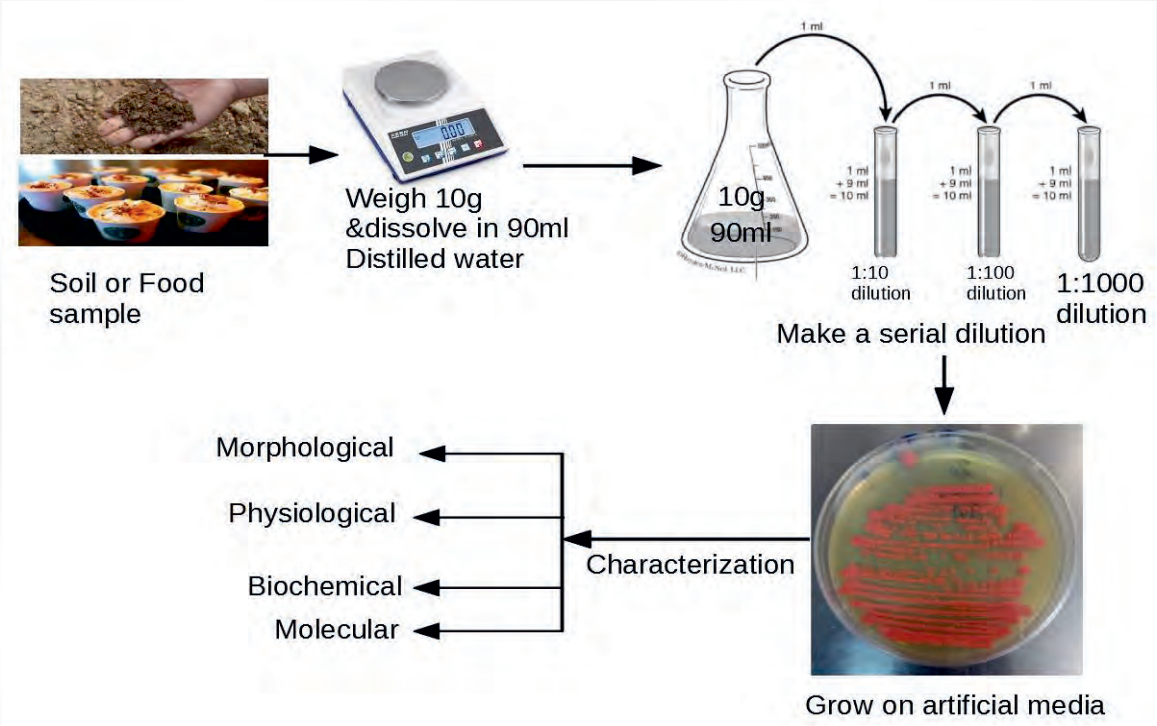


Figure 2.41. Technique for microbial isolation from environmental or food samples

2.11. Renowned Microbiologists in Ethiopia

Activity 2. 10

Search for, study the works of a renowned microbiologist/parasitologist in Ethiopia, and evaluate the contribution of his/her research to the world of science.

Unit summary

Microorganisms are very diverse and are found in all three domains of life: Archaea, Bacteria, and Eukarya. Archaea and bacteria are classified as prokaryotes because they lack a cellular nucleus. Archaea differ from bacteria in evolutionary history, genetics, metabolic pathways, and cell wall and membrane composition. Archaea inhabit nearly every environment on earth, but no archaea have been identified as human pathogens. Eukaryotes studied in microbiology include algae, protozoa, fungi, and helminths. Algae are plant-like organisms that can be either unicellular or multicellular, and derive energy via photosynthesis. Protozoa are unicellular organisms with complex cell structures; most are motile. Microscopic fungi include

molds and yeasts. Helminths are multicellular parasitic worms. They are included in the field of microbiology because their eggs and larvae are often microscopic. Viruses are acellular microorganisms that require a host to reproduce. The field of microbiology is extremely broad. Microbiologists typically specialize in one of many subfields, but all health professionals need a solid foundation in clinical microbiology

Unit review questions

I. Choose the correct answer from A to D for the following equations.

1. Which of the following is present in both gram-positive and gram negative cell walls?

   a. an outer membrane b. peptidoglycan c. teichoic acid d. lipopolysaccharides

2. Bacterial endospore function in,

   Reproduction b. protein synthesis c. survival d. storage

3. Bacterial arrangement in packets of eight cells is described as a----------.

   a. micrococcus b. tetrad c. diplococcus d. sarcina

4. Archaea differ from bacteria on the basis of,

   a. structure of envelope b. size c. the archaea having a nucleus d. type of locomotor structures

5. Which of the following terms refers to a prokaryotic cell that is comma shaped?

   a. Coccus b. coccobacilli c. vibrio d. spirillum

6. Which bacterial structures are important for adherence to surfaces? (Select all that apply.)

   a. endospores b. cell walls c. fimbriae d. capsules e. flagella

II. Fill in the Blank Spaces

1. The bacterium that causes syphilis is called _____.
2. Bacteria in the genus Rhodospirillum that use hydrogen for oxidation and fix nitrogen are _____ bacteria.

3. Streptococcus is the _____ of bacteria that is responsible for many human diseases.

4. The length of the branches of the evolutionary tree characterizes the evolutionary _____ between organisms.

5. The deeply branching bacteria are thought to be the form of life closest to the last universal _____ _____.

6. Nonseptate hyphae are also called _____.

7. Unicellular fungi are called _____

8. Some fungi have proven medically useful because they can be used to produce _____.

9. A virus that infects a bacterium is called a/an _____.

10. A/an _____ virus possesses characteristics of both a polyhedral and helical virus

11. A virus containing only nucleic acid and a capsid is called a/an _____ virus or _____ virus.

12. Viruses can be diagnosed and observed using a(n) _____ microscope.

III. Short Answer

1. Which genera of fungi are common dermatophytes (fungi that cause skin infections)?

2. What is a dikaryotic cell?

3. Discuss the geometric differences among helical, polyhedral, and complex viruses.

4. Briefly explain the various methods of culturing viruses.

5. Does a prion replicate? Explain.

6. What are kinetoplastids?

7. Aside from a risk of birth defects, what other effect might a toxoplasmosis infection have?

Unit 3: Energy Transformation

UNIT 3. ENERGY TRANSFORMATION

Unit learning outcomes

At the end of this unit, student will be able to:

- Discuss the process of energy transformation in cells .
- Summarize the process of photosynthesis using chemical equation.
- Analyze an absorption spectra of chlorophyll a and chlorophyll b using graph.
- Discuss the mechanism as to how CO_2 is fixed in C_3,C_4 plants, and CAM Plants.
- Justify the reason why the rate of photorespiration is less in C4 plants as compared to C3 plants.
- Demonstrate whether or not organic molecules (such as starch) present in leaves.
- Discuss how energy is harvested during aerobic respiration.
- Differentiate between substrate-level phosphorylation and oxidative phosphorylation
- Show the mechanism electron transport system in mitochondria.
- Discuss the significances of fermentation by microorganisms in our daily life.
- Apply the concept of cellular respiration to calculate amount of energy yield from a given molecule of glucose.
- Justify how energy transformations in cells contribute for the maintenance CO_2 and O_2 balance in the atomosphere.
- Appreciate the mechanism of energy transformation in cells.

Appreciate the role of photosynthesis and cellular respiration for continuity of life

Overview

Living cells are in constant activity. Macromolecules are assembled and broken down, substances are transported across cell membranes, and genetic instructions are transmitted. All of these cellular activities require energy. Living organisms are unique in that they can extract energy from their environments and use it to carry out life activities such as movement, growth and development, and reproduction. But the basic question is how living organisms or, their cells extract energy from their environments, and how cells use this energy to synthesize macromolecules? The answers to these questions lie in the enzyme-mediated chemical reactions that take place in living

matter (metabolism). In metabolism, series of chemical reactions are taking place in the cells of organisms. These reactions may aid in the transformations of energy from one form to another in cells.

3.1. Energy

By the end of this section, the student will be able to:

- Define cellular metabolism.
- Explain anabolic and catabolic pathways in cellular metabolism.

Metabolism is the sum of chemical reactions that takes place within each cell of an organism. The chemical reactions enable cells to produce energy for vital processes and also synthesize new organic materials. Broadly, these reactions can be divided into catabolic reactions that convert nutrients to energy and anabolic reactions that lead to the synthesis of larger biomolecules.

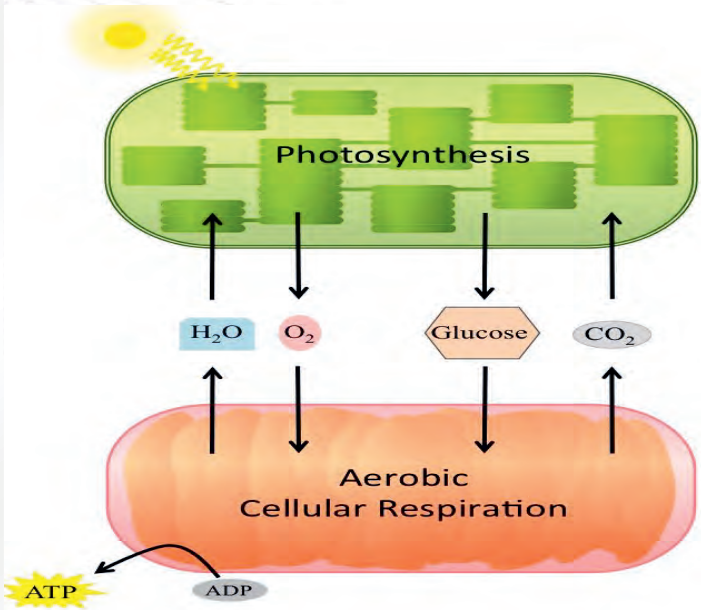
Anabolism is the set of reactions involved in the synthesis of complex molecules, starting from the small molecules inside the cells of an organism. Anabolic reactions help in the building of macromolecules like proteins, nucleic acids, and polysaccharides.

Catabolism is the set of reactions involved in the breakdown of complex molecules like proteins, gacids, glucose, and fatty acids, respectively. It is also the breakdown of monomers into carbon

The **main difference** between anabolism and catabolism is that anabolism is a constructive process whereas catabolism is a descrctive process.

Table 3.1. Comparison of anabolic and catabolic pathways

No.	Anabolism	Catabolism
1.	It is the constructive phase of metabolism.	It is the destructive phase of metabolism.
2.	It is the process whereby simpler substances are joined together to form complex macromolecules.	It is the process whereby complex macromolecules are broken down to form simpler substances or monomers.
3.	The process requires energy to construct substances.	The process releases energy as a result of the breakdown of molecules.
4.	It is an endergonic (energy-absorbing) reaction.	It is an exergonic (energy-releasing) reaction.
5.	It occurs during photosynthesis.	It occurs during cellular respiration.



Self-questioning

? What does metabolism mean in a cellular sense?

? How are anabolism and catabolism differ?

Figure 3.1. Metabolic pathways

3.2. Photosynthesis

By the end of this section, you will be able to:

Define photosynthesis.

Explain the process of dark and light reaction of photosynthesis.

Describe the external and internal structure of a leaf.

Describe the different chlorophyll pigments involved in the absorption of light energy.

Analyze an absorption spectra of the chlorophyll pigments from the graph.

Explain the the roles of grana and and stroma in the process of photosynthesis.

Describe cyclic and non-cyclic photophosphorylation.

Show an absorption spectra of chlorophyll a and chlorophyll b using graph.

Explain the process of Calvin cycle.

Compare cyclic and non-cyclic photophosphorylation.

Differentiate between C3 , C4 plants, and CAM Plants.

Justify the reason why the rate of photorespiration is less in C4 plants as compared to C3 plants.

Summarize the process of photosynthesis using chemical equation.

Conduct an experiment to show the release of energy during photosynthesis.

Conduct an experiment to show the presence of starch in green leaves.

Explain the significances of photosynthesis.

Photosynthesis is a series of chemical reactions that use light energy to assemble CO2 into glucose (C6H12O6) and other carbohydrates. The plant uses water in the process and releases oxygen gas (O2) as a byproduct. The reactions of photosynthesis are summarized as follows:

Self-questioning

- How is an autotroph different from a heterotroph?
- Write the chemical equation of photosynthesis.
- Why is photosynthesis

$$6\,CO_2 + 12\,H_2O \xrightarrow[\text{Chlorophyll}]{\text{Light energy}} C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$$

Carbon dioxide — Water — Glucose — Oxygen — Water

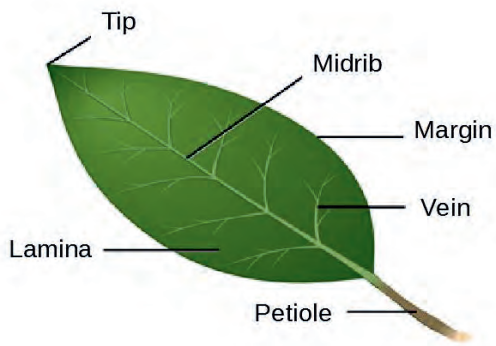
This process provides not only food for the plant but also the energy, raw materials, and O_2 that are used to support most **heterotrophs**. Oxygen is a byproduct of photosynthesis. Furthermore, photosynthesis is important because it is the number one source of oxygen in the atmosphere; it contributes to the carbon cycle among the earth, the oceans, plants and animals; it contributes to the symbiotic relationship among plants, humans and animals; it directly or indirectly affects most living things on earth; it serves as the primary energy process for plants. Plants, multicellular algae, some protists, Cyanobacteria, and Purple sulfur bacteria are Photoautotrophs.

External and Internal Structure of the Leaf

The outer leaf layer is known as the epidermis. The epidermis secretes a waxy coating called the cuticle that helps the plant retain water. A leaf has three main parts– Leaf base, leaf lamina, and petiole (Figure 3.2). Epidermis is a continuous layer on the outside of the plant, one cell thick, that provides protection .In stems and leaves it is covered with a waxy cuticle

which is waterproof and helps to protect the organ from drying out and from infection. In leaves, it also has pores called stomata which allow entry of carbon dioxide for photosynthesis. The mesophyll is made up of specialised parenchyma cells found between the lower and upper epidermis of the leaf. They are specialised for photosynthesis and therefore contain chloroplasts. They are of two types, palisade mesophyll and spongy mesophyll. Spongy mesophyll is so-called because in three

dimensions it is spongy in appearance, because it has many large air spaces between the cells. Palisade mesophyll cells are near the upper surface of the leaf where they receive more sunlight. They therefore contain more chloroplasts than spongy mesophyll cells (Figure 3.3).



A). External Structure of a Leaf

*Explain each external part of a leaf

*Most leaves are flat. What is the advantage of flatness for the plant?

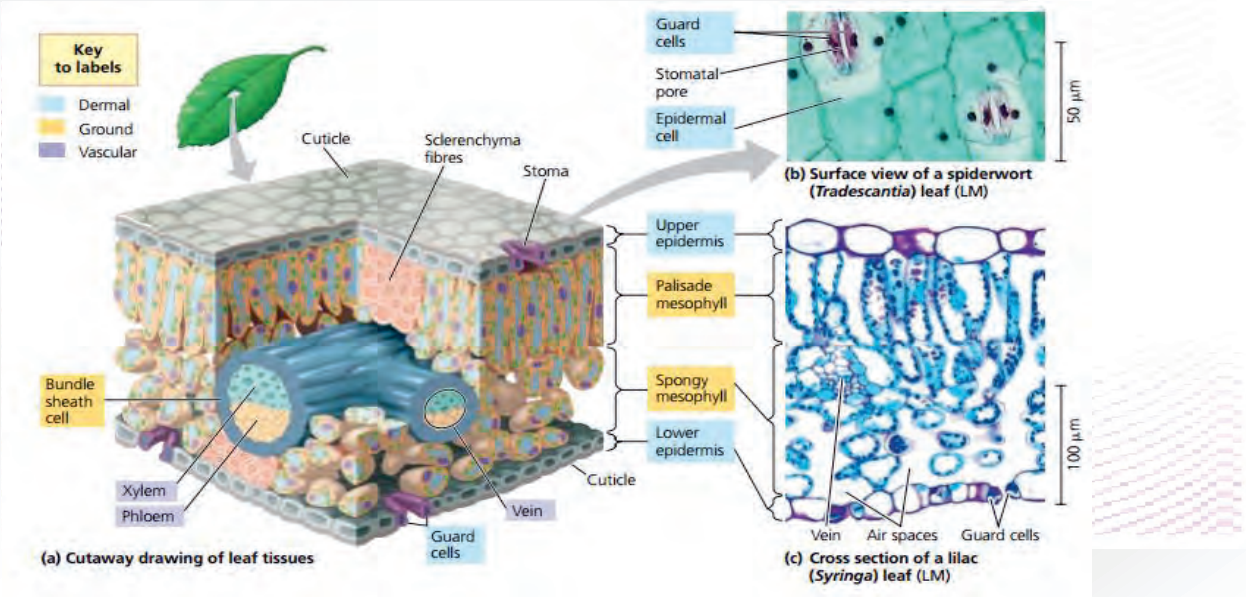
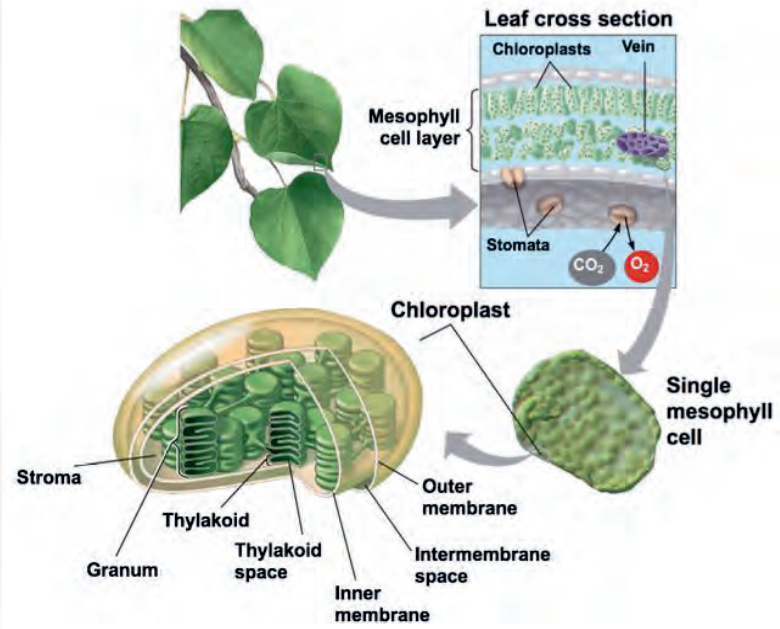


Figure 3.3. Internal structure of leaf

The site of photosynthesis

In plants, the highest density of chloroplasts is found in the mesophyll cells of leaves. A double membrane surrounds chloroplast, where the outer membrane faces the cytoplasm of the plant cell on one side and the intermembrane space of the chloroplast on the other.

The inner membrane separates the narrow intermembrane space from the aqueous interior of the chloroplast, called the **stroma**. The stroma is a fluid-filled matrix where the light-independent stage of photosynthesis takes place. Within the stroma are a number of other structures such as search grains.Within the stroma, another set of membranes form disk-shaped compartments known as **thylakoids (**Figure 3.4**)**. The interior of a thylakoid is called the **thylakoid lumen**. In most plant species, the thylakoids are interconnected to form stacks called **grana**. The grana are stacks of up to 100 disc-like structu res called thylakoids where the light-dependent stage of photosynthesis takes place. Within the thylakoids is the photosynthetic pigment called chlorophyll. Some thylakoids have tubular extensions that join up with thylakoids in adjacent grana. These are called imergranal lamellae.



Self-questioning

- Where do you think is the site of photosynthesis in Plants?

Figure 3.4. The structure of a chloroplast and its location within a plant cell and leaf

3.2.1. Photosynthetic pigments

Self-questioning

1. What are the main components of sunlight?
2. How does having multiple types of pigments benefit plants?

Photosynthetic cells contain special pigments that absorb light energy. Different pigments respond to different wavelengths of visible light. Pigments are chemical compounds which reflect only certain wavelengths of visible light. This makes them appear "colorful". Flowers, corals, and even animal skin contain pigments which give them their particular colors. More important than their reflection of light is the ability of pigments to **absorb** certain wavelengths.

Because they interact with light to absorb only certain wavelengths, pigments are useful to plants and other **autotrophs.** In plants, algae, and cyanobacteria, pigments are the means by which the energy of sunlight is captured for photosynthesis. However, since each pigment reacts with only a narrow range of the spectrum, there is usually a need to produce several kinds of pigments, each of a different color, to capture more solar energy. There are three basic classes of pigments.

Chlorophylls are greenish pigments which contain a **porphyrin ring**. This ring has the potential to gain or lose electrons easily and whereby providing energized electrons to other molecules. There are several kinds of chlorophyll, which the most important one is **chlorophyll "a".**

It is a green pigment found in all plants, algae, and cyanobacteria. The second kind of **chlorophyll, chlorophyll "b"** occurs only in "green algae" and in plants. The third form of chlorophyll called **chlorophyll "c"**, is found only in the photosynthetic members of the Chromista and dinoflagellates.

Carotenoids are usually red, orange, or yellow pigments, and they include the familiar compound carotene, which gives carrots their color. Carotenoids cannot transfer sunlight energy directly to the photosynthetic pathway, but must pass their absorbed energy to chlorophyll. For this reason, they are called **accessory pigments**. One very visible accessory pigment is **fucoxanthin,** the brown pigment whose colors keep other brown algae as well as the diatoms.

Activity 3. 4

Discuss in a group and write your answer on your notebook if you could expose plants to just one wavelength of light at a time, would a wavelength of 350 nm, 450 nm, or 600 nm produce the highest photosynthetic rate?

Phycobilins are water-soluble pigments, and are, therefore, found in the cytoplasm, or in the stroma of the chloroplast. They occur only in Cyanobacteria and Rhodophyta.

Absorption spectra of photosynthetic pigments

An absorption spectrum is a graph that shows absorption from a spectrophotometer. **Figure 3.5** shows absorption at wavelengths from 400-700 nm by three pigments; ***Chlorophyll a***, ***Chlorophyll b***, and the carotenoids. ***Chlorophyll a*** absorbs violet-blue and reddish orange-red wavelengths. ***Chlorophyll b*** absorbs mostly blue and yellow light. Both ***Chlorophyll a*** and ***Chlorophyll b*** also absorb light of other wavelengths with less intensity. However, none of them absorbs green, so that the leaf looks green because light is reflected to our eyes instead of being absorbed by the leaf. Carotenoids are ubiquitous and essential pigments in photosynthesis. They absorb in the **blue-green region** (Figure 3.5) of the solar spectrum and transfer the absorbed energy to (bacterio) chlorophylls, and thereby expanding the wavelength range of light that is able to drive photosynthesis. Only absorbed light (largely blue and red) is useful in photosynthesis.

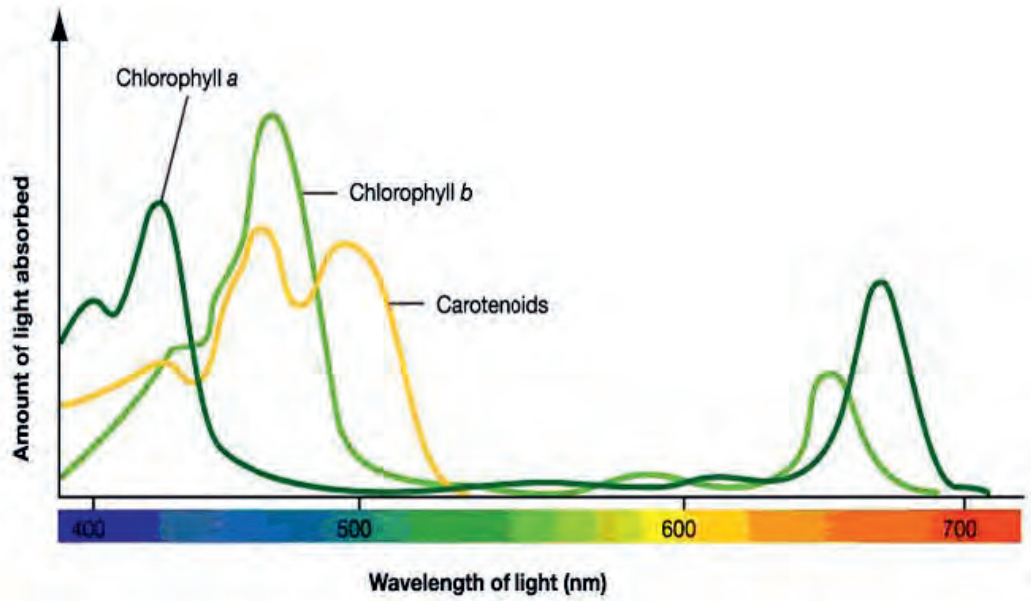


Figure 3.5. Absorption spectrum of *Chlorophyll a*, *Chlorophyll b*, and carotenoids

Practical activity - paper chromatography

3.2.2. Light-dependent and light-independent reactions

Inside a chloroplast, photosynthesis occurs in two stages: the light-dependent reactions and the light-independent (or Calvin Cycle) reactions.

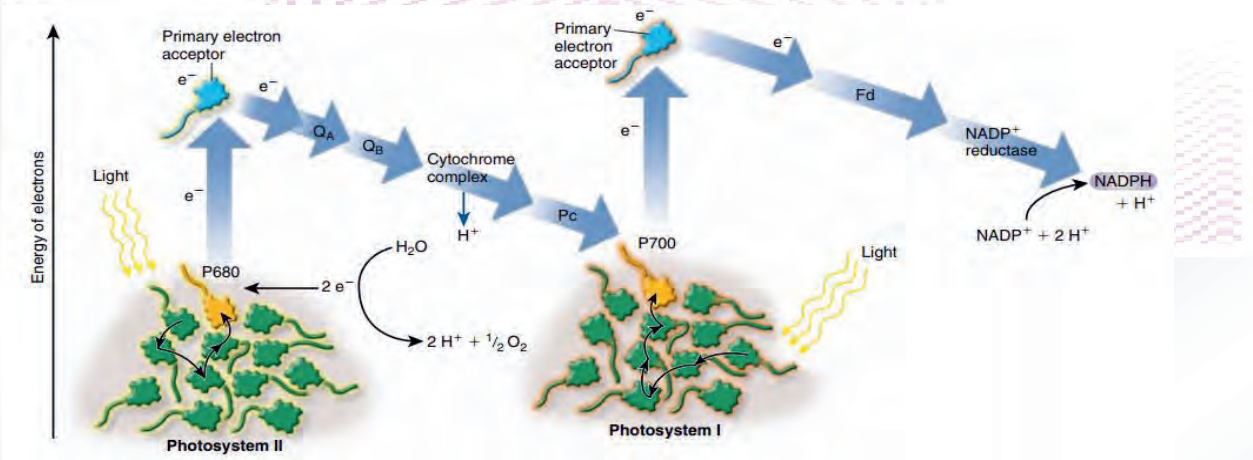
Light-Dependent Reactions(cyclic and non-cyclic photophosphorylation)

The light reactions are the steps of photosynthesis that convert solar energy to chemical energy. Water is split, providing a source of electrons and protons (hydrogen ions, H+) and giving off O_2 as a by-product (Figure 3.6). Light absorbed by chlorophyll drives a transfer of the electrons and hydrogen ions from water to an acceptor called $NADP^+$ (nicotinamide adenine dinucleotide phosphate), where they are temporarily stored. (The electron acceptor NADP + is first cousin to NAD^+, which functions as an electron carrier in cellular respiration; the two molecules differ only by the presence of an extra phosphate group in the NADP + molecule.) The light reactions use solar energy to reduce NADP + to NADPH by adding a pair of electrons along with an H^+. The light reactions also generate do not produce sugar.

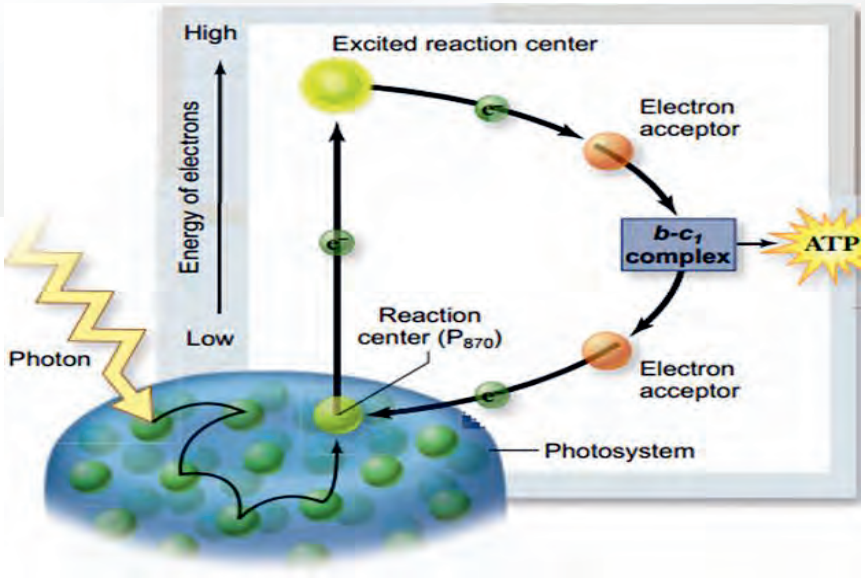
ATP, using chemiosmosis to power the addition of a phosphate group to ADP, a process called photophosphorylation. Thus, light energy is initially converted to chemical energy in the form of two compounds: NADPH and ATP. NADPH, a source of electrons, acts as "reducing power" that can be passed along to an electron acceptor, reducing it, while ATP is the versatile energy currency of cells. Notice that the light reactions

Self-questioning

- What are the two stages at which photosynthesis occurs?
- What happens in each of the two main stages of photosynthesis?
- Where in the chloroplast does each stage occur?



a). Non-cyclic photophosphorylation



b). cyclic photophosphorylation

Figure 3.6. Light dependent reaction of photosynthesis a) Non-cyclic (b) Cyclic photophosphorylation Comparison of cyclic and non-cyclic photophosphorylation

	Non-cyclic	**Cyclic**
Pathway of electrons	Non-cyclic	Cyclic
First electron donor(source of electrons)	Water	photosystemI(P700)
Last electron acceptor destination of electrons)	NADP	Photosystem I(P700)
Products	Useful:ATP, reduced NADP oxygen (byproduct).	Useful:ATP only
Photosystem involved	I and II	I only

Photosystem I and photosystem II

1. Electrons (e^-) in chlorophyll molecules in photosystem II are excited by the energy in photons of light: they become more energetic. Because of the extra energy, they

escape from the chlorophyll and pass to an electron acceptor (the primary electron acceptor).

2. The conditions created in the chloroplast cause the following reaction to occur:

$$2H2O \rightarrow O2 + 4H^{+} + 4e^{-}$$

This light-dependent splitting of water is called photolysis. The electrons replace those lost from the chlorophyll molecule.

3. The primary electron acceptor passes the electrons to the next molecule in an electron transport chain (plastoquinone or 'Pq'). The electrons then pass along a series of cytochromes (similar to those in the mitochondrial electron transport chain) and finally to plastocyanin (Pc) – the last carrier in the chain. The electrons lose energy as they are passed from one carrier to the next.

4. One of the molecules in the cytochromes complex is a proton (hydrogen ion) pump. As electrons are transferred to and then transferred from this molecule, the energy they lose powers the pump which moves protons from the stroma of the chloroplast to the space inside the thylakoid. This leads to an accumulation of protons inside the thylakoid, which drives the chemiosmotic synthesis of ATP.

5. Electrons in chlorophyll molecules in photosystem I are excited (as this photosystem absorbs photons of light) and escape from the molecule. They are replaced by the electrons that have passed down the electron transport chain from photosystem II.

6. The electrons then pass along a second electron transport chain involving ferredoxin (Fd) and NADP reductase. At the end of this electron transport chain, they can react with protons (hydrogen ions) and NADP in the stroma of the chloroplast to form reduced NADP.

Products of Photosynthesis: In summary, the steps of the light reactions of photosynthesis produce three chemical products: O_2, NADPH, and ATP:

1. O_2 is produced in the thylakoid lumen by the oxidation of water by photosystem II. Two electrons are removed from water, which produces 2 H^{+} and 1/2 O_2. The two electrons are transferred to P680 molecules.

2. NADPH is produced in the stroma using high-energy electrons that start in photosystem II and are boosted a second time in photosystem I. Two high-energy electrons and one H+ are transferred to $NADP^{+}$ to produce NADPH.

3. ATP is produced in the stroma via ATP synthase that uses an H^{+} electrochemical gradient.

3.2.5. Light-Independent Reactions (Calvin cycle)

This is the second step in the mechanism of photosynthesis. The chemical processes of photosynthesis occurring independent of light are called dark reactions. It takes place in the stroma of the chloroplast. The dark reaction is purely enzymatic and it is slower than the light reaction. Dark reaction does not require light . In a dark reaction, the sugars are synthesized from CO_2. The energy-poor CO_2 is fixed to energy-rich carbohydrates using the energy-rich compound, ATP, and the assimilatory power, $NADPH_2$ of light reaction. The process is called carbon fixation or carbon assimilation.

C3 Plants are plants capable of fixing CO_2 into a 3-Carbon sugar called Phosphoglycerate (PGA). The energy from ATP and NADPH energy carriers generated by the photosystems is used to phosphorylate the PGA. In this process, carbon dioxide enters a plant through its stomata, and the enzyme Rubisco fixes carbon into sugar using the Calvin cycle. This fixation of carbon dioxide by Rubisco is the first step of the Calvin cycle. The plants that use this mechanism of carbon fixation are called **C3 plants**. Approximately 95% of plants on the earth are C3 plants. Some of the C3 plant examples are wheat, rye, oats, and orchard grass.

The photosynthesis process can take place only when the stomata on leaves are open. C3 plants exhibit the C3 pathway. It is a three-carbon compound (3-PGA). Here the first carbon compound produced has three carbon atoms hence the name "C3 pathway"(Figure 3.7). The light-independent reactions of the Calvin cycle can be organized into three basic stages: fixation, reduction, and regeneration.

1. Carbon fixation: A molecule of carbon dioxide is combined with a carbon acceptor molecule containing five atoms known as ribulose-1,5-bisphosphate (RuBP). This step gives rise to a compound having six carbon atoms that disintegrate into two molecules of a compound containing three carbons called 3-phosphoglyceric acid (3-PGA). This reaction is catalyzed by the enzyme RuBP carboxylase, or rubisco.

2. Reduction: In the second stage of the Calvin cycle, ATP and NADPH molecules are utilized to change the 3-PGA molecules into a sugar molecule containing three carbon atoms called glyceraldehyde-3-phosphate (G3P). This stage has derived its name from the fact that NADPH donates electrons to a three-carbon intermediate to form **G3P**.

3. **Regeneration:** Some **G3P** molecules form glucose, while others need to be recycled so that they can regenerate the RuBP acceptor. Regeneration needs ATP and involves a complex series of reactions called the "carbohydrate scramble."

In the dark reaction, CO_2 is fixed to carbohydrates and the CO_2 acceptor ribulose-diphosphate is regenerated. In the Calvin cycle, $12NADPH_2$ and 18 ATPs are required to fix $6CO_2$ molecules into one hexose sugar molecule (fructose- 6- phosphate).

C_4 plants: In some plants like maize, sorghum, and sugarcane , the first product of carbondioxide fixation is not the three carbon molecule phosphoglycerate but the four carbon compound oxaloacetic acid. Plants that utlize this pathway are commonly called the C4 or four carbon plants. The oxaloacetic acid is formed when carbondioxide is bound to a compound known as phophoenolpyruvate (PEP) in the mesophyll cell. The oxaloacetic acid is reduced to malic acid or converted to aspartic acid; and the malic acid(aspartic acid) is decarboxylated to yield CO_2 and pyruvic acid in the bundle sheth cell (Figure 3.8). Then, CO_2 enters to Calvin cycle.

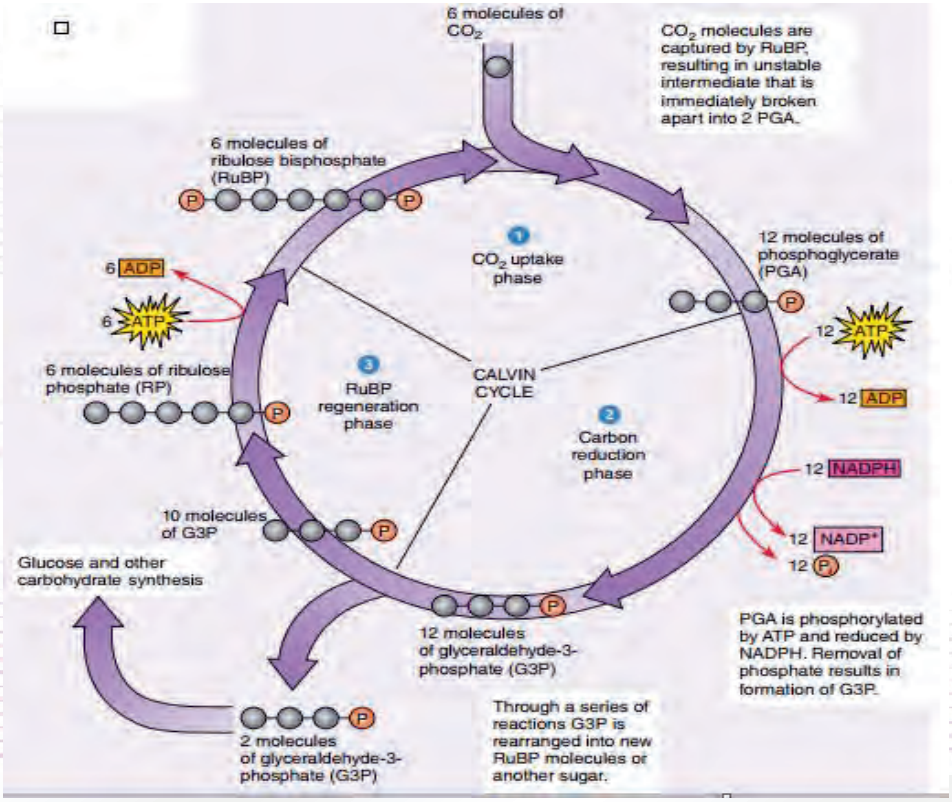


Figure 3.7. Calvincycl (C3 cycle)

The basic C4 cycle consists of four stages:

1. Fixation of CO_2 by the carboxylation of phosphoenol-pyruvate (PEP) in the mesophyll cells to form a C_4 acid.

2. Transport of the C_4 acids to the bundle sheath cells

3. Decarboxylation of the C_4 acids within the bundle sheath cells and generation of CO_2, which is then reduced to carbohydrate via the Calvin cycle.

4. Transport of the C_3 acid (pyruvate or alanine) that is formed by the decarboxylation step back to the mesophyll cell and regeneration of the CO_2 acceptor phosphoenolpyruvate

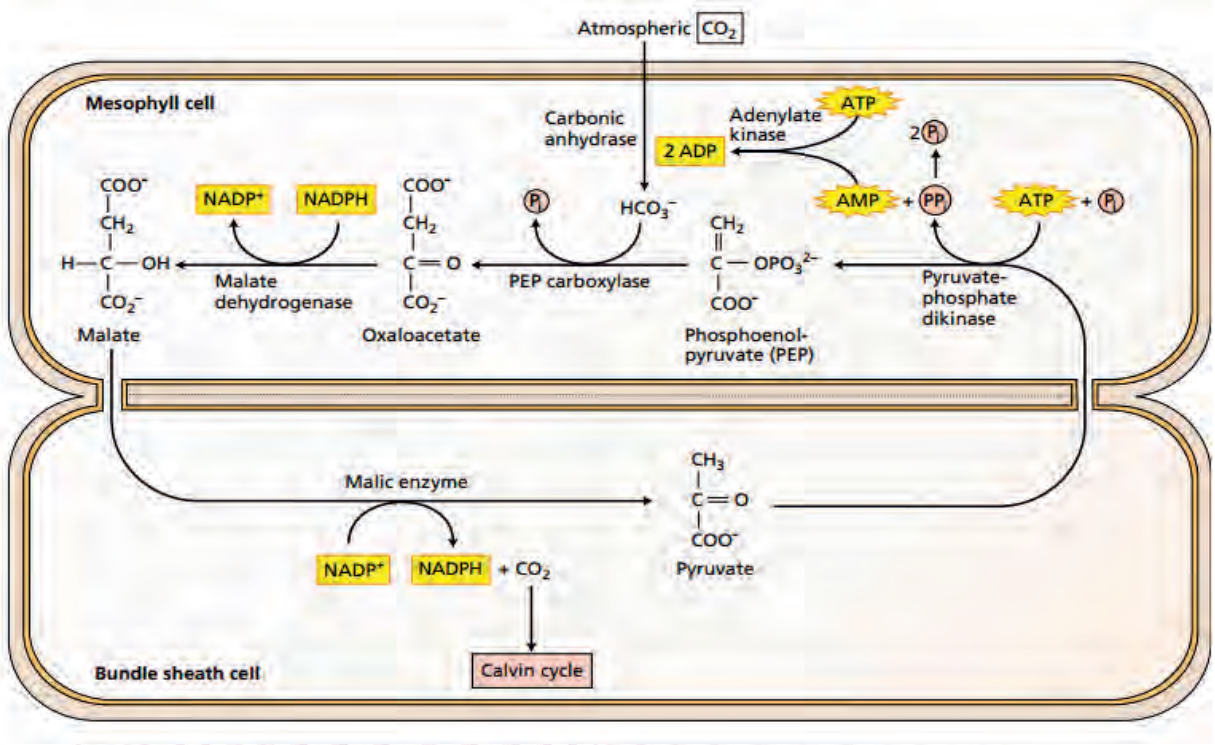


Figure 3.8. The C_4 photosynthetic pathway

CAM (crassulacean acid metabolism)Plants:- The CAM mechanism enables plants to improve water use efficiency. The CAM mechanism is similar in many respects to the C4 cycle. In C_4 plants, formation of the C_4 acids in the mesophyll is spatially separated from decarboxylation of the C_4 acids and from refixation of the resulting CO_2 by the Calvin cycle in the bundle sheath. In CAM plants, formation of the C_4 acids is both temporally and spatially separated.

At night, CO_2 is captured by PEP carboxylase in the cytosol, and the malate that forms from the oxaloacetate product is stored in the vacuole. During the day time, the stored malate is transported to the chloroplast and decarboxylated by NADP-

malic enzyme, the released CO_2 is fixed by the Calvin cycle, and the NADPH is used for converting the decarboxylated triose phosphate product to starch.

CAM plants succulent (water-storing) plants such as cacti achieve their high water use efficiency by opening their stomata during the cool, desert nights and closing them during the hot, dry days. Closing the stomata during the day minimizes water loss, but because H_2O and CO_2 share the same diffusion pathway, CO_2 must then be taken up at night. CO_2 is incorporated via carboxylation of phosphoenolpyruvate to oxaloacetate, which is then reduced to malate. The malate accumulates and is stored in the large vacuoles that are a typical, but not obligatory, anatomic feature of the leaf cells of CAM plants (Figure 3.9).

The accumulation of substantial amounts of malic acid, equivalent to the amount of CO_2 assimilated at night, has long been recognized as a nocturnal acidification of the leaf. With the onset of day, the stomata close, preventing loss of water and further uptake of CO_2. The leaf cells deacidify as the reserves of vacuolar malic acid are consumed. Because the stomata are closed, the internally released CO_2 cannot escape from the leaf and instead is fixed and converted to carbohydrate by the Calvin cycle.

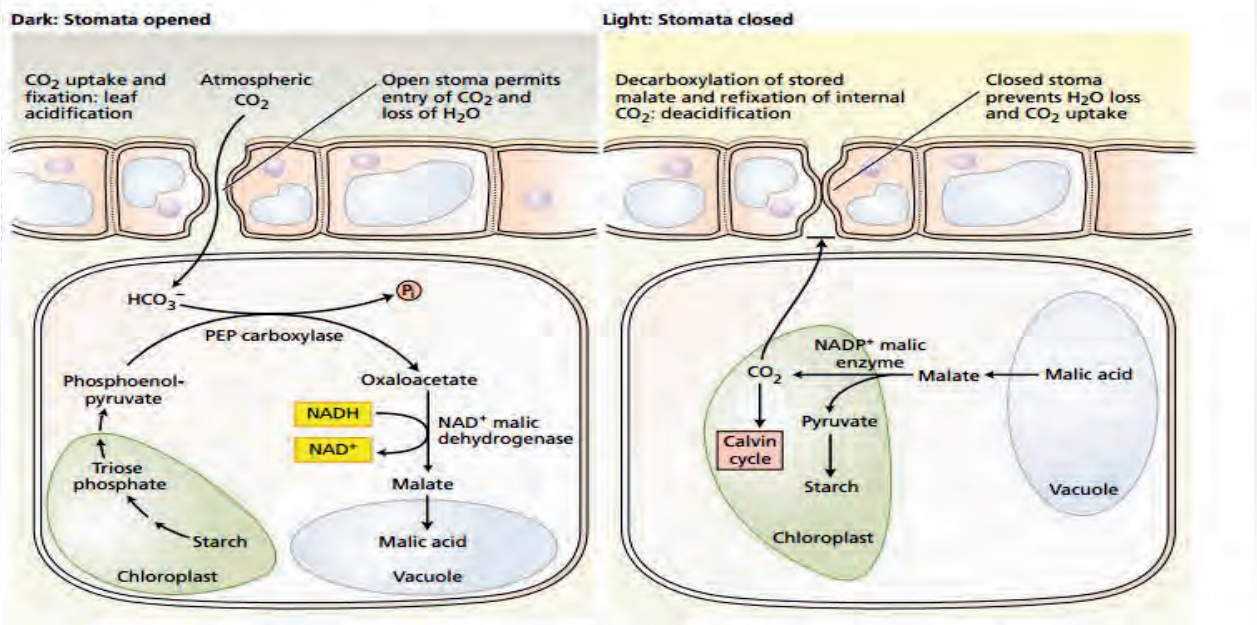


Figure 3.9. CAM cycle

Photorespiration:- Photorespiration is a process which involves oxidation of organic compounds in plants by oxygen in the presence of light. Like ordinary respiration,

this process also releases carbon from organic compounds in the form of CO_2 but does not produce ATP. Thus, apparently it seems to be a wasteful process, but it must have some functions which are still unknown. This process occurs in C3 plants and to some extent in C4. Photorespiratory substrate is glycolate. RuBisco instead of combining with CO_2 it combines with O_2. This type of oxidation of O_2 to RuBP molecule is known as photosynthetic C-oxidation cycle or glycolate pathway or C_2 photorespiratory carbon oxidation cycle. Generally photorespiration is expressed by the term called CO_2 compensation point and it is defined as CO_2 concentration at which rate of uptake will be equal to the rate of photosynthetic respiratory CO_2 released. Photorespiration is common in C3 plants but highly reduced in C4 plants, and absent in CAM plants (Figure 3.10).

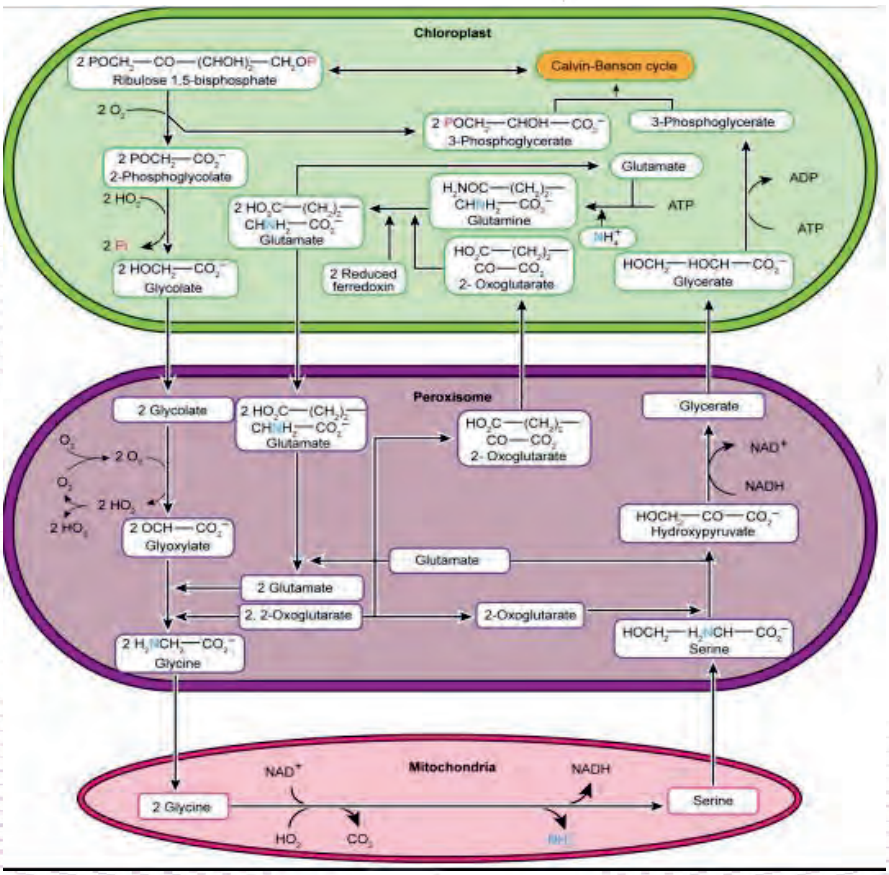


Figure 3.10 The photorespiratory cycle.

Self- questioning

- What is the difference between light-dependent and light-independent reactions?
- Why is the Calvin cycle also called the C3 pathway?
- How does photorespiration counter photosynthesis?
- Describe how a C4 plant minimizes photorespiration.
- How is the CAM pathway like C4 metabolism, and how is it different?

Practical activities 3.3. Testing a leaf for starch

Iodine solution (yellow/brown) and starch (white) form a deep blue colour when they mix. The test for starch, therefore, is to add iodine solution to a leaf to see if it goes blue. However, a living leaf is impermeable to iodine and the chlorophyll in the leaf masks any colour change. So, the leaf has to be treated as follows:

- Heat some water to boiling point in a beaker and then **turn off the Bunsen flame**. Use forceps to dip a leaf in the hot water for about 30 seconds. This kills the cytoplasm, denatures the enzymes and makes the leaf more permeable to iodine solution.
- **Note**: make sure the Bunsen flame is extinguished before starting the next part of the procedure, as ethanol is flammable.
- Push the leaf to the bottom of a test-tube and cover it with ethanol (alcohol). Place the tube in the hot water.
- The alcohol will boil and dissolve out most of the chlorophyll. This makes colour changes with iodine easier to see.
- Pour the green alcohol into a spare beaker, remove the leaf and dip it once more into the hot water to soften it.

- Spread the decolorized leaf flat on a white tile and drop iodine solution on to it. The parts containing starch will turn blue; parts without starch will stain brown or yellow with iodine.

Practical Activity 3.4. Test for Chlorophyll in leaves

Equipment ■ a few spinach leaves	■ rubbing alcohol	■ glass jar
■ filter paper	■ bowl	■ hot water

Method

- Cut up the spinach leaves into very fine pieces.
- Pour some rubbing alcohol into the glass jar and transfer a small quantity of the leaves into the jar.
- Pour some hot water into the bowl and stand the covered jar in it. Stir the leaves in the alcohol from time to time.
- After about an hour, insert a strip of filter paper into the glass jar so that it dips into the green coloured liquid.
- As the liquid rises up the filter paper, draw students' attention to the yellow and orange bands that form on the filter paper.
- **Precaution**

  Alcohol is flammable.

Practical Activity 3.4. To show that oxygen is produced during photosynthesis

in the equation for photosynthesis, we saw that oxygen is given off as a waste product.
When the light intensity is high (bright sunshine), much oxygen is produced.

The oxygen can easily be collected from water since it is not very soluble in water.
This explains why we use an aquatic plant.

Materials and chemicals

- local pond weed
- a 250 cm3 beaker
- a glass funnel
- a test tube
- small stones
- sodium hydrogen carbonate
- water or pond water

Method

1. Place the pond weed in a large beaker filled with about 150 cm3 of pond water. You may add sodium hydrogen carbonate to the water to produce more carbon dioxide.

2. Carefully invert a funnel over the pond weed.

3. Fill a test tube with the same amount of water as in the beaker. Carefully and without allowing any water out of the test tube, invert the test tube over the stem of the funnel.

4. You will need to put some small stones at the bottom of the beaker to support the funnel so that it is raised above the bottom of the beaker. This arrangement allows free water circulation.

5. Place the apparatus in bright sunlight for 3–4 hours.

6. After this period, move the inverted test tube from the inverted funnel stem while still under water. Then cover the test tube with your thumb before removing it from the beaker.

7. Remove your thumb from the test tube mouth and quickly plunge a glowing splint into the test tube.

Activity questions

1. What did you observe during the four hours?
2. What happens to the glowing splint when plunged into the test tube?
3. What is the identity of the gas?
4. What conclusion can you make from this activity?

3.3. Contributions of photosynthesis for the continuity of life, for O_2 and CO_2 balance and global warming

The oxygen in the air comes from photosynthesis. Plants continue to replenish oxygen in the air. All of our food comes directly or indirectly from photosynthesis. Human beings are also dependent on ancient products of photosynthesis (fossil fuels, natural gas, coal & petroleum);

needed for modern industrial energy; complex mix of hydrocarbons; represent remains of organisms that relied on photosynthesis millions of years ago; carbon, oxygen, and hydrogen atoms are recycled in the environment where a constant input of solar energy is needed for energy to continue flowing to support life remove carbon dioxide from the atmosphere (inhibit global warming) (Figure 3.11).

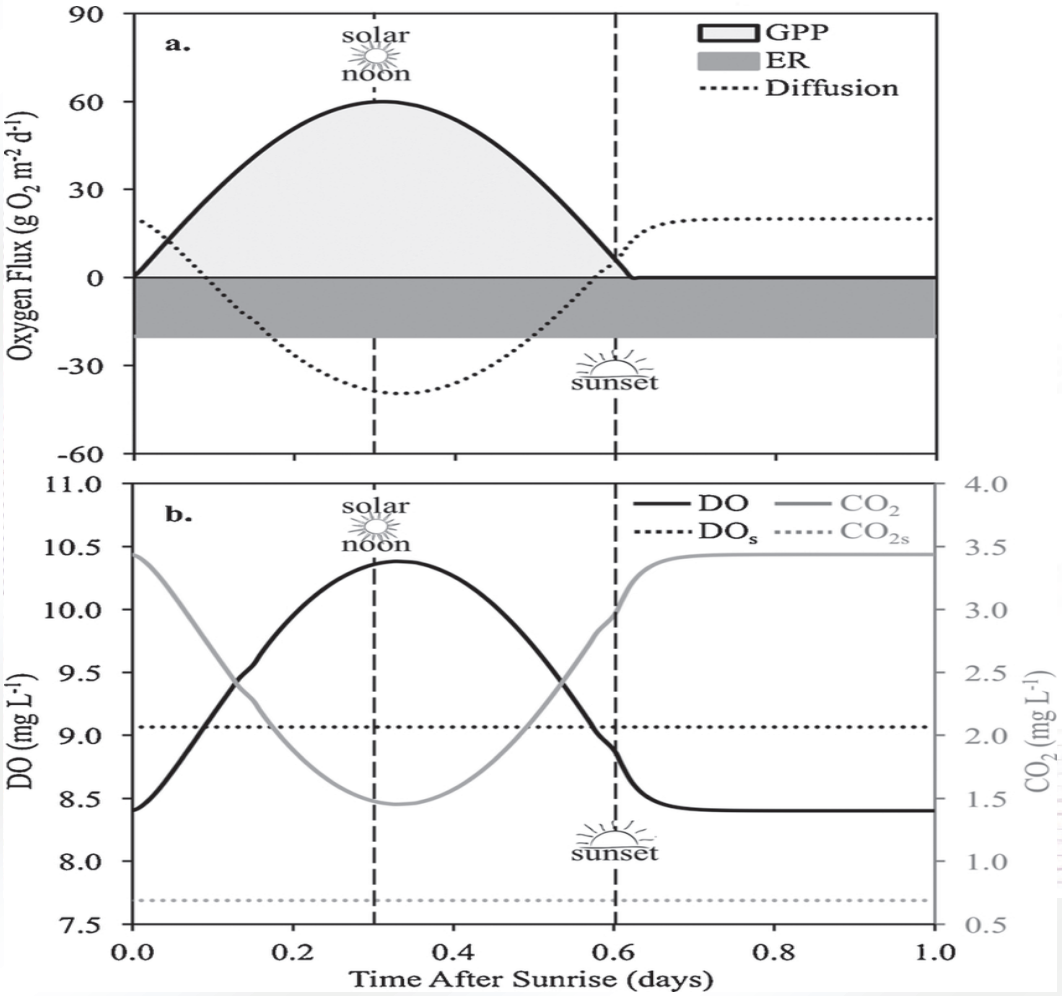


Figure 3.11. Carbondioxide –oxygen balance in the atomospher (Idealized daily rates of photosynthesis (GPP), respiration (ER), and diffusion (a.) with resultant dissolved O 2 concentration and saturated (DO s) concentration at 20 °C (b.)).

Review questions

I. Choose the correct answer for the following questions.

1. Where does the energy come from to drive photosynthesis?

A. Chloroplast C. Sun

B. ATP D. Glucose

4. Only high-energy light can penetrate the ocean and reach photosynthetic organisms in coral reefs. What color light would you predict these organisms use?

a. Red c. Blue b. Yellow d. Orange

5. Which molecules are used to connect the light reactions and the carbon reactions?

a. ATP and ADP c. Glucose b. NADPH and NADP d. Both a and b are correct.

6. A plant that only opens its stomata at night is a

a. C2 plant. c. C4 plant. b. C3 plant. d. CAM plant.

7. Choose the pair of terms that correctly completes this sentence: Catabolism is to anabolism as _____ is to _____.

a. exergonic; spontaneous b. exergonic; endergonic c. free energy; entropy

d. work; energy

8. Most cells cannot harness heat to perform work because

a. heat does not involve a transfer of energy.

b. cells do not have much thermal energy; they are relatively cool.

c. temperature is usually uniform throughout a cell.

d. heat can never be used to do work.

9. Which of the following metabolic processes can occur without a net influx of energy from some other process?

a. $ADP + {\sim}Pi \rightarrow ATP + H_2O$ b. $C_6H_{12}O_6 + 6\ O_2 \rightarrow 6\ CO_2 + 6\ H_2O$

c. $6\ CO_2 + 6\ H_2O \rightarrow 6H_{12}O_6 + 6\ O_2$ d. Amino acids → Protein

10. If an enzyme in solution is saturated with substrate, the most effective way to obtain a faster yield of products is to,

a. add more of the enzyme b. heat the solution to 90°C c. add more substrate d. add a noncompetitive inhibitor.

11. Some bacteria are metabolically active in hot springs because

a. they are able to maintain a lower internal temperature.

b. high temperatures make catalysis unnecessary.

c. their enzymes have high optimal temperatures.

d. their enzymes are completely insensitive to temperature.

3.4. Cellular Respiration

At the end of this lesson, the student will be able to:

- define cellular respiration
- describe the process of glycolysis.
- list the products formed at the end of glycolysis.
- describe the structure of mitochondria.
- explain how energy is harvested in aerobic respiration.
- describe the different stages of aerobic respiration.
- compare and contrast aerobic and anaerobic respiration.
- write the chemical equation of aerobic respiration.
- state the products of alcoholic fermentation by yeast.
- show the mechanism electron transport system in mitochondria.
- explain the difference between substrate-level phosphorylation and

Cellular respiration

It is the process by which cells produce energy from glucose in the form of energy storing compound called ATP(AdenosineTri-Phosphate) for various cellular activities. The energy released by cellular respiration is temporarily captured by the formation of Adenosine Triphosphate (ATP) within the cell. It is often referred to as the energy currency of the cell, and this can be compared to depositing cash in a bank. ATP can be used to store energy for future reactions or be withdrawn to pay for reactions when energy is required by the cell. Animals store the energy obtained from the breakdown of food as ATP. Likewise, plants capture and store the energy they derive from light during photosynthesis in ATP molecules. ATP is a nucleotide consisting of an adenine base attached to a ribose sugar, which is attached to three phosphate groups (**see Figure 3.12**). These three phosphate groups are linked to one another by two high-energy bonds called phosphoanhydride bonds. When one phosphate group is removed by breaking a phosphoanhydride bond in a process called hydrolysis, energy is released, and ATP is converted to adenosine diphosphate (ADP). Similarly, energy is also released when a phosphate is removed from ADP to form adenosine monophosphate (AMP). This free energy can be transferred to other molecules to make unfavorable reactions in a cell favorable.

AMP can then be recycled into ADP or ATP by forming new phosphoanhydride bonds to store energy once again. In the cell, AMP, ADP, and ATP are constantly interconverted as they involve in biological reactions.

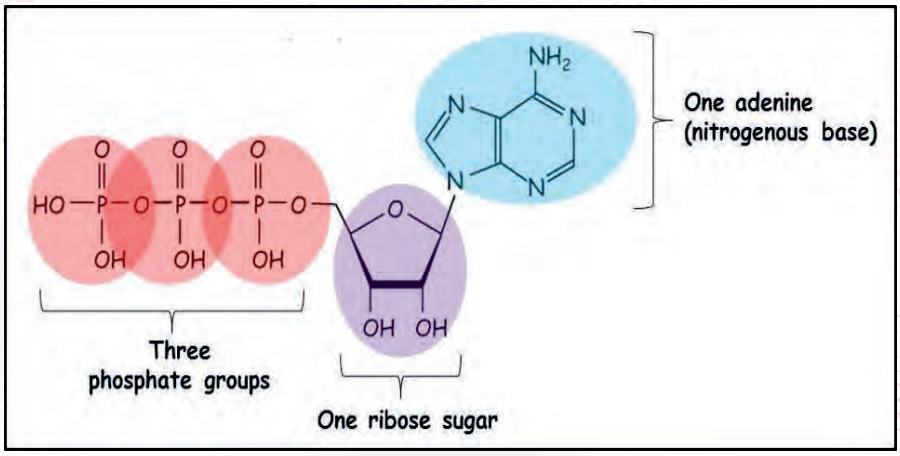


Figure 3.12. Adenosine triphosphate

Coupled Reactions

Many biochemical reactions in which energy is given off- (is called **exothermic**), whereas many others reactions that require energy (are called **endothermic**). In order for both processes

to be carried out efficiently, they must be "coupled". Usually, a **coupled reaction** will involve ATP (Figure 3.13) or some similar molecules. A coupled reaction is carried out when two reactions occur nearly simultaneously. The first reaction must be exothermic and that gives off energy. The second reaction is endothermic, which immediately uses the energy produced from the first reaction.

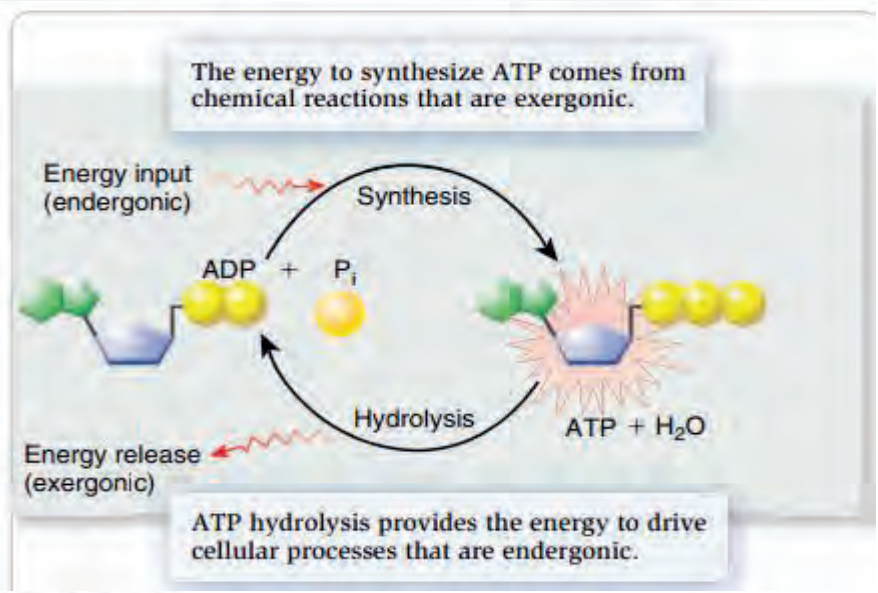


Figure 3.13. ATP cycle

An example of a coupled reaction is the hydrolysis of ATP and the contraction of muscle tissue. Two proteins, actin and myosin, form a loose complex called actomyosin. When ATP is added to isolated actomyosin, the protein fibers contract. The hydrolysis of ATP releases energy which is used by muscles to contract. The coupled reaction is:

A. $ATP + H_2O \rightarrow\rightarrow ADP + P + energy$

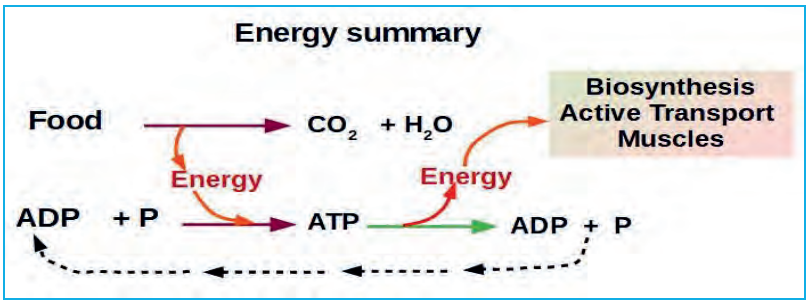
B. Relaxed muscle + energy $\rightarrow\rightarrow$ contracted muscle

When the ATP is used up by the muscles, a further supply of energy is released from creatine phosphate. Another example of a coupled reaction is the hydrolysis of creatine phosphate to release energy which in turn is used for the formation of more ATP. The coupled reaction is:

A. Creatine --- $PO_3 + H_2O \rightarrow\rightarrow$ creatine H + HPO_4^{-3} + energy

B. $ADP + HPO_4^{-3} + energy \rightarrow\rightarrow ATP + H_2O$

During periods of low muscular activity, the reactions are reversed to replenish the supplies of ATP and creatine phosphate. The energy for the formation of ATP is supplied by other metabolic reactions.



3.4.1. The site of cellular respiration

Glycolysis occurs in the cytosol of the cell and does not require oxygen, whereas the Krebs cycle and electron transport occur in the mitochondria and requires oxygen. **Cellular respiration** is carried out by both prokaryotic and eukaryotic cells. In prokaryotic cells, it is carried out in the **cell cytoplasm**, whereas in eukaryotic cells it begins in the **cytosol** then is carried out in the **mitochondria**. In eukaryotes, the four stages of cellular respiration include glycolysis, transition reaction (pyruvate oxidation), the Krebs cycle (also known as the citric acid cycle), and oxidative phosphorylation through the electron transport chain.

Self-questioning

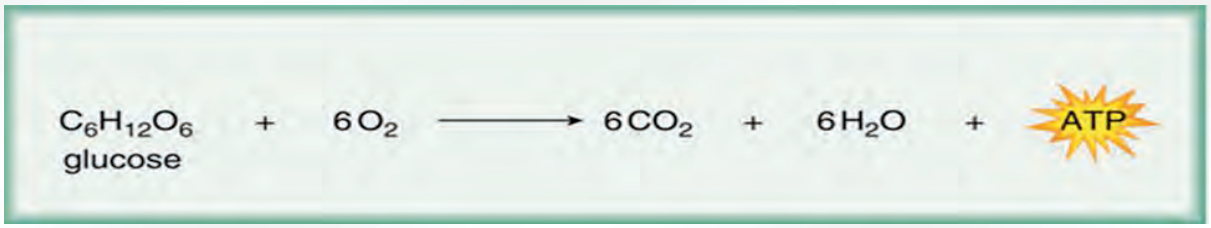
- Why is cellular respiration important?
- What is metabolism on a cellular level?
- Which reactions require energy input and which release energy?
- Why oxidation and reduction reactions always are linked?

3.4.2. Stages of respiration

Cellular respiration involves many chemical reactions. The reactions can be summed up in this equation

Self-questioning

- Where does cellular respiration take place?



Cellular respiration consists of a sequence of many chemical reactions that vary during aerobic and anaerobic conditions. Aerobic respiration is divided into main stages: Glycolysis, Citric acid cycle, and Electron transport chain (Figure 3.14).

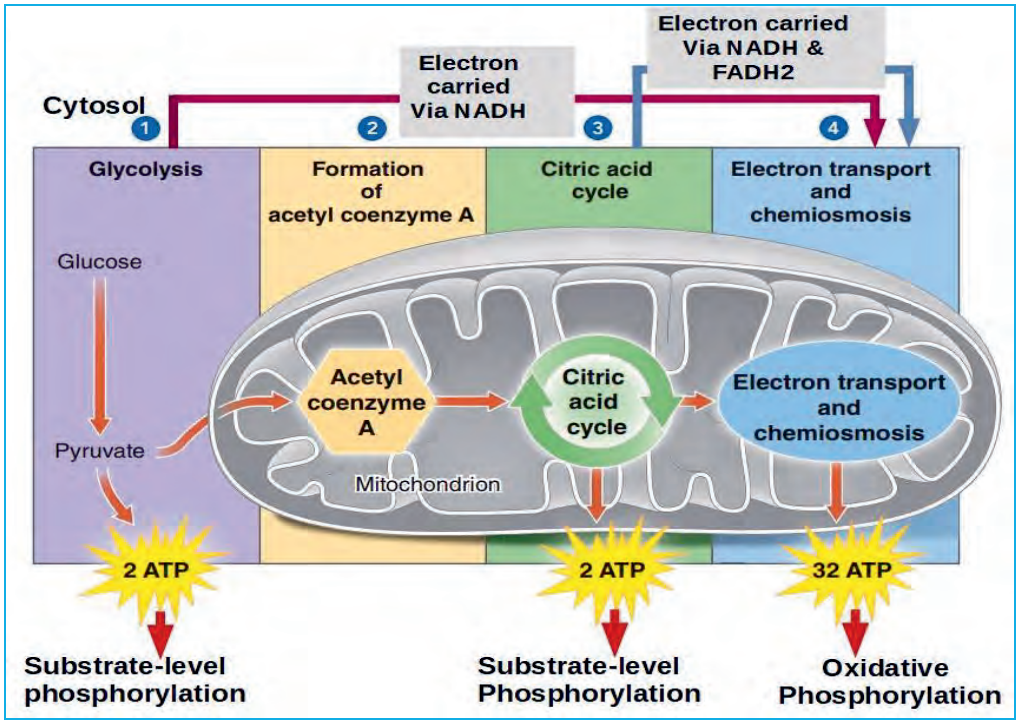


Figure 3.14. Stages of cellular respiration

Stage I: Glycolysis

Glycolysis occurs in the cytosol of the cell in anaerobic condition. It is the common pathway of both aerobic and anaerobic respiration. Glycolysis begins cellular respiration by breaking glucose into two molecules of a three-carbon compound called pyruvate (**Figure 3.14**).

Activity: 3.6. Group work

- Identify the reduced and oxidized compounds from the following equation?

$$C_4H_6O_5 + NAD^+ \rightleftharpoons C_4H_4O_5 + NADH + H^+$$

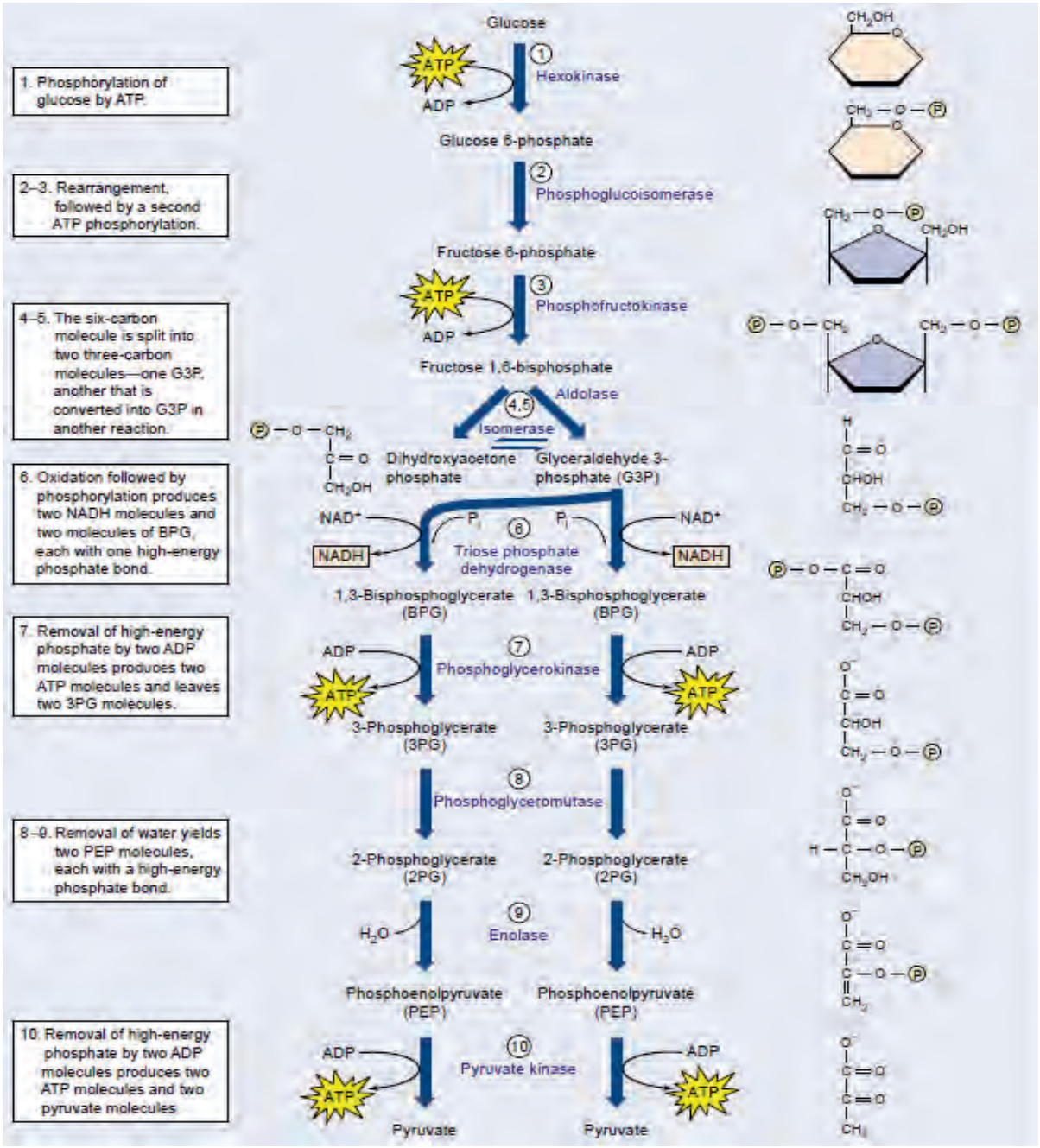


Figure 3.15. Glycolysis pathway.

The 10 steps of glycolysis can be grouped into three phases:

The first phase (steps 1–3) involves an energy investment. Two ATP molecules are hydrolyzed, and the phosphates from those ATP molecules are attached to glucose, which is converted to fructose-1,6-bisphosphate(Figure 3.15). The energy investment phase raises the free energy of glucose, thereby allowing later reactions to be exergonic.

• The cleavage phase (steps 4–5) breaks this six-carbon molecule into two molecules of glyceraldehyde-3-phosphate.

• The energy liberation phase (steps 6–10) produces four ATP, two NADH, and two molecules of pyruvate. Because two molecules of ATP are used in the energy investment phase, the net yield is two molecules of ATP.

Net reaction of glycolysis

$$Glucose + 2ADP + 2NAD^{+} + 2Pi \rightarrow 2Pyruvate + 2ATP + 2NADH + 2H^{+} + 2H_2O$$

Stsge II: Pyruvate oxidation (link reaction)

In order to oxidize to pyruvate, which is the product of glycolysis and enter the next pathway, it must undergo several changes to become acetyl Coenzyme A (acetyl CoA). Acetyl CoA is a molecule that is further converted to oxaloacetate, which enters the citric acid cycle (Krebs cycle). The conversion of pyruvate to acetyl CoA is a three-step process (**Figure 3.16**). Step 1. A carboxyl group is removed from pyruvate, releasing a molecule of carbon dioxide into the surrounding medium. (Note: carbon dioxide is one carbon attached to two oxygen atoms and is one of the major end products of cellular respiration).

Step 2. The hydroxyethyl group is oxidized to an acetyl group, and the electrons are picked up by NAD+, forming NADH (the reduced form of NAD^{+}). The high- energy electrons from NADH will be used later by the cell to generate ATP for energy.

Step 3. The enzyme-bound acetyl group is transferred to CoA, producing a molecule of acetyl CoA. This molecule of acetyl CoA is then further converted to be used in the next pathway of metabolism, or the citric acid cycle.

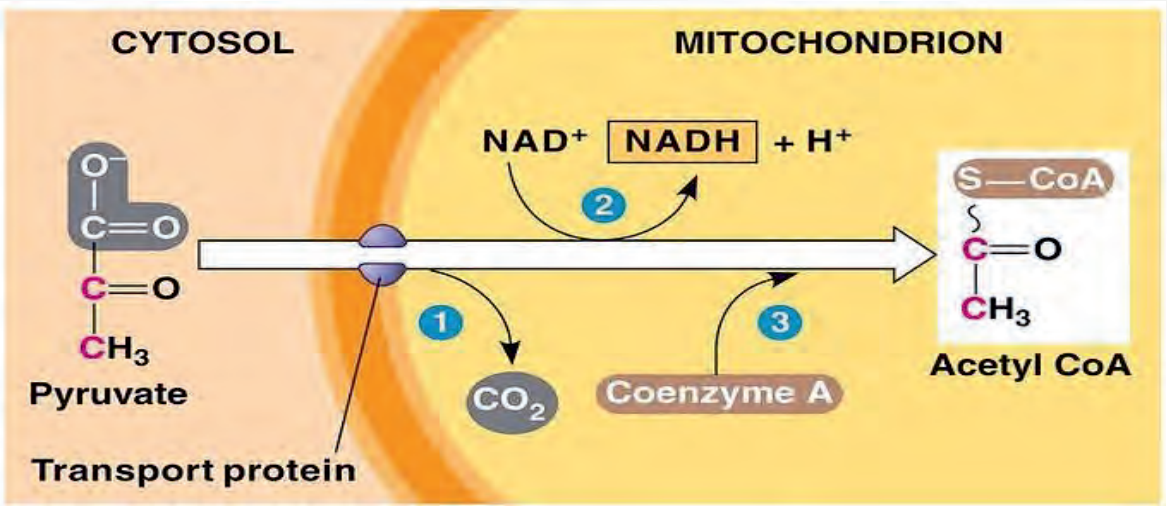


Figure 3.16. Pyruvate oxidation (link reaction)

The overall reaction:

2pyruvate + 2NAD+ + 2 CoA --> 2 acetyl-CoA + 2NADH + 2H+ + 2CO2.

The Acetyl-coA molecules enter the Kreb cycle, NADH goes to the electron transport chain to produce ATP. Carbon dioxide diffuses out of the cell as a waste product. The protons (2H+) stay in the matrix.

Net reaction:

$3NAD+ + FAD + GDP + Pi + acetyl\text{-}CoA + 2\ H_2O \rightarrow 3NADH + FADH_2 + GTP + 2CO_2 + 4H^+$

Stage III: Kreb cycle

The Krebs cycle itself actually begins when acetyl-CoA combines with a four-carbon molecule called OAA (oxaloacetate) (Figure 3.17). This produces citric acid, which has six carbon atoms. This is why the Krebs cycle is also called the citric acid cycle. After citric acid forms, it goes through a series of reactions that release energy. The energy is captured in molecules of NADH, ATP, and FADH2, another energy-carrying compound. Carbon dioxide is also released as product of these reactions. The final step of the Krebs cycle regenerates OAA, the molecule that began the Krebs cycle. This molecule is needed for the next turn through the cycle. Two turns are needed because glycolysis produces two pyruvate molecules when it splits glucose.

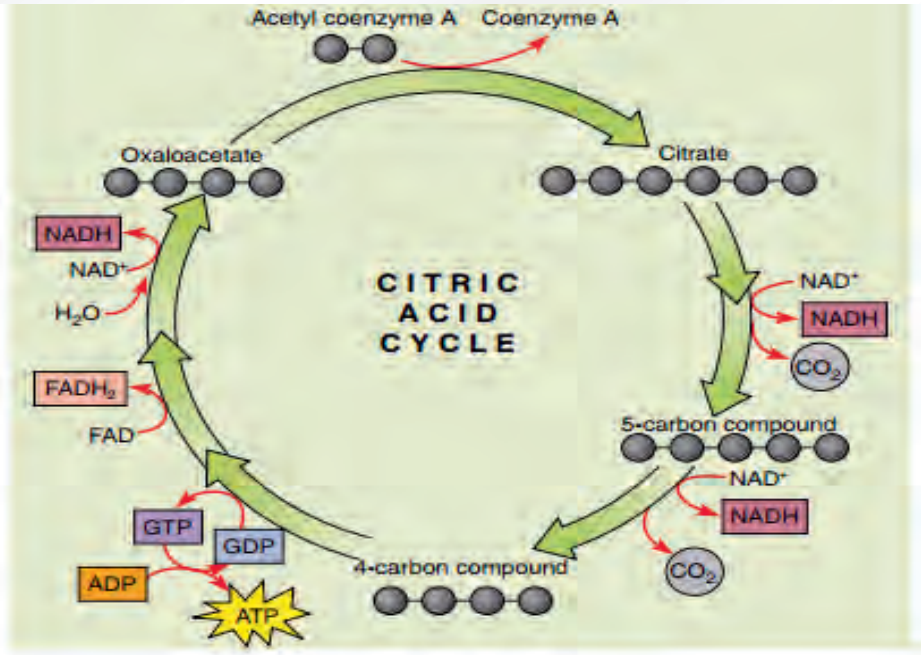


Figure 3.17 citric acid cycle (Krebs cycle)

Activity 3.7. Group work

Discuss in your groups and prepare a presentation on:

- In the citric acid cycle shown in **Figure 3.17**, what molecules capture energy from the redox reactions? How is ATP produced?
- What processes in your cells produce the CO_2 that you exhale?

Stage IV: Oxidative phosphorylation

It is the process in which ATP is formed as a result of the transfer of electrons from NADH or $FADH_2$ to O_2 by a series of electron carriers. This process, which takes place in mitochondria, is the major source of ATP in aerobic organisms. Oxidative phosphorylation generates 26 out of the 30 molecules of ATP that are formed when glucose is completely oxidized to CO_2 and H_2O. The three major steps in oxidative phosphorylation are:

(a) oxidation-reduction reactions involving electron transfers between specialized proteins embedded in the inner mitochondrial membrane;

(b) the generation of a proton (H+) gradient across the inner mitochondrial membrane (which occurs simultaneously with step (a)); and

(c) the synthesis of ATP using energy from the spontaneous diffusion of electrons down the proton gradient generated in step (b) (**Figure 3.18**).

The NADH and $FADH_2$, formed during glycolysis, the link reaction, and the TCA cycle, give up their electrons to reduce molecular O_2 to H_2O. Electron transfer occurs through a series of protein electron carriers, the final acceptor being O_2 and the pathway is called the electron transport chain (ETC). The function of ETC is to facilitate the controlled release of free energy that was stored in reduced cofactors during catabolism. Energy is released when electrons are transported from higher energy NADH/$FADH_2$ to lower energy O_2. This energy is used to phsophorylate ADP. There are 3 sites of the chain that can give enough energy for ATP synthase. These sites are:

Site I between FMN and Coenzyme Q at enzyme complex I.

Site II between cyt b and cyt C1 at enzyme complex III

Site III between cyt a and cyt a3 at enzyme complex IV

Because energy generated by the transfer of electrons through the electron transport chain to O_2 is used in the production of ATP, the overall process is known as oxidative phosphorylation. This oxidation process refers to the coupling of the electron transport in respiratory chain with phosphorylation of ADP to form ATP. It is a process by which the energy of biological oxidation is ultimately converted to the chemical energy of ATP. Oxidative phosphorylation is responsible for 90% of the total ATP synthesis in the cell. Oxidative phosphorylation is the process in which ATP is formed as a result of the transfer of electrons from NADH or $FADH_2$ to O_2 by a series of electron carriers (Figure 3.17).

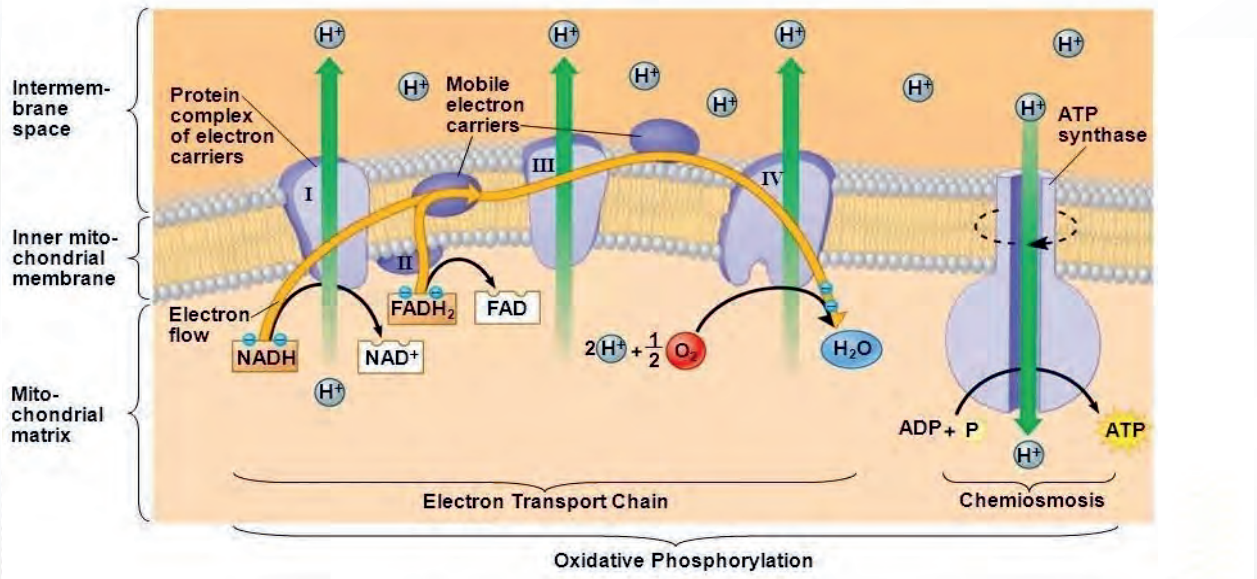
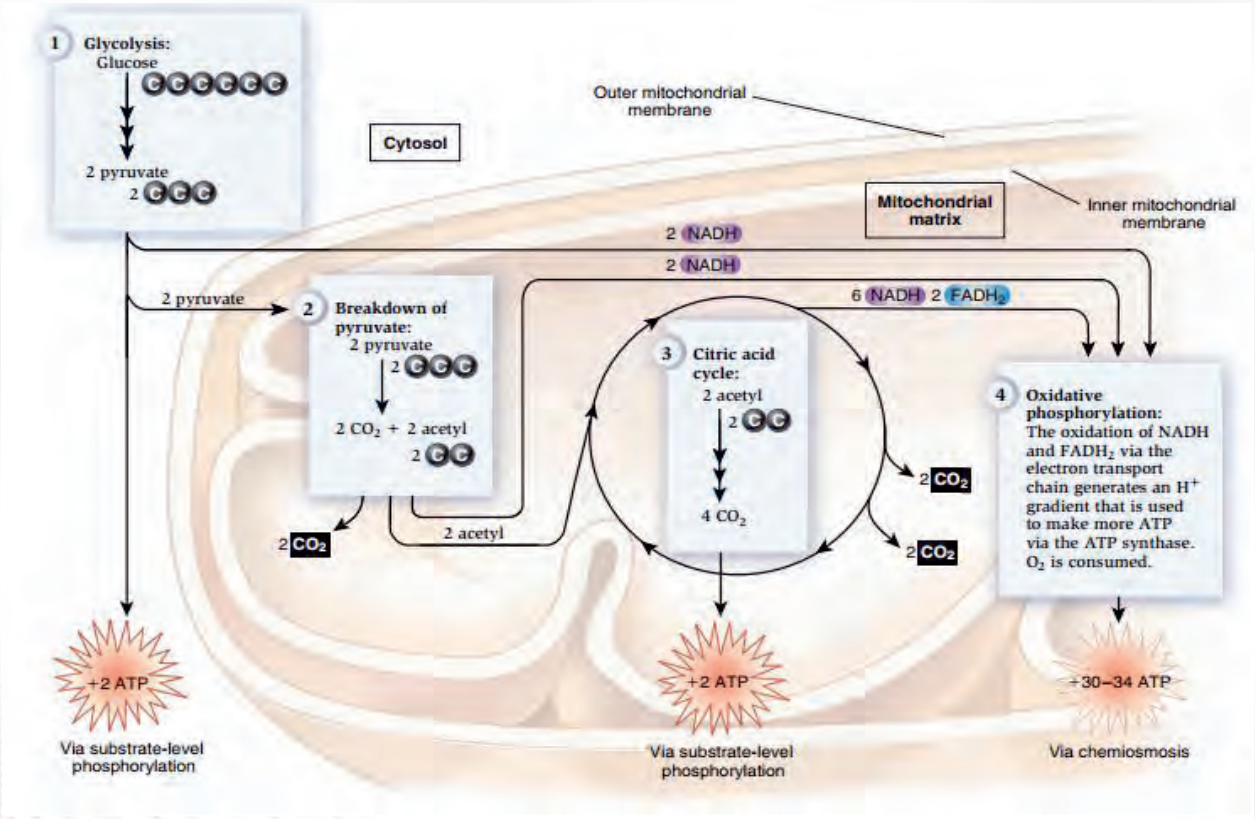


Figure 3.18. Electron transport and oxidative phosphorylation

S



Mechanism

Mechanism suggests that the transfer of electrons through the electron transport chain causes protons to be translocated (pumped out) from the mitochondrial matrix to the intermembrane space at the three sites of ATP production (i.e. it acts as a proton pump) resulting in an electrochemical potential difference across the inner mitochondrial membrane.

The electrical potential difference is due to the accumulation of the positively charged hydrogen ions outside the membrane, whereas the chemical potential difference is due to the difference in pH when it is more acidic outside the membrane. This electrochemical potential difference drives (forces) ATP synthase to generate ATP from ADP and inorganic phosphate.

(Figure 3.19).

The energy budget of one glucose molecule

As from one glucose molecule, two pyruvate molecules are formed so that two cycles will be formed for the complete breakdown. So, the total yield will be 6NADH, 2FADH2, and 2 ATP

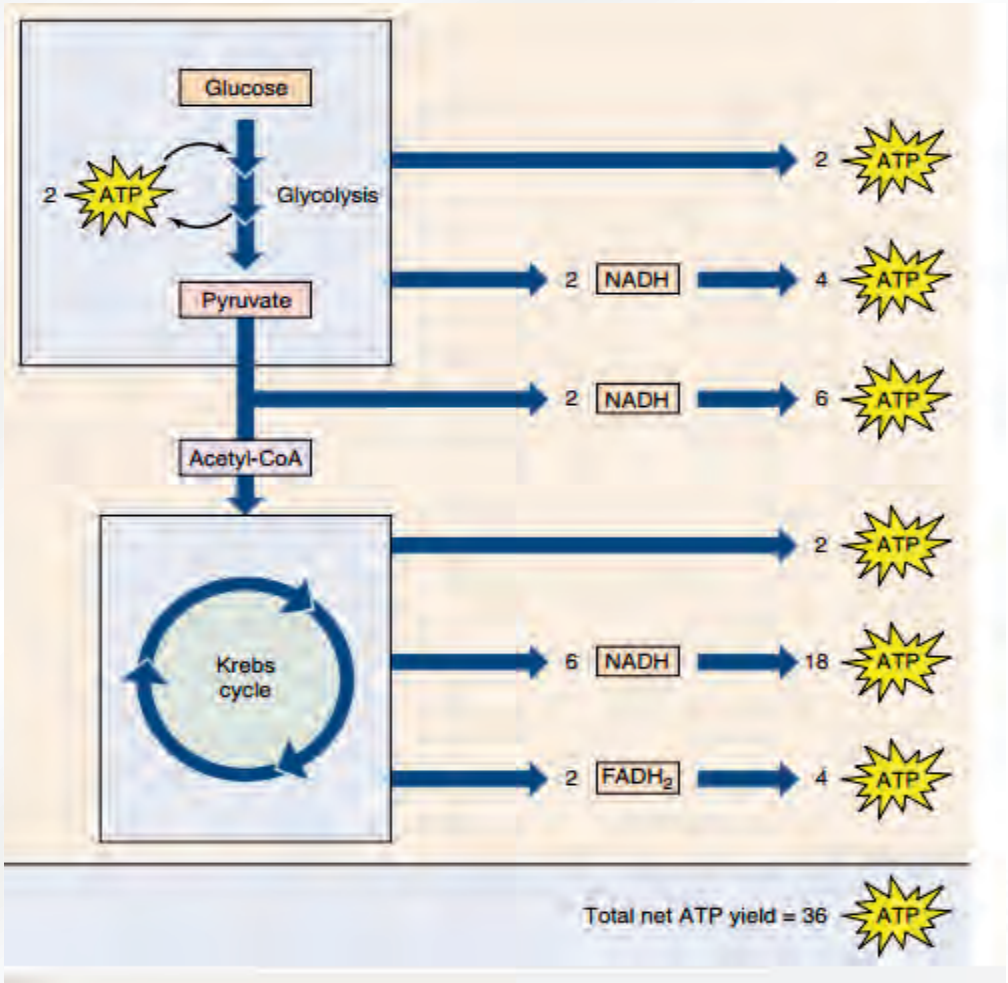


Figure 3. 19. Energy yield per glucose molecule breakdown.

Self-questioning

What couples the electron transport chain to ATP synthesis?

Energy from non-carbohydrate sources

We obtain most of our energy in the form of fats, proteins, sucrose and other disaccharides, and starch, a polysaccharide. All these organic molecules in food can be used by cellular respiration to make ATP **(Figure 3.20)**.

Glycolysis can accept a wide range of carbohydrates for catabolism. In the digestive tract, starch is hydrolyzed to glucose, which can then be broken down in the cells by glycolysis and the citric acid cycle. Similarly, glycogen, the polysaccharide that humans and many other animals store in their liver and muscle cells, can be hydrolyzed to glucose between meals as fuel for respiration. The digestion of disaccharides, including sucrose, provides glucose and other monosaccharides as fuel for respiration.

Proteins can also be used for fuel, but first they must be digested to their constituent amino acids. Many of the amino acids are used by the organism to build new proteins. Amino acids present in excess are converted by enzymes to intermediates of glycolysis and the citric acid cycle. Before amino acids can feed into glycolysis or the citric acid cycle, their amino groups must be removed, a process called *deamination*. The nitrogenous refuse is excreted from the animal in the form of ammonia (NH3), urea, or other waste products.

Catabolism can also harvest energy stored in fats obtained either from food or from storage cells in the body. After fats are digested to glycerol and fatty acids, the glycerol is converted to glyceraldehyde 3-phosphate, an intermediate of glycolysis. Most of the energy of a fat is stored in the fatty acids. A metabolic sequence called **beta oxidation** breaks the fatty acids down to two-carbon fragments, which enter the citric acid cycle as acetyl CoA.

NADH and FADH2 are also generated during beta oxidation; they can enter the electron transport chain, leading to further ATP production. Fats make excellent fuels, in large part due to their chemical structure and the high energy level of their electrons (equally shared between carbon and hydrogen) compared to those of carbohydrates. A gram of fat oxidized by respiration produces more than twice as much ATP as a gram of carbohydrate. Unfortunately, this also means that a person trying to lose weight must work hard to use up fat stored in the body because so many kilojoules are stockpiled in each gram of fat.

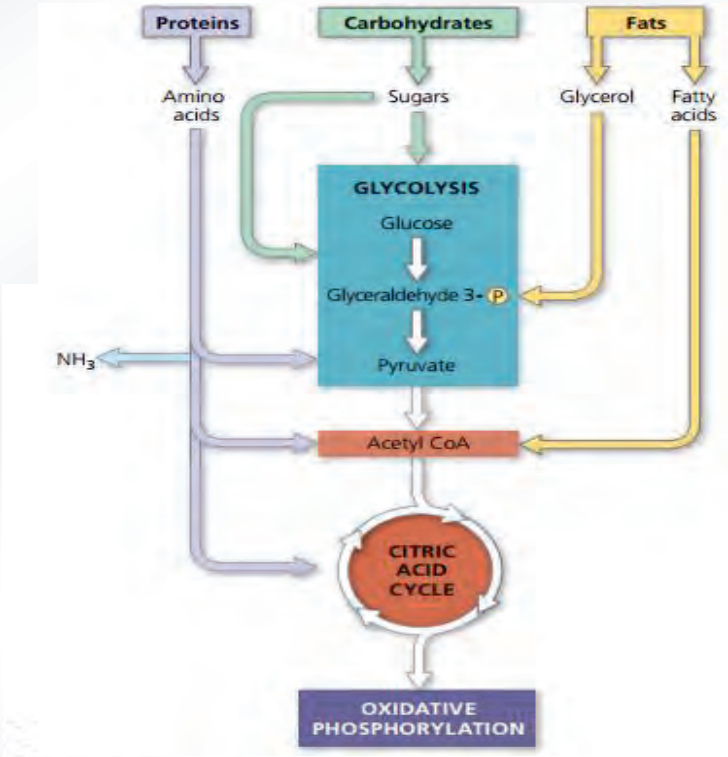


Figure 3. 20. The cataboism of various molecules from food.

Activity 3.8. Group work

Discuss in your groups and prepare a presentation on:

- What effect would an absence of O_2 have on the process shown in Figure 2.29?
- In the absence of O_2, as in question 1, what do you think would happen if you decreased the pH of the intermembrane space of the mitochondrion? Explain your answer.
- Membranes must be fluid to function properly. How does the operation of the electron transport chain support that assertion?

Two ideas to keep in mind

1. The idea of net gain of ATP is like the profit a business person makes. It invests in money materials, advertising and building the capacity of staff. He/ She sell his/her product where the extra money is profit – net gain. In a similar vein, glycolysis 'invests' in two molecules of ATP to make the glucose reactive, then, later, produces **four molecules of ATP** – a net gain of two molecules of ATP.

2. There are two molecules of pyruvate made from each molecule of glucose. So, all the gains of ATP and the reduced NAD and reduced FAD that accrue from each pyruvate must be doubled to give the gain from each molecule of glucose.

Key words

Coenzyme A coenzyme is derivedfrom a pantothenic acid needed for respiration

Acetyl coenzyme A is producedby the reaction of coenzyme Awith a molecule of pyruvate

Dehydrogenation refers to the process of removinghydrogen from a molecule

Decarboxylation means removing carbon from a molecule

Terminal electron acceptor is the final molecule at the end of the electron transport chain to accept an electron

Activity 3.9 group work

Make a large annotated wall chart **showing** glycolysis and Krebs cycle and how they are linked together. Make sure that you show the different compounds and where ATP is formed. This should be as accurate as possible so that it can form the basis of your revision of this complex topic.

The summary equation for aerobic respiration is:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6H_2O + 6CO_2$ + energy released

Fermentation

What happens in the anaerobic pathway?

In the process of glycolysis, a net profit of two ATP was produced, two NAD+ were reduced to two NADH + H+, and glucose was split into two pyruvate molecules. When oxygen is not present, pyruvate will undergo a process called fermentation. In the process of fermentation, the NADH + H+ from glycolysis will be recycled back to NAD+ so that glycolysis can continue. In the process of glycolysis, NAD+ is reduced to form NADH + H+. If NAD+ is not present, glycolysis will not be able to continue. During aerobic respiration, the NADH formed in the glycolysis will be oxidized to reform NAD+ for use in glycolysis again. When oxygen is not present or if an organism is not able to undergo aerobic respiration, pyruvate will undergo a process called fermentation. Fermentation does not require oxygen and is therefore anaerobic.

Self-questioning

- How do respirometers work?
- What are respirometers used for?
- How does a respirometer measure oxygen uptake?
- How do you use a respirometer a level biology?

Fermentation will replenish NAD+ from the NADH + H+ produced in glycolysis.

One type of fermentation is alcohol fermentation. First, pyruvate is decarboxylated (CO_2 leaves) to form acetaldehyde. Hydrogen atoms from NADH + H+ are then used to help convert acetaldehyde to ethanol where NAD+ results. Facultative anaerobes are organisms that can undergo fermentation when they are deprived of oxygen. Yeast is an example of a

facultative anaerobe that will undergo alcohol fermentation (Figure 3.22).

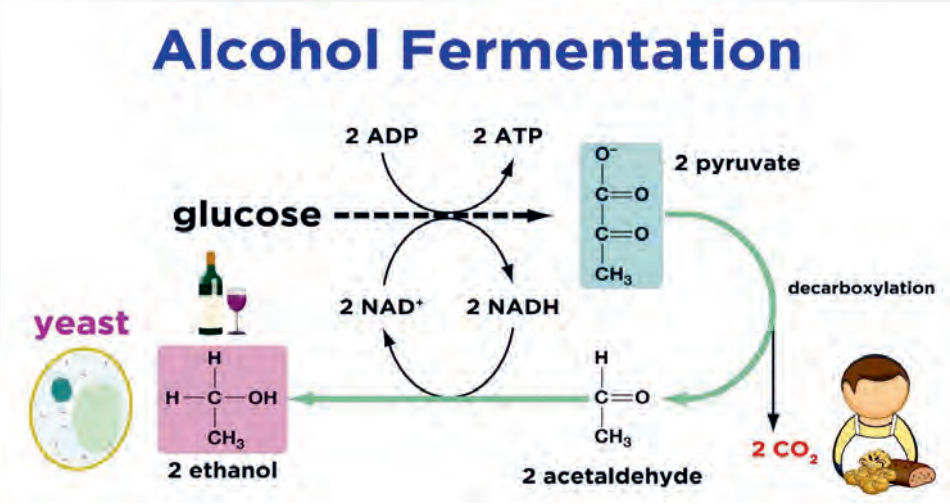


Figure 3.22. Alcohol fermentation

Lactic acid fermentation

Lactic Acid Fermentation: is the process by which pyruvate molecules are converted to lactic acid in the muscle cells of humans, and also in the cells of bacteria.

During lactic acid fermentation, the pyruvate molecules from glycolysis are used to oxidize NADH and convert it back to NAD^+. During the process, lactic acid or lactate is produced as a byproduct. Most animals and some bacteria can carry out lactic acid fermentation. Animals use the process to regenerate NAD+ in the absence of oxygen (Figure 3.23). Anaerobic respiration doesn't produce enough ATP to power the entire organism, but can be used to supplement the ATP levels in tissues (like muscle) where oxygen levels may drop quickly. The products of bacterial lactic acid fermentation have been used by humans to produce food products such as yogurt, sour cream, and buttermilk.

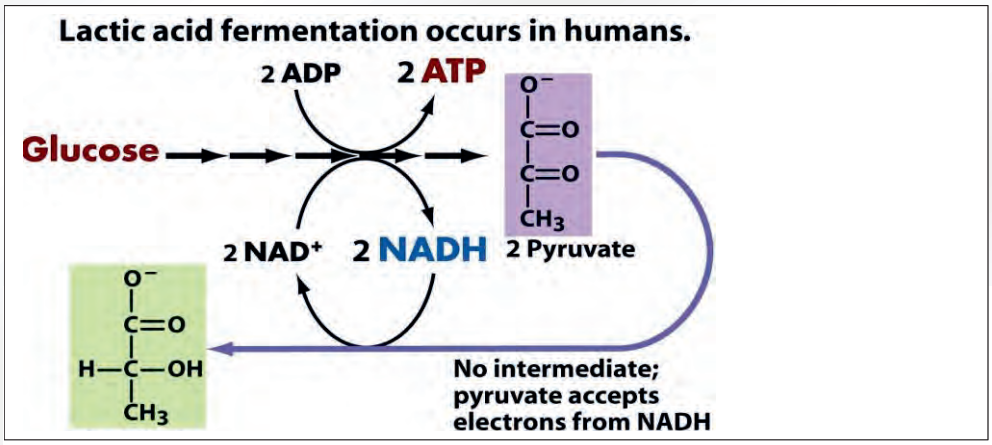


Figure 3.23. Lactic acid fermentation

Activity 3.10. Investigation of the rate of fermentation in yeast

There are many different ways of carrying out this investigation, that range from those using only basic equipment to sophisticated electronically monitored fermenters.

The test tube containing the yeast and glucose can be held in a water bath at the desired temperature and the number of bubbles collected per minute recorded. However, rate of bubbling is not the most accurate way of measuring rate of respiration. Are you sure that all the bubbles are the same volume? The method is improved if the gas syringe replaces test tube of water.

Using this basic equipment, devise experiments to investigate:

- the effect of temperature on the rate of fermentation the effect of different substrates (different sugars) on the rate of fermentation
- the effect of substrate concentration on the rate of fermentation

In your plans, you should make clear:

- The independent variable
- The dependent variable
- Other variables that you intend to control as well as:
  - Why you need to control them, and
  - How you intend to control them.
- More sophisticated fermenters control all the conditions inside the fermenter and monitor the changes in the concentration of oxygen, carbon dioxide and ethanol. Other sensors could also monitor the concentration of the sugar being fermented.
  - Can you explain the changes in the concentrations of the various substances as fermentation proceeds?

Table 1. Results from Part A.

Tube #	Sugar	Start Time	End Time	Duration (Min)	Volume of CO_2 (ml)	Fermentation rate(ml CO2/min)
1	Glucose					
2	Fructose					
3	Sucrose					
4	Lactose					

Introduction for Part B – Effect of Temperature on Fermentation

In this experiment you will be investigating the effects of temperature on the fermentation rate of yeast. As in Part A, CO2 production will be used as a measure of fermentation rate. However, instead of manipulating the sugar types, only one sugar (glucose) will be used in all the fermentation tubes. The fermentation tubes will be placed in water baths of different temperatures (0°C, ~22°C, 37°C, and 70°C) to see how the temperatures affect the fermentation rates.

Procedure for Part B:

1. Label 4 clean fermentation tubes (1- 4). Using the graduated cylinder in your tray, measure and pour 5 ml of the yeast suspension into each of your four fermentation tubes. Be sure to mix the suspension before dispensing. You may need to use a pipette to accurately bring the volume of the graduated cylinder to exactly 5 ml.

2. using the graduated cylinder, add 7 ml of glucose solution to each of the fermentation tubes.

3. Tip the fermentation tubes so that the vertical column of each tube fills with the liquid.

4. Place tube #1 in the 0° C ice-water bath; tube #2 in the room temperature bath (record the exact temperature in Table 2); tube #3 in the 37° C water bath; and tube #4 in the 70° C water bath. Record the time in Table 2.

5. Monitor the amount of CO_2 produced. This may take some time. When ONE of the tubes is half filled with CO_2, note the time, and remove ALL tubes from the water bath.

6. Measure the volumes (in ml) of gas in each tube and record them in the table below.

Calculate the fermentation time by subtracting the starting time from the ending time, and record it in the table. Calculate the fermentation rates at each of the temperatures from these data.

7. BEFORE YOU THROW ANYTHING AWAY!!!...Take tubes #1 (0° C) and #4 (70° C) and place them both in the 37° C water bath for 5 minutes to let them both equilibrate to that temperature. After 5 minutes, tip the tubes to mix and remove the air, and leave them in the bath to incubate. Record the time.

8. Monitor the amount of CO2 produced. This may take some time. When ONE of the tubes is half filled with CO2, note the time, and remove BOTH tubes from the water bath.

9. Measure the volume (in ml) of gas in each tube and record them it Table 2 below. Calculate the fermentation time by subtracting the starting time from the ending time, and record it in the table. Calculate the fermentation rate at each of the temperatures from these data.

10. WASH AND RINSE ALL OF YOUR TUBES (contents can be poured down the sink). Tip the tubes as you wash them to thoroughly clean them.

Table 2. Results from Part B.

Tube #	Temp. ^{o}C	Temp. ^{o}F	Start Time	End Time	Duration (min)	Volume of CO_2 (ml)	Fermentation rate (ml CO_2/min)
1	0						
2							
3	37						
4	70						
1	0→37						
4	70→37						

Unit Summary

Metabolism is the sum of chemical reactions that takes place within each cell of an organism. The chemical reactions enable cells to produce energy for vital processes and also synthesize new organic materials. Broadly, these reactions can be divided into catabolic reactions that convert nutrients to energy and anabolic reactions that lead to the synthesis of larger biomolecules.

Photosynthesis is the process by which photosynthetic organisms convert the energy of sunlight into chemical energy stored in sugars. Inside a chloroplast, photosynthesis occurs in two stages: the light-dependent reactions and the light-independent (or Calvin Cycle) reactions.

The light dependent reactions, a light-dependent series of reactions which occur in the grana, and require the direct energy of light to make energy-carrier molecules (NADPH and ATP) that are used in the light independent phase.

The light-independent reactions, a light-independent series of reactions which occur in the stroma of the chloroplasts, when the products of the light reaction, ATP and NADPH, are used to make carbohydrates from carbon dioxide (reduction); initially glyceraldehyde 3-phosphate (a 3-carbon atom molecule) is formed.

Cellular respiration is the process by which cells produce energy from glucose in the form of energy storing compound called ATP(AdenosineTri-Phosphate) for various cellular activities. Cellular respiration can take place in the presence as well as in the absence of oxygen. When it utilizes oxygen, it reffered to as aerobic respiration. Aerobic respiration takes place in the mitochondria. It is more efficient as it yields about 36-38ATP per mole gilucose consumed. During glycolysis, glucose molecules (six-carbon molecules) are split into two pyruvates (three-carbon molecules) during a sequence of enzyme-controlled reactions. Glycolysis takes place in the cytosol or cytoplasm.

Anaerobic respiration occurs in the absence of oxygen. Without oxygen, pyruvate is converted to lactic acid in animals or ethanol in plants and yeast. It produces only about 10% of the energy released in the complete oxidation of glucose.

Anaerobic respiration in humans takes place when muscle undergoes extreme contraction as in vigorous exercise. When oxygen is limited the oxidation of NADH to NAD^+ by the electron transport chain is insufficient to maintain glycolysis. Under these conditions NAD^+ is regenerated by the reduction of pyruvate to lactate.

Unit review questions

I. Choose the correct answer for the following questions.

1. Where does the energy come from to drive photosynthesis?
   A. Chloroplast C. Sun
   B. ATP D. Glucose
2. Which colors light is predominantly absorbed by the chlorophyll?
   A. Red B. Green C. Yellow D. Orange
3. Which plant usually opens its stomata during night time?
   A. C2 plant. B. C3 plant. C. C4 plant. D. CAM plant.
4. Which of the following processes takes place during the dark reaction of photosynthesis?
   A. Photolysis of water C. Production of ATP
   B. Fixation of CO_2 D. Production of NADPH
5. Where do the enzymatic reactions of the Calvin cycle take place?
   A. stroma of the chloroplast. C. cytoplasm of plant cell
   B. thylakoid membranes D. outer membrane of the chloroplast
6. Why are C4 plants able to photosynthesize with no apparent photorespiration?
   A. They are capable of excluding oxygen from their tissues.
   B. They have a mechanism to use PEP carboxylase to initially fix CO_2.
   C. They have special adaptations to cold and wet climatic conditions.
   D. They are capable of conserving water more efficiently.
7. The equation below represents a summary of a biological process.
   Glucose + Oxygen → Carbon dioxide + Water + Energy
   Where does this process take place in the cell?
   A. Cytoplasm C. Thylakoid of chloroplast
   B. Mitochondrial matrix D. Stroma of Chloroplast
8. What is the role of oxygen in cellular respiration?
   A. It is directly invoved in the oxidation of glucose during glycolysis.
   B . It accepts electrons during oxidation of pyruvate to acetyl CoA.
   C. It is directly involved in the oxidation process of the citric acid cycle.
   D. It accepts electrons at the end of the electron transport chain.
9. Which metabolic pathway is common to both cellular respiration and fermentation?
   A. The citric acid cycle

B. The oxidative phosphorylation.

C. The process of glycolysis,

D. The oxidation of pyruvate to acetyl CoA.

10. How does photorespiration reduce the efficiency of photosynthesis?

A. By preventing the formation of carbon dioxide molecules.

B. By preventing the formation 3-phosphoglycerate molecules

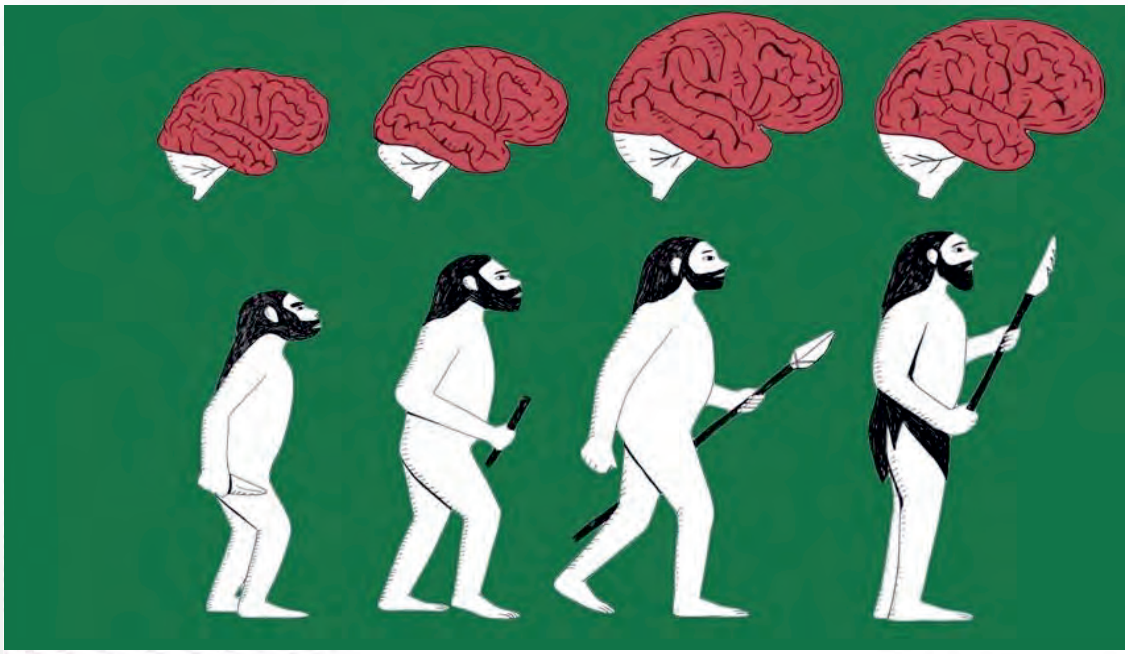
C. By enhancing the formation of oxygen molecules.

D. By enhancing the formation of RuBP carboxylase molecules.

II. Answer the following questions

1. Distinguish between anabolic and metabolic pathways.
2. Describe the process of cellular respiration.
3. Justify why oxidation of glucose involving mitochondrion is more efficient.
4. During a race how can your muscles cells produce ATP if oxygen deficiency occurs?
5. What are the chlorophyll pigments responsible for absorption of light spectra?
6. Compare cyclic and non-cyclic photophosphorylation
7. What are the products of the light reacttions subsequently used by the Calvin cycle?
8. Where do the enzymatic reactions of the Calvin cycle take place?
9. Compare the mechanism of carbon fixation C3 and C4 plants.
10. What is the primary function of the Calvin cycle?
11. Compare the process of photosynthesis and cellular respiration.
12. Justify how the processes of cellular respiration and photosynthesis help to maintain the balance of CO_2 and O_2 in the atmosphere.

UNIT 4: EVOLUTION



4.1. EVOLUTION

After successful completion of this unit, the learner will be able to:

- Analyze different views about the origin of life and evaluate views from a scientific explanations point of view
- Describe scientific evidence to support that modern living things have evolved from a common ancestor.
- Explain the major causes of evolution that lead to variation in organisms
- Appreciate the role of natural selection and survival of the fittest in evolution and day to day life

4.1.1. Definition

By the end of this section, the learner will be able to:

- Define evolution.
- Debate on the different thoughts of the origin of life.

Self-Assessment Questions

- What do you understand by the word 'evolution'?
- Have you ever asked yourself how did all millions of new species appear on our planet? Explain

Evolution is a change in genetic composition of a population over successive generations, which may be caused by meiosis, hybridization, natural selection or mutation. This leads to a sequence of events by which the population diverges from other populations of the same species and may lead to the origin of a new species.

Keyword

Evolution- The theory of evolution describes how the various forms of life on Earth (including humans) emerged and evolved.

Theories of the origin of life

The origin of life means the emergence of heritable and evolvable self-reproduction. "Origin of Life" is a very complex subject, and oftentimes controversial. Two opposing scientific theories that existed on this complex subject for a long time were the so called **intelligent design** and **creationism**. The **big bang theory of the origin of the Universe** gave new ideas about the topic of biological evolution. In the theory it has been hypothesized that complex life-forms on Earth, including humans, arose over a period of time from simple bacteria like tiny cells by a process of self-organization similar to the evolution of the Universe of simple

Keyword

Creationism (or special creationism) is a theory claiming that the different forms of life on Earth were created by a supreme being.

Self-questioning

What basic differences can you draw from the following regarding the origin of life?

- Religious thoughts and science
- Observation, provable events and empirical facts against opinions about morality or purpose

material structures toward more and more complex structures. There are several theories about the origin of life. Some of them are mentioned below.

1. Special creationism

Special creationism explained that the formation of life on earth may have been taken place due to **supernatural** or **divine forces**. However, acceptance of evolution is linked to scientific thinking. There are fundamental differences between special reationism and scientific thinking in that it is nlikely that the difference between the two will ever be resolved. Special creation states that at some stage, some supreme being created life on Earth. There are many different versions of special creation, linked with different religions. Often, there is considerable variation as to how rigidly the special creation theory is interpreted within a religion.

Activity 4.1

Make a small group and read from different sources about the following forms of creationism theories and reflect your idea to the class.

-Young Earth creationism
- Old Earth creationism
- Day-age and gap creationism
- Progressive creationism
- Theistic evolution/
- Evolutionary creationism
- Intelligent design

2. Spontaneous generation (abiogenesis)

Spontaneous generation suggests that some life can evolve spontaneously' from non-living objects. The Greek philosopher Aristotle (384-322) was one of the earliest scholars to articulate the theory. It was once believed that life could come from nonliving things, such as mice from corn, flies from bovine manure, maggots from rotting meat, and fish from the mud of previously dry lakes.

Keyword

Spontaneous generation- a theory that claimed that some types of organism could come into being almost instantly from non-living materials

In 1668, Francesco Redi, an Italian scientist, designed a scientific experiment to test the spontaneous creation of maggots by placing fresh meat in jars (Figure 4.1). One jar was left open; the others were covered with a cloth. Days later, the open jar contained maggots, whereas the covered jars contained no maggots. He did note that maggots were found on the exterior surface of the cloth that covered the jar. Redi successfully demonstrated that the maggots came from fly eggs and thereby helped to disprove spontaneous generation.

In 1748, the English priest John Needham (1713–1781) reported the results of his experiments on spontaneous generation. Needham boiled mutton broth and then tightly stoppered the flasks. Eventually many of the flasks became cloudy and contained microorganisms. He thought organic matter contained a vital force that could confer the properties of life on nonliving matter.

A few years later, the Italian priest and naturalist Lazzaro Spallanzani (1729–1799) improved on Needham's experimental design by first sealing glass flasks that contained water and seeds. If the sealed flasks were placed in boiling water for 3/4 of an hour, no growth took place as long as the flasks remained sealed. He proposed that air carried germs to the culture medium, but also commented that the external air might be required for growth of animals already in the medium. The supporters of spontaneous generation maintained that heating the air in sealed flasks destroyed its ability to support life.

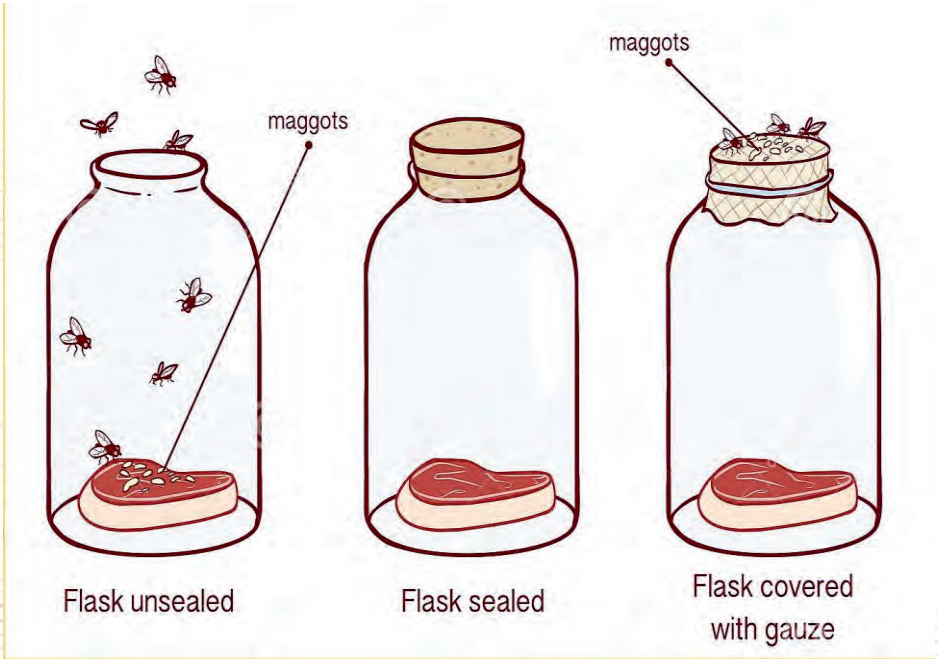


Figure 4.1 Fransesco Redi's experiment set up.

Louis Pasteur, the notable French scientist, showed that broth (or wine) only went sour if micro-organisms were allowed to enter. Also no micro-organisms appeared in the broth unless they were allowed to enter from the outside (Figure 4.2). Pasteur made a series of flasks with long, twisted necks ("swan-neck" flasks), in which he boiled broth to sterilize it (Figure 4.2). His design allowed air inside the flasks to be exchanged with air from the